CAMBRIDGE INTERNATIONAL · AS & A LEVEL PHYSICS

9702

PAPER 1

24 / 25

MULTIPLE CHOICE · COMPLETE COLLECTION 2024–2025

WITH ANSWER KEYS

PHYSICS (9702) · 15 PAPERS

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March 2024 · Paper 12

9702_m24_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **February/March 2024**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which row shows a physical quantity and its base unit in the SI system?

	quantity	unit
A	current	A
B	force	N
C	mass	g
D	temperature	°C

2 A car of mass 850 kg is travelling in a horizontal straight line. The diagram shows the two horizontal forces acting on the car in opposite directions.

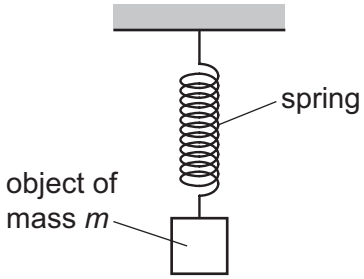


One force has magnitude 1200 N, and the other force has magnitude 1600 N.

What is the magnitude of the acceleration of the car?

A $0.47\,\text{m s}^{-2}$ **B** $1.4\,\text{m s}^{-2}$ **C** $1.9\,\text{m s}^{-2}$ **D** $3.3\,\text{m s}^{-2}$

3 An object of mass m is suspended by a spring from a fixed point.



The spring has spring constant k. The object is set into vertical oscillations of period T.

Which equation for T is homogeneous with respect to base units?

A $T = 2\pi\left(\frac{k}{m}\right)$ **B** $T = 2\pi\left(\frac{m}{k}\right)$ **C** $T = 2\pi\sqrt{\frac{k}{m}}$ **D** $T = 2\pi\sqrt{\frac{m}{k}}$

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4 An object of fixed mass is initially at rest at point P. The object then moves away from point P with uniform acceleration.

Which statement describes the resultant force acting on the object when it is moving?

A It increases uniformly with respect to time.

B It is constant but not zero.

C It is proportional to the displacement from point P.

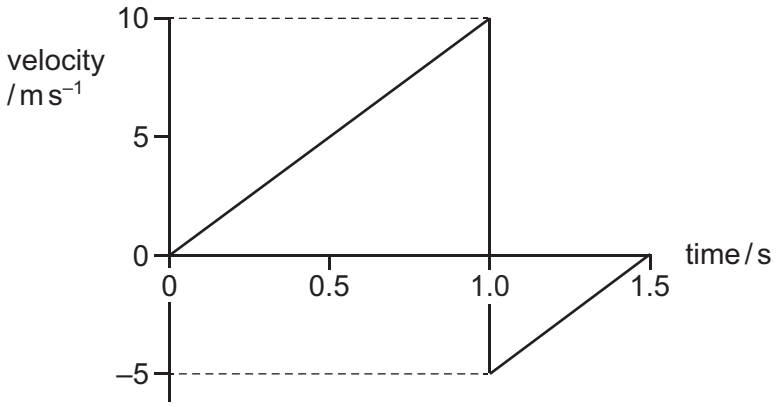
D It is zero.

5 A projectile is launched at an angle of 25° to the horizontal with a horizontal component of velocity of $13\,m\,s^{-1}$.

What is the vertical component of the velocity of the projectile when it is launched?

A $5.5\,m\,s^{-1}$ **B** $6.1\,m\,s^{-1}$ **C** $12\,m\,s^{-1}$ **D** $14\,m\,s^{-1}$

6 A ball is released from rest at position X at time zero. At 1.0 s, it bounces inelastically from a horizontal surface and rebounds, reaching the top of its first bounce at 1.5 s.



What is the total displacement of the ball from its original position X at 1.5 s?

A 1.25 m **B** 3.75 m **C** 5.00 m **D** 6.25 m

7 What is the definition of acceleration?

A change in velocity per unit time

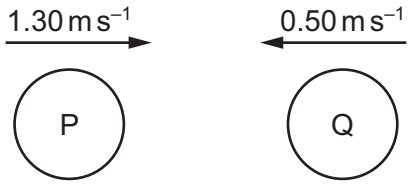
B rate of change of speed per unit time

C rate of change of velocity per unit time

D resultant force per unit mass

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8 Two balls P and Q, of equal mass, move along a straight line directly towards each other as shown.

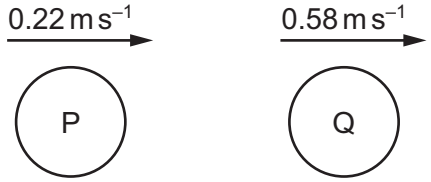


Ball P has velocity $1.30\,m\,s^{-1}$ to the right. Ball Q has velocity $0.50\,m\,s^{-1}$ to the left.

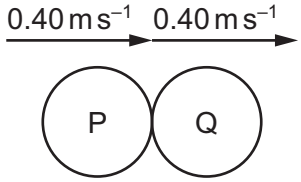
P and Q collide with one another. The collision is perfectly elastic and the total momentum is conserved.

Which diagram correctly shows the motion of P and Q after the collision?

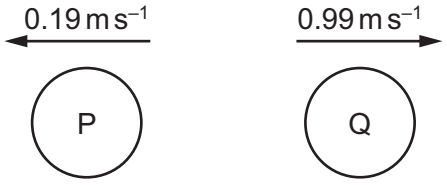
A



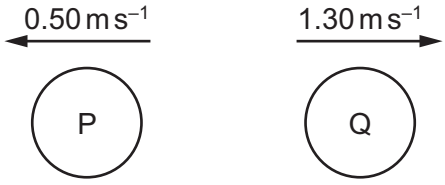
B



C



D



9 A basketball player hits a ball vertically downwards with a speed of $2.4\,m\,s^{-1}$ from a height of 0.90 m.

Air resistance is negligible.

What is the speed of the ball as it hits the ground?

A $4.2\,m\,s^{-1}$ **B** $4.8\,m\,s^{-1}$ **C** $18\,m\,s^{-1}$ **D** $23\,m\,s^{-1}$

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10 A ball falls through a liquid at a constant speed. It is acted upon by three forces: an upthrust, a drag force and its weight.

The liquid has a uniform density.

Which statement is correct?

A The drag force increases with increasing depth.

B The drag force is equal to the sum of the upthrust and weight.

C The upthrust is constant with increasing depth.

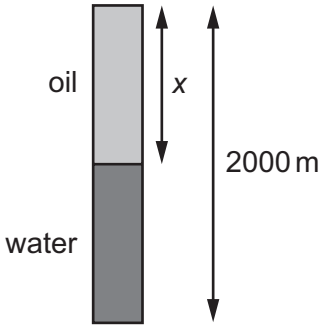
D The weight is greater than the sum of the drag force and the upthrust.

11 Some small solid cubes each have mass 1.0 kg and sides of length 5.0 cm. These small cubes are stacked together to form a large solid cube with sides of length 2.0 m.

What is the weight of the large cube?

A 0.39 kN **B** 0.39 MN **C** 0.63 MN **D** 0.63 GN

12 A borehole of depth 2000 m contains both oil and water, as shown. The pressure due to the liquids at the bottom of the borehole is 17.5 MPa. The density of the oil is 830 kg m^{-3} and the density of the water is 1000 kg m^{-3}.



What is the depth x of the oil?

A 907 m **B** 1000 m **C** 1090 m **D** 1270 m

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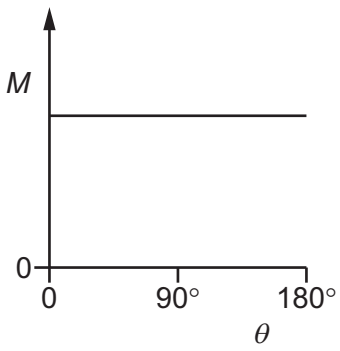
13 A rod is pivoted at one end. Initially the angle θ of the rod to the horizontal is 0°.

The weight of the rod causes a moment M about the pivot.

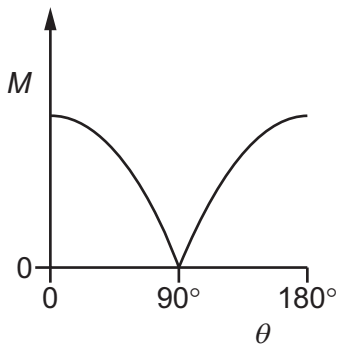
The rod is then rotated in the vertical plane so that the angle θ of the rod increases from 0° to 180°.

Which graph shows the variation of M with θ?

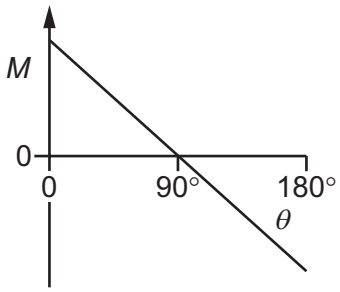
A



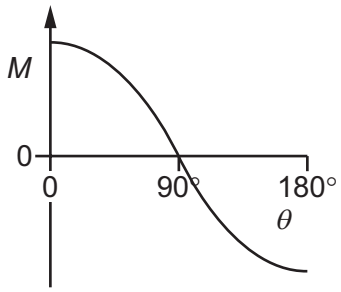
B



C

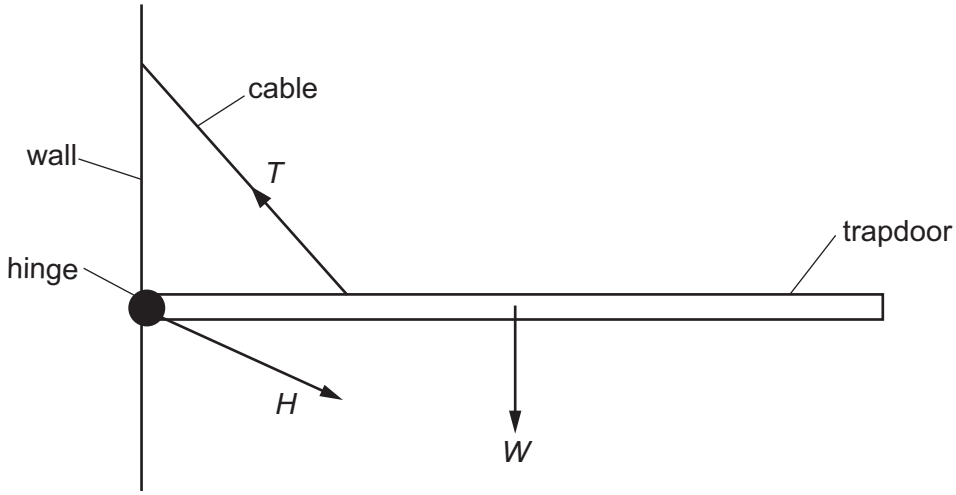


D



14 A hinged trapdoor is held closed in the horizontal position by a cable.

Three forces act on the trapdoor: the weight W of the door, the tension T in the cable and the force H at the hinge.



Which list gives the three forces in **increasing** order of magnitude?

A H, T, W **B** T, H, W **C** W, H, T **D** W, T, H

15 What is the definition of force?

A the product of mass and acceleration

B the product of mass and velocity

C the rate of change of momentum

D the rate of transfer of energy

16 A boat moves at a constant velocity v through still water.

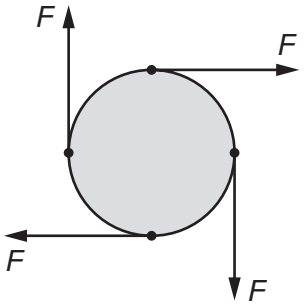
A constant drag force F acts on the boat.

What is the power used by the boat to move through the water?

A $\frac{1}{2}Fv$ **B** Fv **C** $\frac{1}{2}Fv^2$ **D** Fv^2

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17 The diagram shows four forces acting on a circular disc.



Each force has magnitude *F*. Two of the forces act vertically and the other two forces act horizontally.

All four forces act in the same plane as the disc. No other forces act on the disc.

The disc has diameter *d*.

Which statement is correct?

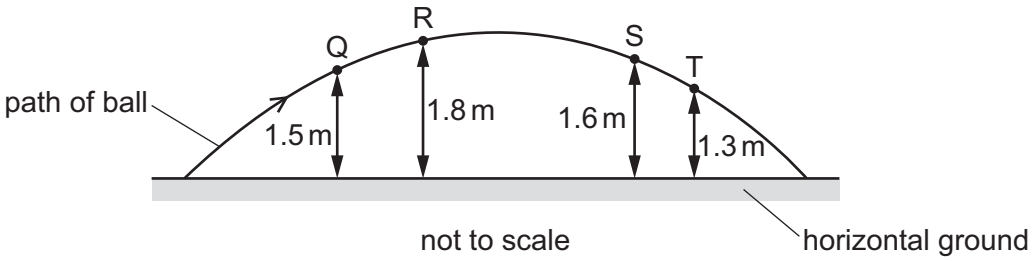
A The disc is in equilibrium because the resultant force is zero.

B The disc is **not** in equilibrium because the resultant force is 4*F*.

C The disc is in equilibrium because the resultant torque is zero.

D The disc is **not** in equilibrium because the resultant torque is 2*Fd*.

18 A ball is projected into the air from horizontal ground and follows the path shown in the diagram.



At points Q, R, S and T, the ball has kinetic energies E_Q, E_R, E_S and E_T respectively. The heights above the ground of these four points are shown.

Air resistance is negligible.

Which difference in kinetic energies is the smallest?

A $E_Q - E_S$ **B** $E_S - E_R$ **C** $E_T - E_Q$ **D** $E_T - E_R$

19 The battery of a small tablet computer is initially uncharged. It is connected to a constant 10 W power supply for 2.0 hours to charge the battery.

The efficiency of the charging process is 80%.

What is the total energy stored in the battery?

A 1.6×10^1 J **B** 1.6×10^3 J **C** 5.8×10^4 J **D** 5.8×10^6 J

20 An initially stationary firework explodes and splits into two fragments that move horizontally in opposite directions.

The total kinetic energy transferred to the fragments by the explosion is E.

One fragment has mass m and the other one has mass $2m$.

What is the speed of the fragment of mass m immediately after the explosion?

A $\sqrt{\frac{E}{m}}$ **B** $\sqrt{\frac{2E}{m}}$ **C** $\sqrt{\frac{2E}{3m}}$ **D** $\sqrt{\frac{4E}{3m}}$

21 A spring is fixed at one end and extended by applying force F to the other end. The spring has extension x and elastic potential energy E_P. The spring constant is k.

The spring obeys Hooke's law.

Which relationship is correct for this spring?

A $E_P \propto F$ **B** $E_P \propto x$ **C** $E_P \propto k$ **D** $E_P \propto x^2$

22 A force–extension graph is produced for a metal wire.

What **must** describe the limit of proportionality of the wire?

A the point at which the wire breaks

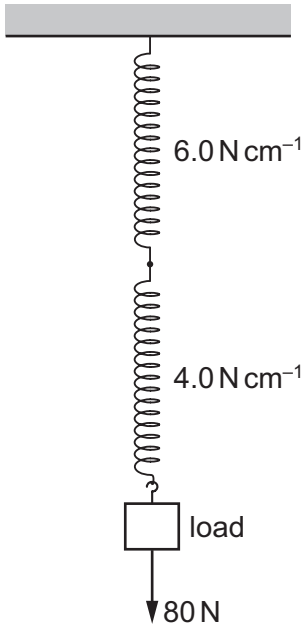
B the point beyond which Hooke's law is not obeyed

C the point beyond which the wire cannot return to its original length

D the point beyond which the wire starts to deform plastically

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23 A spring has a spring constant of $6.0\,N\,cm^{-1}$. It is joined to another spring whose spring constant is $4.0\,N\,cm^{-1}$. A load of 80 N is suspended from this composite spring.



What is the extension of this composite spring?

A 8.0 cm **B** 16 cm **C** 17 cm **D** 33 cm

24 The range of frequencies of sound waves emitted by blue whales is 10 Hz to 40 Hz.

The speed of sound in seawater is approximately $1.5\,km\,s^{-1}$.

What is the approximate range of wavelengths of the sound waves emitted by blue whales?

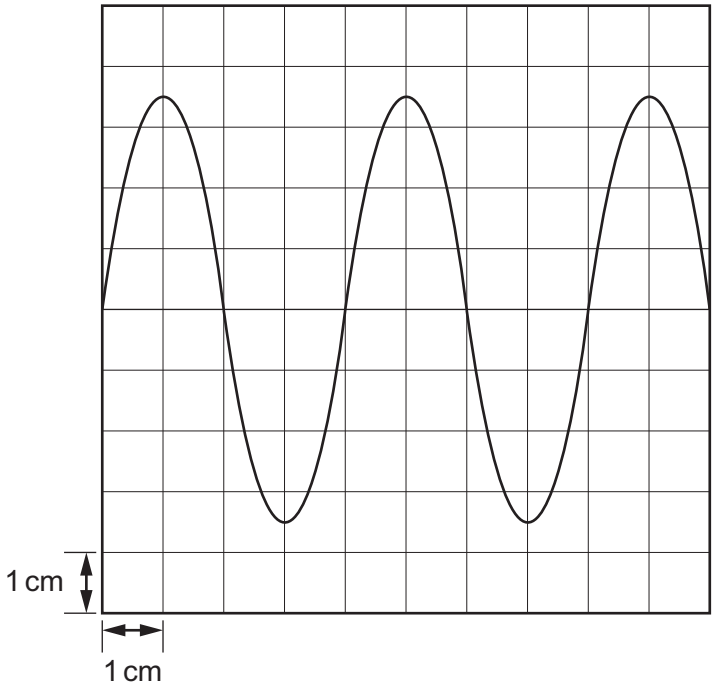
A 6.7 mm to 27 mm

B 3.8 cm to 15 cm

C 6.7 m to 27 m

D 38 m to 150 m

25 A sound wave is detected by a microphone and displayed on the screen of a cathode-ray oscilloscope (CRO).



The frequency of the wave is 2.5 kHz.

What is the setting on the time-base of the CRO?

A $0.1\,\text{ms}\,\text{cm}^{-1}$ **B** $0.4\,\text{ms}\,\text{cm}^{-1}$ **C** $0.1\,\text{s}\,\text{cm}^{-1}$ **D** $0.4\,\text{s}\,\text{cm}^{-1}$

26 A source of sound waves with constant frequency moves towards a stationary observer.

The observer compares the sound waves arriving at the observer's position with the waves emitted by the source of sound.

What is detected by the observer?

A a decreased frequency of the sound waves

B no change in frequency of the sound waves

C a decreased wavelength of the sound waves

D no change in wavelength of the sound waves

27 Which type of waves **cannot** be polarised?

A radio waves

B sound waves

C ultraviolet waves

D X-rays

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28 A beam of light with power P has an area of cross-section A.

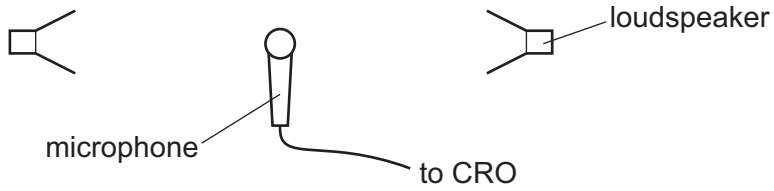
The amplitude of the light waves in the beam is X.

The beam of light is then changed to one with the same frequency but with an increased amplitude of $4X$ and an area of cross-section reduced to $\frac{A}{3}$.

What is the power of the new beam?

A $1.3P$ **B** $5.3P$ **C** $12P$ **D** $48P$

29 Two loudspeakers are connected to the same signal generator. The signal generator produces a single frequency. The loudspeakers face each other so that a stationary sound wave is set up in the region between the loudspeakers.



A microphone is connected to a cathode-ray oscilloscope (CRO) and positioned between the two loudspeakers.

The microphone is moved along a line joining the two loudspeakers.

The signal on the CRO shows 5 maximum amplitudes as the microphone moves. The microphone moves a distance of 2.0 m from the position that gives the first maximum to the position that gives the fifth maximum.

What is the wavelength of the sound wave?

A 0.40 m **B** 0.50 m **C** 0.80 m **D** 1.0 m

30 Two wave sources emit coherent waves.

Which condition **must** be correct for the coherent waves?

A The waves are emitted in phase.

B The waves are emitted and move in opposite directions.

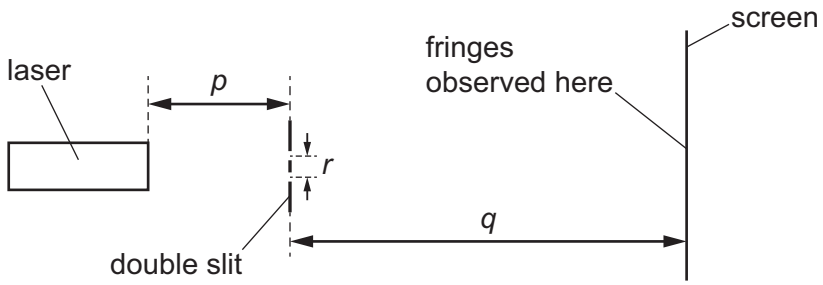
C The waves are emitted with a constant phase difference.

D The waves are emitted with the same amplitude.

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31 A student sets up an experiment to investigate double-slit interference.

The student uses light of a single wavelength from a laser to illuminate a double slit so that a pattern of interference fringes is observed on the screen.



The student finds that the fringes are very close together.

What could the student **decrease** in order to increase the separation of the fringes on the screen?

A the distance *p* from the laser to the double slit

B the distance *q* from the double slit to the screen

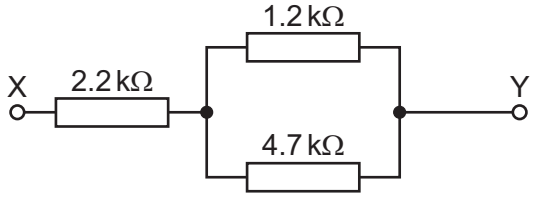
C the separation *r* of the slits

D the wavelength of the light from the laser

32 What are the definitions of potential difference (p.d.) and electromotive force (e.m.f.), in terms of energy transfer *W* and charge *q*?

	p.d.	e.m.f.
A	$\frac{W}{q}$	$\frac{W}{q}$
B	$\frac{W}{q}$	Wq
C	Wq	$\frac{W}{q}$
D	Wq	Wq

33 The diagram shows a network consisting of three resistors.



What is the combined resistance of the network between terminal X and terminal Y?

A 0.67 kΩ **B** 1.6 kΩ **C** 3.2 kΩ **D** 8.1 kΩ

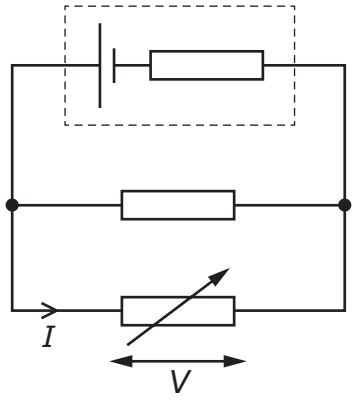
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34 A resistor dissipates 25 W of power when there is a potential difference (p.d.) of 4.0 V across it.

What is the resistance of the resistor?

A 0.16 Ω **B** 0.64 Ω **C** 100 Ω **D** 400 Ω

35 The diagram shows a cell with internal resistance connected in parallel with a fixed resistor and a variable resistor.



The resistance of the variable resistor is decreased.

What happens to the potential difference V across the variable resistor and the current I in the variable resistor?

	V	I
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

36 A radioactive source produces a beam of α-particles in a vacuum. The average current caused by the α-particles in the beam is 1.5×10^{-9} A.

The beam is incident on a metal target.

What is the average number of α-particles hitting the metal target in a time of 3.0 s?

A 4.7×10^{9} **B** 9.4×10^{9} **C** 1.4×10^{10} **D** 2.8×10^{10}

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37 The charge carriers in a metal wire are free electrons.

Which statement about the charge of each free electron is correct?

A The magnitude of the charge increases with the potential difference across the wire.

B The magnitude of the charge is zero unless there is a potential difference across the wire.

C The sign and magnitude of the charge do **not** depend on the potential difference across the wire.

D The sign of the charge depends on the potential difference across the wire.

38 Which flavours of quark have charge $+\frac{2}{3}e$?

	charm	strange	top	bottom
A	✓	✗	✓	✗
B	✓	✗	✗	✗
C	✗	✓	✓	✓
D	✗	✓	✗	✓

key

✓ = has charge $+\frac{2}{3}e$

✗ = does not have charge $+\frac{2}{3}e$

39 An unstable nucleus of an element decays by emitting an α-particle or a β^- particle to become a nucleus of a different element. This nucleus is also unstable and emits an α-particle or a β^- particle. The process continues until an isotope of the original element is produced.

What is the minimum possible number of these particles emitted?

A 2 **B** 3 **C** 4 **D** 5

40 A nucleus of carbon-10, ${}^{10}_{6}\mathrm{C}$, decays by beta-emission to form a nucleus of boron-10, ${}^{10}_{5}\mathrm{B}$.

For this decay process, what is the change to a nucleon and what is the change in the quark composition of the nucleon?

	change to nucleon	change in quark composition
A	proton to neutron	down becomes up
B	proton to neutron	up becomes down
C	neutron to proton	down becomes up
D	neutron to proton	up becomes down

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CAMBRIDGE
International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **February/March 2024**

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the February/March 2024 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

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Question	Answer	Marks
1	A	**1**
2	A	**1**
3	D	**1**
4	B	**1**
5	B	**1**
6	B	**1**
7	A	**1**
8	D	**1**
9	B	**1**
10	C	**1**
11	C	**1**
12	D	**1**
13	D	**1**
14	C	**1**
15	C	**1**
16	B	**1**
17	D	**1**
18	A	**1**
19	C	**1**
20	D	**1**
21	D	**1**
22	B	**1**
23	D	**1**
24	D	**1**
25	A	**1**
26	C	**1**
27	B	**1**
28	B	**1**

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Question	Answer	Marks
29	D	**1**
30	C	**1**
31	C	**1**
32	A	**1**
33	C	**1**
34	B	**1**
35	B	**1**
36	C	**1**
37	C	**1**
38	A	**1**
39	B	**1**
40	B	**1**

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June 2024 · Paper 11

9702_s24_qp_11

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2024**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which unit is an SI base unit?

A ampere

B coulomb

C degree Celsius

D gram

2 Which of the following could have the same units as force?

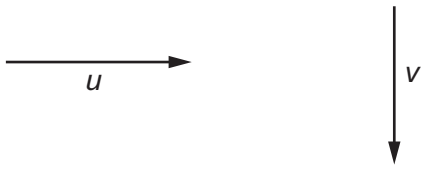
A $\frac{\text{energy}}{\text{distance}}$

B $\frac{\text{energy}}{\text{time}}$

C momentum × distance

D momentum × time

3 The velocity of an object changes from an initial velocity u to a final velocity v. The vectors represent these velocities.



Which single vector represents the change in velocity of the object?

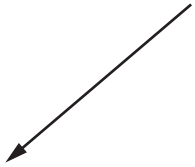
A



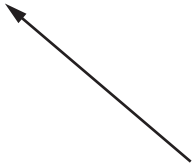
B



C



D



4 An object is moving with initial velocity u. The object then moves with uniform acceleration a for time t until it reaches final velocity v.

Which equation describes the motion of the object?

A $u = v - 2at$ **B** $u = v - at$ **C** $v = u + at^2$ **D** $v = u + 2at^2$

5 Which calculation produces a vector quantity?

A current × time

B final displacement – initial displacement

C $\frac{\text{work done}}{\text{time}}$

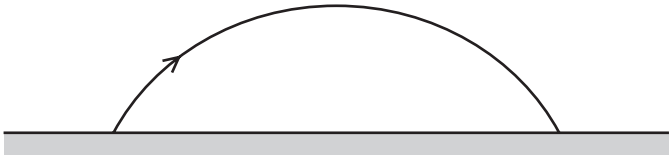
D $\frac{1}{2} \times \text{mass} \times (\text{speed})^2$

6 A thermometer can be read to an accuracy of ± 0.5 °C. This thermometer is used to measure a temperature rise from 40 °C to 100 °C.

What is the percentage uncertainty in the measurement of the temperature rise?

A 0.5% **B** 0.8% **C** 1.3% **D** 1.7%

7 The diagram shows the path of a golf ball.



Which row describes changes in the horizontal and vertical components of the golf ball's velocity when air resistance is ignored?

	horizontal	vertical
A	constant deceleration	constant acceleration downwards
B	constant deceleration	acceleration decreases upwards then increases downwards
C	constant velocity	constant acceleration downwards
D	constant velocity	acceleration decreases upwards then increases downwards

8 An aircraft flies from London to Sydney in a time of 21 hours 40 minutes.

The distance travelled is 17 000 km.

What is the average speed of the aircraft?

A $2.2\,\text{m}\,\text{s}^{-1}$

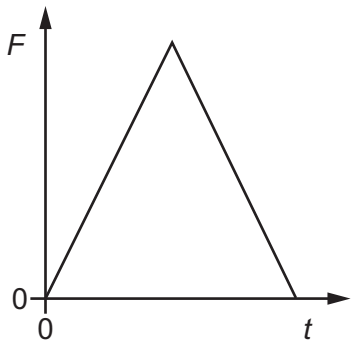
B $2.2 \times 10^{7}\,\mu\text{m}\,\text{s}^{-1}$

C $2.2 \times 10^{11}\,\text{nm}\,\text{s}^{-1}$

D $2.2 \times 10^{6}\,\text{mm}\,\text{s}^{-1}$

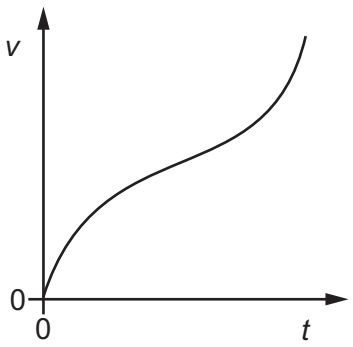
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9 A golf club hits a golf ball. The graph shows how the force F on the ball varies with time t.

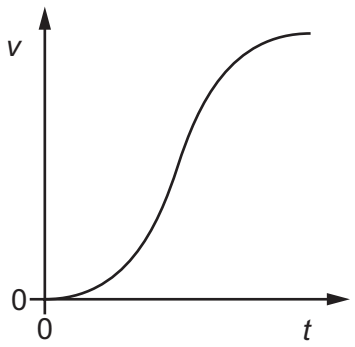


Which graph shows how the velocity v of the ball varies with time t?

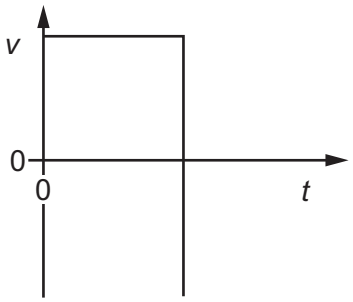
A



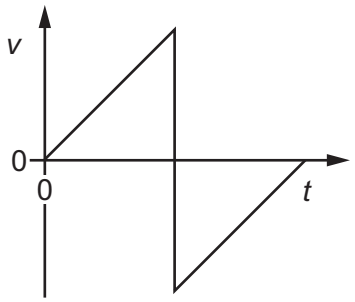
B



C



D



10 What is meant by the mass and by the weight of an object on the Earth?

	mass	weight
A	its momentum divided by its velocity	the work done in lifting it one metre
B	the gravitational force on it	the property that resists its acceleration
C	the pull of the Earth on it	its mass divided by the acceleration of free fall
D	the property that resists its acceleration	the pull of the Earth on it

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11 A thin horizontal plate of area $0.036\,m^2$ is beneath the surface of a liquid of density $930\,kg\,m^{-3}$. The force on one side of the plate due to the pressure of the liquid is 290 N.

What is the depth of the plate beneath the surface of the liquid?

A 0.88 m **B** 1.1 m **C** 1.8 m **D** 8.7 m

12 Spheres X and Y form an isolated system. The mass of Y is greater than the mass of X.

Sphere Y is initially stationary.

Sphere X collides elastically with sphere Y.

The speed of sphere X before the collision is u.

Which statement **must** be correct?

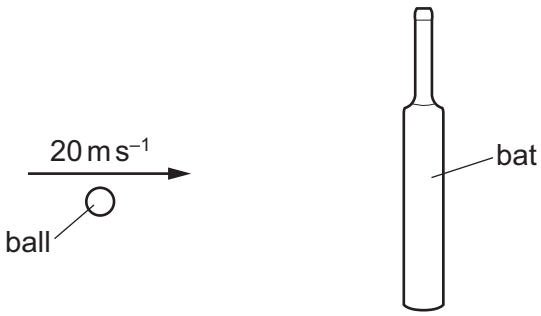
A Sphere X rebounds with a speed that is greater than u, and sphere Y moves off with a speed that is less than u.

B Sphere X rebounds with a speed that is less than u, and sphere Y moves off with a speed that is also less than u.

C Sphere X rebounds with speed u, and sphere Y remains stationary.

D Sphere X remains stationary, and sphere Y moves off with a speed that is less than u.

13 A ball of mass 0.10 kg is thrown towards a stationary vertical bat. The ball hits the bat with a horizontal velocity of $20\,m\,s^{-1}$.



The ball rebounds and leaves the bat with a horizontal velocity of $15\,m\,s^{-1}$.

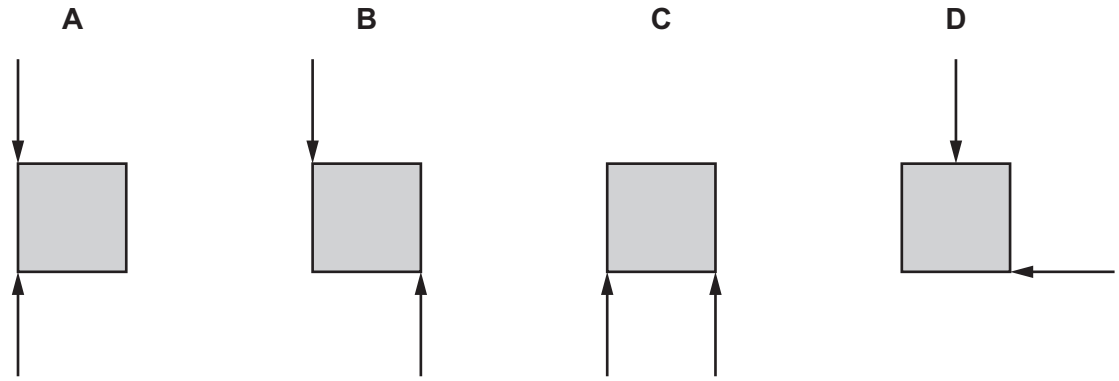
What is the change in momentum of the ball?

A 0.20 N s **B** 0.50 N s **C** 1.5 N s **D** 3.5 N s

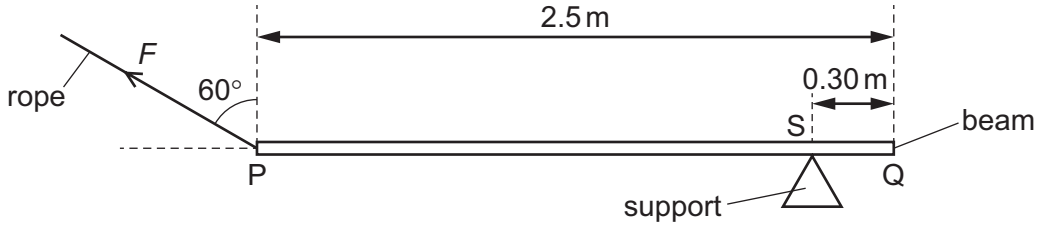
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14 An isolated object of negligible weight is acted on by two coplanar forces of the same magnitude.

In which diagram is the object in equilibrium?



15 A uniform beam PQ rests horizontally on a support at point S.

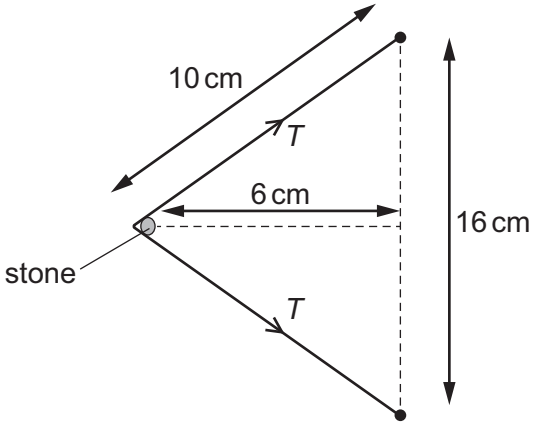


A rope is attached at one end of the beam. The rope is at an angle of 60° to the vertical and exerts a force F, in newtons, on the beam.

What is the moment, in N m, of the force F about the point S?

A $1.1F$ **B** $1.3F$ **C** $1.9F$ **D** $2.2F$

16 The diagram shows the dimensions of an elastic cord used to project a stone. The tension in the cord is T when the cord is pulled into the shape shown.



Which force does the elastic cord exert on the stone?

A $\frac{3}{5}T$ **B** $\frac{6}{5}T$ **C** $\frac{8}{5}T$ **D** $2T$

17 A box of weight 40 N is pushed with a horizontal force of 20 N along level ground for a distance of 2.4 m.

The box is then lifted at constant velocity through a height of 1.6 m by a vertical force.

What is the total work done on the box by the two forces?

A 80 J **B** 110 J **C** 120 J **D** 160 J

18 Which statement about efficiency is correct?

A Efficiency does **not** have a unit.

B The joule is a unit of efficiency.

C The metre is a unit of efficiency.

D The watt is a unit of efficiency.

19 A plane wave of amplitude A is incident on a surface of area S placed so that it is perpendicular to the direction of travel of the wave. The energy per unit time reaching the surface is E.

The amplitude of the wave is increased to $2A$ and the area of the surface is reduced to $\frac{1}{2}S$.

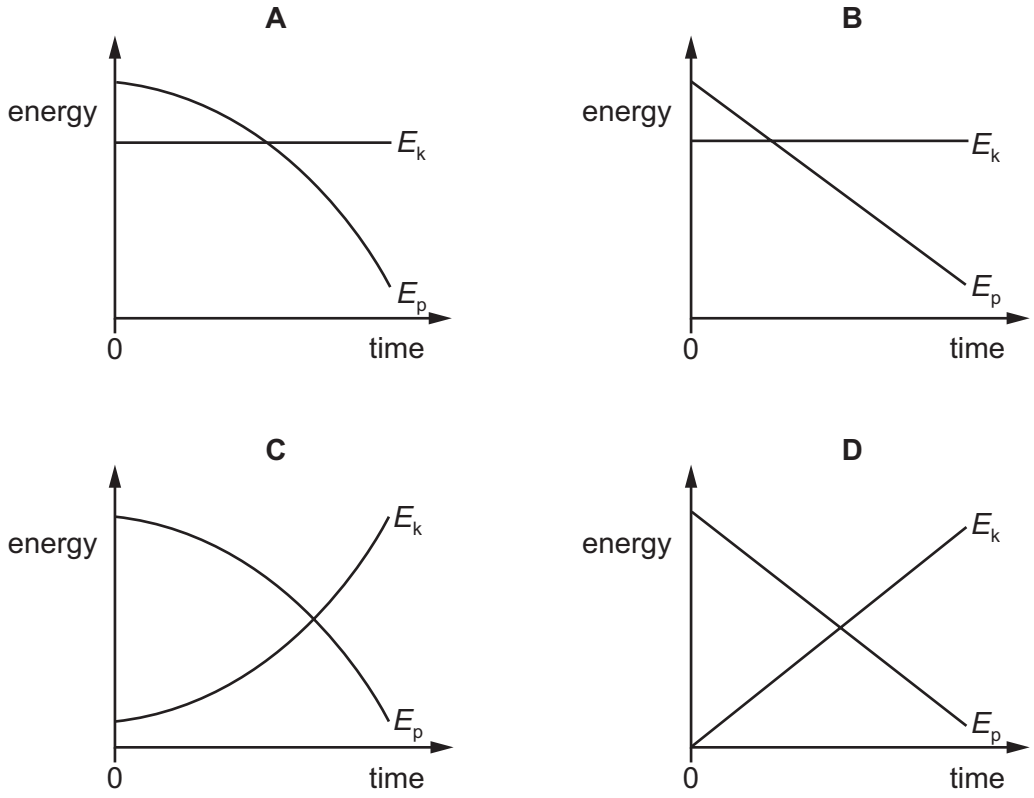
How much energy per unit time reaches this smaller surface?

A $4E$ **B** $2E$ **C** E **D** $\frac{1}{2}E$

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20 A steel ball is falling at constant speed in oil.

Which graph shows the variation with time of the gravitational potential energy E_p and the kinetic energy E_k of the ball?



21 When a force of 0.80 N is applied to a spring, the length of the spring is 90 mm.

When a force of 1.30 N is applied to the same spring, its length is 115 mm.

The spring obeys Hooke's law.

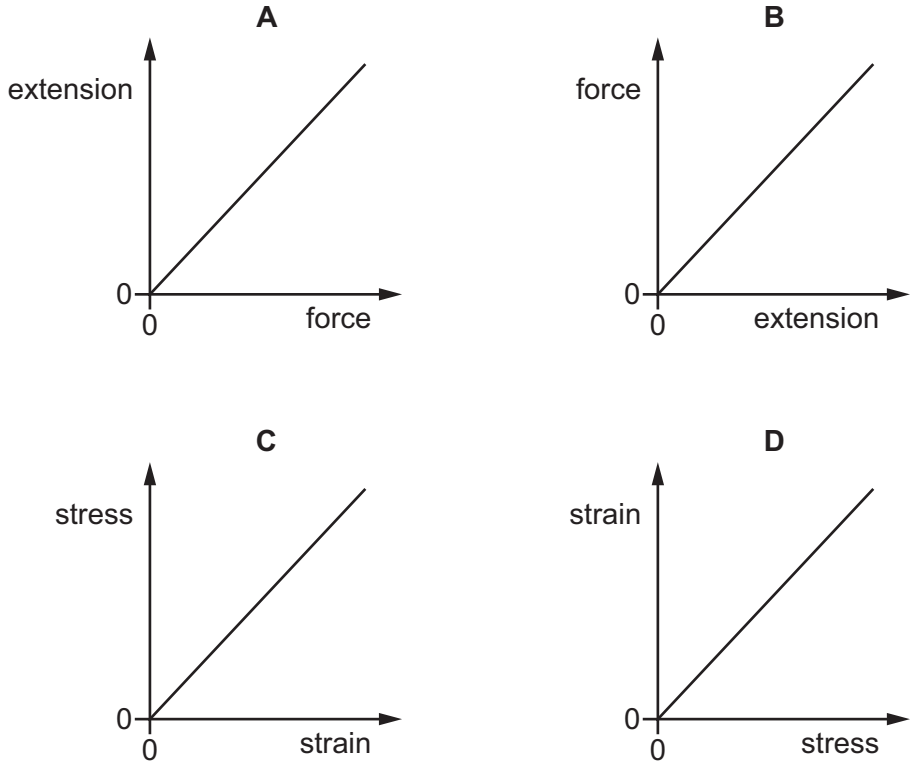
What is the spring constant of the spring?

A $8.9\,N\,m^{-1}$ **B** $10\,N\,m^{-1}$ **C** $11\,N\,m^{-1}$ **D** $20\,N\,m^{-1}$

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22 An experiment is carried out using a metal wire to investigate how it responds to a varying tensile force. The cross-sectional area of the wire is constant.

Which graph has a gradient that is equal to the Young modulus of the metal?



23 For a wire, Hooke's law is obeyed for a tension F and extension x. The Young modulus for the material of the wire is E.

Which expression represents the elastic potential energy stored in the wire?

A $\frac{1}{2}Ex$ **B** Ex **C** $\frac{1}{2}Fx$ **D** Fx

24 A plane polarised wave has amplitude A. The wave is incident normally on a polarising filter.

The transmission axis of the filter is at angle θ to the plane of polarisation of the incident wave.

What is the amplitude of the wave that emerges from the filter?

A $A\cos\theta$ **B** $A\cos^2\theta$ **C** $A^2\cos\theta$ **D** $A^2\cos^2\theta$

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25 An electromagnetic wave is travelling through a vacuum.

What could be the wavelength and period of the electromagnetic wave?

	wavelength	period
A	1.2×10^{-10} Tm	2.5 Ms
B	1.2 pm	2.5×10^{11} Gs
C	1.2×10^{2} pm	4.0×10^{-10} ns
D	1.2×10^{3} μm	4.0 ns

26 Light of frequency 6.7×10^{14} Hz in a vacuum is incident normally on a diffraction grating that contains 4.0×10^{5} lines m^{-1}.

What is the angle between the adjacent second and third order intensity maxima?

A 12° **B** 21° **C** 33° **D** 54°

27 The siren of a moving police car emits a sound wave with a frequency of 440 Hz. A stationary observer hears sound of frequency 494 Hz. The speed of sound in the air is 340 $m\,s^{-1}$.

What could be the speed and the direction of movement of the car?

A 37 $m\,s^{-1}$ directly away from the observer

B 37 $m\,s^{-1}$ directly towards the observer

C 42 $m\,s^{-1}$ directly away from the observer

D 42 $m\,s^{-1}$ directly towards the observer

28 The diagram shows the shape at one instant in time of part of a stretched string as a wave travels along it from left to right.



What are the directions of the velocities of the points 1, 2 and 3 on the string at this instant in time?

	point 1	point 2	point 3
A	→	→	→
B	→	←	→
C	↑	↓	↑
D	↓	↑	↓

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29 Which wave **cannot** be a longitudinal wave?

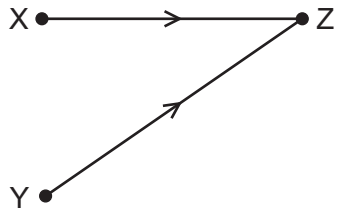
A a diffracted wave

B a polarised wave

C a reflected wave

D a stationary wave

30 Microwaves are emitted from two sources at points X and Y. The two waves meet at point Z. The diagram shows the paths of the two waves.



The waves emitted from points X and Y are coherent.

What is a direct consequence of the two waves being coherent?

A There is a constant difference in the path lengths YZ and XZ.

B There is a constant difference in phase between the two waves at Z.

C There is a constant non-zero difference in frequency of the two waves at Z.

D There is a constant non-zero difference in amplitude of the two waves at Z.

31 What is the unit of resistivity?

A $\Omega\,m^{-2}$ **B** $\Omega\,m^{-1}$ **C** Ω **D** $\Omega\,m$

32 A kettle is connected to a 250 V mains supply.

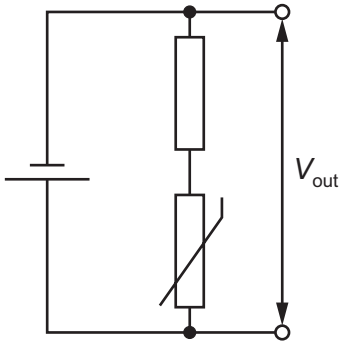
What are possible values for the power of the kettle and the current in the kettle?

	power / W	current / A
A	500	0.5
B	500	5.0
C	2500	0.1
D	2500	10

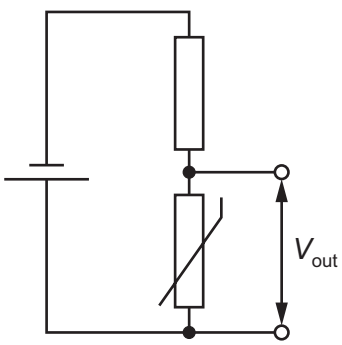
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33 Which circuit results in output voltage V_{out} increasing with increasing temperature?

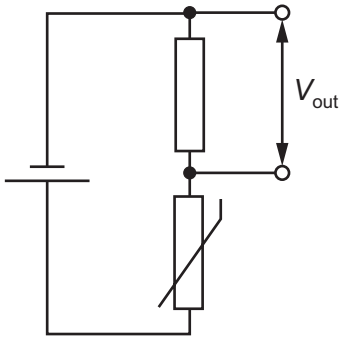
A



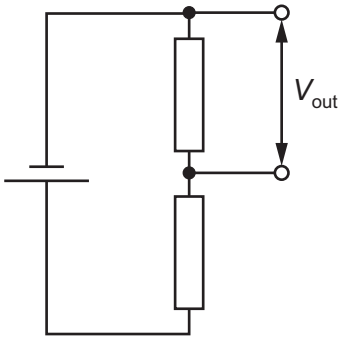
B



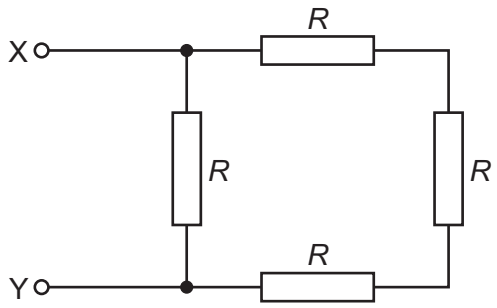
C



D



34 Four resistors, each of resistance R, are connected as shown.



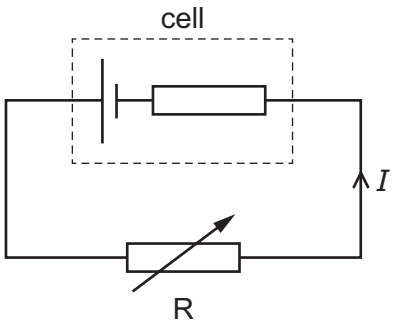
The total resistance between point X and point Y is 120 Ω.

What is the magnitude of the resistance R?

A 30 Ω **B** 90 Ω **C** 160 Ω **D** 480 Ω

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35 A cell with internal resistance is connected to a variable resistor R as shown.



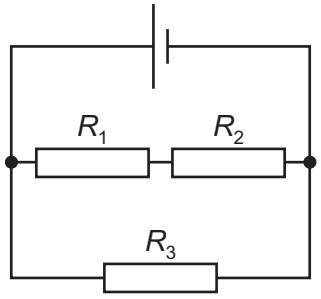
The resistance of R is gradually decreased.

How do the current I and the terminal potential difference (p.d.) across the cell change?

	current I	terminal p.d. across cell
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

36 The diagram shows a circuit with a cell and three resistors with resistances R_1, R_2 and R_3.

The cell has negligible internal resistance.



The total resistance of the circuit is R_T.

Which equation for R_T is correct?

A $R_T = R_1 + R_2 + R_3$

B $R_T = \frac{1}{R_1 + R_2} + \frac{1}{R_3}$

C $\frac{1}{R_T} = \frac{1}{R_1 + R_2 + R_3}$

D $\frac{1}{R_T} = \frac{1}{R_1 + R_2} + \frac{1}{R_3}$

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37 Hydrogen and deuterium can be represented by the nuclide symbols ${}^{1}_{1}\mathrm{H}$ and ${}^{2}_{1}\mathrm{H}$ respectively.

What is a difference between hydrogen and deuterium?

A The deuterium atom has twice the number of electrons as the hydrogen atom.

B The deuterium nucleus has a charge, but the hydrogen nucleus has no charge.

C The deuterium nucleus has less mass than the hydrogen nucleus.

D The deuterium nucleus has half the charge per unit mass of the hydrogen nucleus.

38 A radioactive sample decays by emitting β^- particles.

The energy released in the decay process is the same for each nucleus that decays, but the β^- particles emitted have a continuous range of kinetic energies.

Which statement explains why the β^- particles are emitted with a continuous range of kinetic energies?

A Some of the energy released is given to the remaining nucleons in the nucleus.

B Some of the energy released is taken by an emitted antineutrino.

C Some of the energy released is used to create the β^- particle.

D Some of the energy released is used to create a new nucleon.

39 Which particle is **not** a fundamental particle?

A electron

B neutrino

C neutron

D top quark

40 What is the charge of an anti-top quark?

A $-\frac{2}{3}e$ **B** $-\frac{1}{3}e$ **C** $+\frac{1}{3}e$ **D** $+\frac{2}{3}e$

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CAMBRIDGE
International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2024**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	A	**1**
2	A	**1**
3	C	**1**
4	B	**1**
5	B	**1**
6	D	**1**
7	C	**1**
8	C	**1**
9	B	**1**
10	D	**1**
11	A	**1**
12	B	**1**
13	D	**1**
14	A	**1**
15	A	**1**
16	B	**1**
17	B	**1**
18	A	**1**
19	B	**1**
20	B	**1**
21	D	**1**
22	C	**1**
23	C	**1**
24	A	**1**
25	C	**1**
26	A	**1**
27	B	**1**
28	D	**1**

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Question	Answer	Marks
29	B	**1**
30	B	**1**
31	D	**1**
32	D	**1**
33	C	**1**
34	C	**1**
35	C	**1**
36	D	**1**
37	D	**1**
38	B	**1**
39	C	**1**
40	A	**1**

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June 2024 · Paper 12

9702_s24_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2024**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

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[Turn over

Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What is a scalar quantity?

A a quantity that can be represented as two perpendicular components

B a quantity that does **not** require a unit

C a quantity without a direction

D a quantity without a magnitude

2 The value of quantity X has a percentage uncertainty of 2%.

The value of quantity Y has a percentage uncertainty of 4%.

The value of a quantity W is calculated from the values of X and Y.

The value of W has a percentage uncertainty of 8%.

What could be the relationship between W, X and Y?

A $W = XY$ **B** $W = 2XY$ **C** $W = \frac{X}{Y^2}$ **D** $W = \frac{Y}{X^2}$

3 A football is kicked so that it moves vertically upwards through the air.

What is the variation in the air resistance and the resultant force acting on the ball as it moves vertically upwards?

	air resistance	resultant force
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

4 Which statement is **not** correct?

A Acceleration can be determined from the gradient of a velocity–time graph.

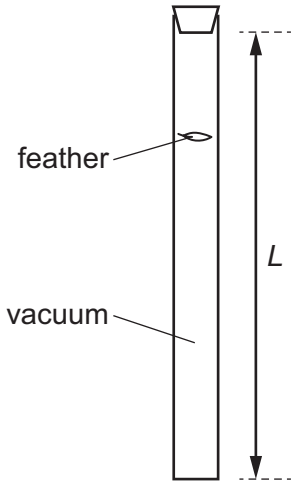
B Acceleration is the rate of change of velocity.

C Displacement can be determined from the area under a velocity–time graph.

D Velocity is the rate of change of distance.

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5 The diagram shows a laboratory experiment in which a feather falls from rest in a long evacuated vertical tube of length *L*.

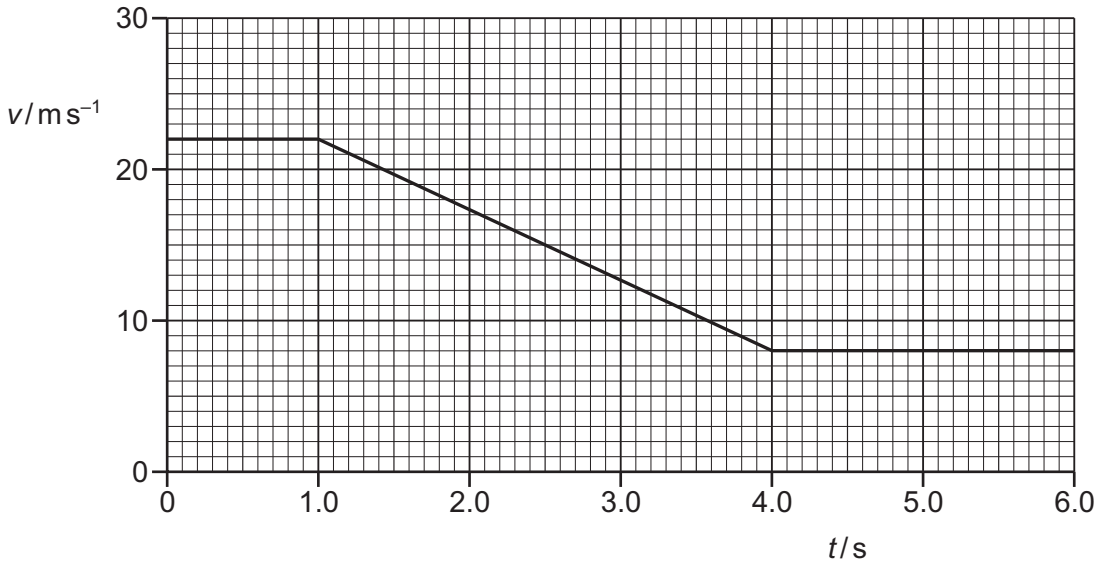


The feather takes time *T* to fall from the top to the bottom of the tube.

How far does the feather fall from the top of the tube in time 0.50*T*?

A 0.13*L** **B** 0.25*L** **C** 0.38*L** **D** 0.50*L*

6 A car travels along a straight horizontal road. The graph shows the variation of the velocity *v* of the car with time *t* for 6.0 s of its journey.



The brakes of the car are applied from *t* = 1.0 s to *t* = 4.0 s.

How far does the car travel while the brakes are applied?

A 21 m **B** 45 m **C** 67 m **D** 83 m

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7 Two satellites in deep space collide inelastically.

What happens to the total kinetic energy and total momentum?

	total kinetic energy	total momentum
A	conserved	conserved
B	conserved	reduced
C	reduced	conserved
D	reduced	reduced

8 What is a reasonable estimate of the momentum of a family car travelling at 25 kilometres per hour?

A $1 \times 10^4\,\text{kg m s}^{-1}$

B $1 \times 10^5\,\text{kg m s}^{-1}$

C $1 \times 10^6\,\text{kg m s}^{-1}$

D $1 \times 10^7\,\text{kg m s}^{-1}$

9 A ball collides with a wall. Before the collision, the ball moves with velocity $8\,\text{m s}^{-1}$ to the right. After the collision, it moves with velocity $3\,\text{m s}^{-1}$ to the left.

What is the change in velocity of the ball during the collision?

A $5\,\text{m s}^{-1}$ to the left

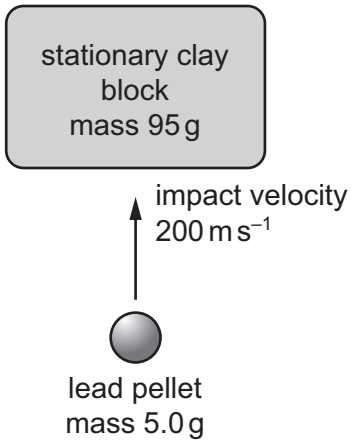
B $5\,\text{m s}^{-1}$ to the right

C $11\,\text{m s}^{-1}$ to the left

D $11\,\text{m s}^{-1}$ to the right

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10 A lead pellet is shot vertically upwards into a clay block that is stationary at the moment of impact, but is able to rise freely after impact.



The mass of the pellet is 5.0 g and the mass of the clay block is 95 g.

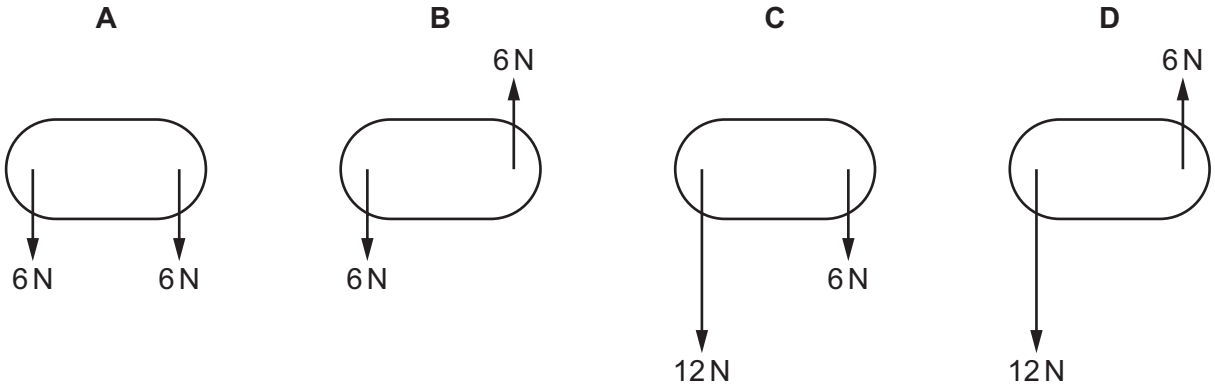
The pellet hits the block with an initial vertical velocity of 200 m s^{-1}. It embeds itself in the block and does not emerge.

How high above its initial position will the block rise?

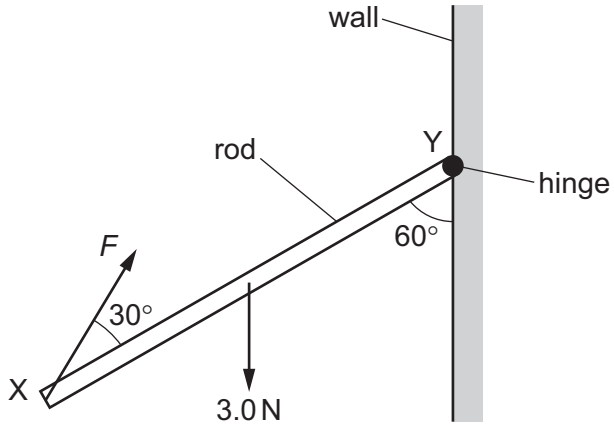
A 5.1 m **B** 5.6 m **C** 10 m **D** 100 m

11 Two forces act on an object.

Which diagram represents a couple?



12 The diagram shows a uniform rod, XY, that is freely hinged to a vertical wall at end Y. The rod is at an angle of 60° to the wall.



not to scale

A force *F* acts at an angle of 30° to the rod at end X. The rod has a weight of 3.0 N and is in equilibrium.

What is the magnitude of force *F*?

A 0.87 N **B** 1.5 N **C** 2.6 N **D** 5.2 N

13 Water has a density of $1.0\,g\,cm^{-3}$.

Glycerine has a density of $1.3\,g\,cm^{-3}$.

A student measures out a volume of $40\,cm^3$ of glycerine into a container.

The student adds water to the container to make a mixture of water and glycerine. Assume that the total volume of water and glycerine does not change when the two liquids are mixed.

Which volume of water needs to be added to make a mixture of density $1.1\,g\,cm^{-3}$?

A $4.0\,cm^3$ **B** $8.0\,cm^3$ **C** $34\,cm^3$ **D** $80\,cm^3$

14 The force resisting the motion of a car is proportional to the square of the car's speed. The magnitude of the force at a speed of $20.0\,m\,s^{-1}$ is 800 N.

What useful output power is required from the car's engine to maintain a steady speed of $40.0\,m\,s^{-1}$?

A 32 kW **B** 64 kW **C** 128 kW **D** 512 kW

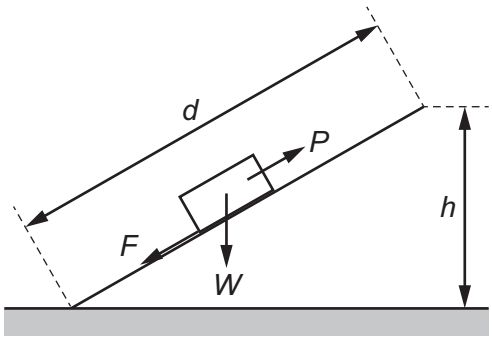
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15 A box of weight W is pulled by a force P along a slope.

The length of the slope is d, and the box rises a height h.

The frictional force between the box and the slope is F.

The diagram shows the directions of the forces.



The purpose of the slope is to raise the box vertically.

Which expression gives the efficiency of the slope?

A $\frac{Fd}{Wh}$ **B** $\frac{Pd}{Wh}$ **C** $\frac{Wh}{Fd}$ **D** $\frac{Wh}{Pd}$

16 The kinetic energy of a particle is increased by a factor of 4.

By what factor does its speed increase?

A 2 **B** 4 **C** 8 **D** 16

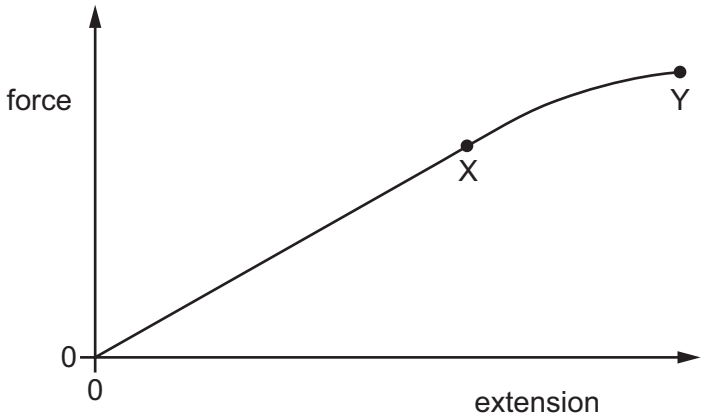
17 A mass of 28 g is raised vertically upwards through a distance of 4.6 m.

What is the change in gravitational potential energy of the mass?

A 0.13 J **B** 1.3 J **C** 130 J **D** 1300 J

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18 A sample of metal is subjected to a force which increases to a maximum value and then decreases back to zero. A force–extension graph for the sample is shown.



When the sample contracts, it follows the same force–extension curve as when it was being stretched.

What is the behaviour of the metal between X and Y?

A both elastic and plastic

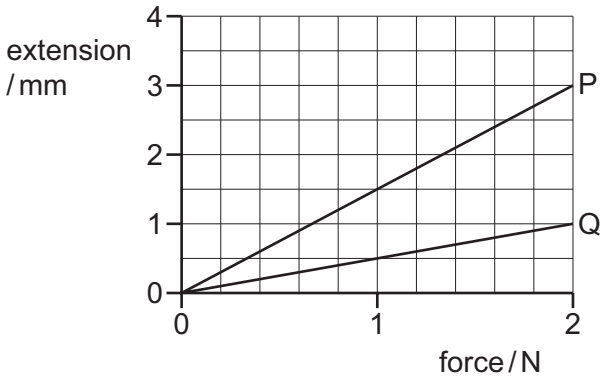
B **not** elastic and **not** plastic

C elastic but **not** plastic

D plastic but **not** elastic

19 Two wires, P and Q, made of the same material, are stretched with an increasing force.

A graph is plotted of the variation with force of the extension of each wire.

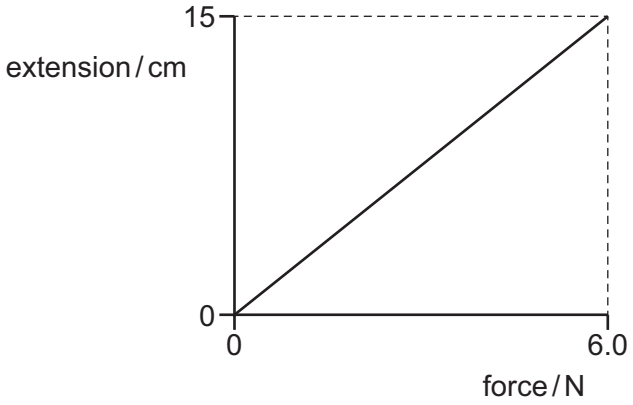


The wires have the same original length but different diameters.

What is the ratio $\frac{\text{diameter of wire Q}}{\text{diameter of wire P}}$?

A $\frac{1}{3}$ **B** $\frac{1}{\sqrt{3}}$ **C** $\sqrt{3}$ **D** 3

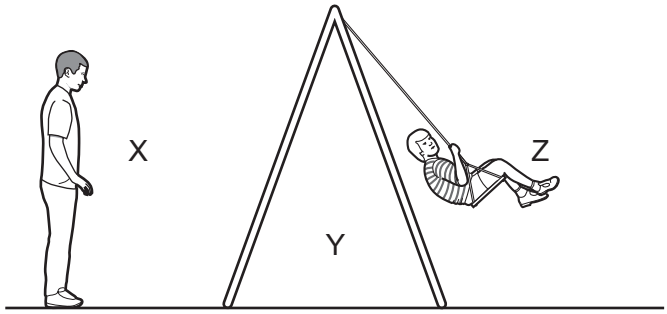
20 An extension–force graph for a spring is shown.



What is the spring constant of the spring?

A $0.025\,N\,m^{-1}$ **B** $0.40\,N\,m^{-1}$ **C** $2.5\,N\,m^{-1}$ **D** $40\,N\,m^{-1}$

21 A man stands stationary in front of a swing. A child sits and swings.



The child blows a whistle that emits a sound at a constant frequency.

The man observes the frequency of the sound when the swing is at positions X, Y and Z.

When will the man hear the highest frequency?

A when the swing is at X

B when the swing is at Y and moving away from the man

C when the swing is at Y and moving towards the man

D when the swing is at Z

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22 A loudspeaker is playing music in a room. The door to the room is open and has a width of 0.80 m.

Sound waves of many different frequencies pass through the doorway and diffract. The speed of sound in air is 340 m s^{-1}.

Which frequency of sound wave diffracts the most as it passes through the doorway?

A 2.4×10^{-3} Hz

B 8.0×10^{-1} Hz

C 2.7×10^{2} Hz

D 4.3×10^{2} Hz

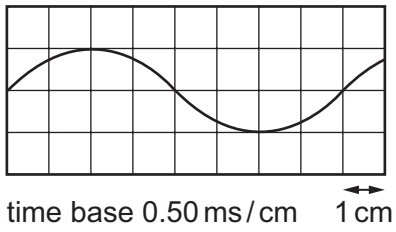
23 A stationary sound wave is set up between a loudspeaker and a wall.

A microphone is connected to a cathode-ray oscilloscope (CRO) and is moved along a line directly between the loudspeaker and the wall. The amplitude of the trace on the CRO rises to a maximum at a position X, falls to a minimum and then rises once again to a maximum at a position Y.

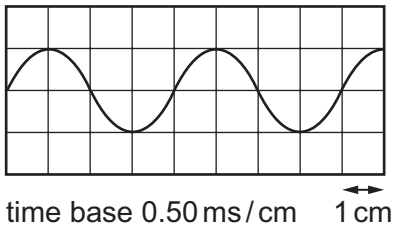
The distance between X and Y is 33 cm. The speed of sound in air is 330 m s^{-1}.

Which diagram could represent the CRO trace of the sound received at X?

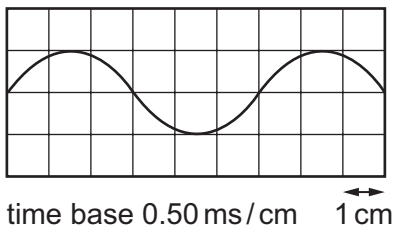
A



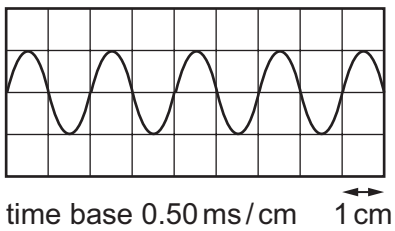
B



C



D



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[Turn over

24 Polarisation is associated with certain waves.

Which waves **cannot** be polarised?

A radio waves from a transmitter

B sound waves from a moving source

C ultraviolet rays from the Sun

D X-rays from an X-ray emitter

25 A guitar string is plucked.

Which statement describes the resulting waves?

A Longitudinal waves on the string cause longitudinal waves in the air.

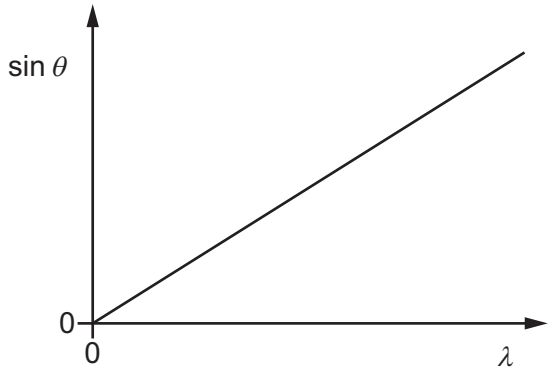
B Longitudinal waves on the string cause transverse waves in the air.

C Transverse waves on the string cause longitudinal waves in the air.

D Transverse waves on the string cause transverse waves in the air.

26 A diffraction grating with N lines per metre is used to diffract light of various wavelengths λ.

The graph shows the relation between the diffraction angle θ and λ for different wavelengths in the nth order interference pattern.



What is the gradient of the graph?

A Nn **B** $\frac{N}{n}$ **C** $\frac{n}{N}$ **D** $\frac{1}{Nn}$

27 A stationary sound wave is formed in the air column inside a tube that is open at both ends.

The stationary wave has three nodes.

How many antinodes does it have?

A 1 **B** 2 **C** 3 **D** 4

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28 Interference fringes of separation x are observed on a screen at a distance of 1.00 m from a double slit that is illuminated by yellow light of wavelength 600 nm.

At which distance from the double slit would interference fringes of the same separation x be observed when using blue light of wavelength 400 nm?

A 0.33 m **B** 0.67 m **C** 0.75 m **D** 1.50 m

29 What is the definition of potential difference across a component?

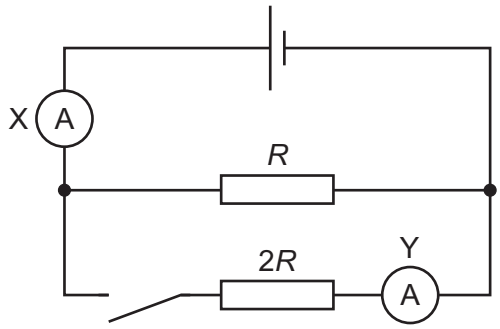
A energy transferred per unit charge

B energy transferred per unit current

C energy transferred per unit distance

D energy transferred per unit time

30 The diagram shows a cell of negligible internal resistance connected to a switch and two resistors of resistances R and $2R$.



The circuit also contains two ammeters X and Y.

The reading on X is 4.0 A when the switch is open.

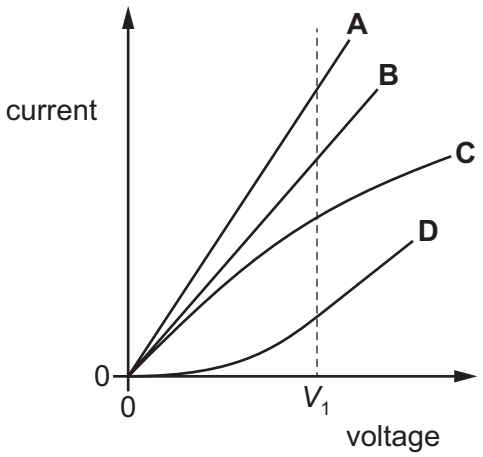
What are the readings on X and Y after the switch is closed?

	reading on X / A	reading on Y / A
A	4.0	1.3
B	4.0	2.7
C	6.0	2.0
D	6.0	4.0

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31 The I–V characteristics for four components, **A**, **B**, **C** and **D**, are shown.

Which component has the greatest resistance when the potential difference across it is V_1?



32 A cylindrical wire has cross-sectional area A and number density of free electrons n.
The wire has current I and the free electrons have average drift speed v.

A second cylindrical wire has cross-sectional area $0.5A$ and number density of free electrons $2n$.
In this wire, the free electrons have average drift speed $2v$.

What is the current in the second wire?

A $0.5I$ **B** I **C** $2I$ **D** $4I$

33 An electric current is formed by moving charge carriers.

What is **not** a possible charge on a charge carrier?

A -4.8×10^{-19} C

B -2.4×10^{-19} C

C $+1.6 \times 10^{-19}$ C

D $+3.2 \times 10^{-19}$ C

34 Which description of Kirchhoff's first law is correct?

A It considers the currents at a junction in a circuit and is a consequence of the conservation of charge.

B It considers the currents at a junction in a circuit and is a consequence of the conservation of energy.

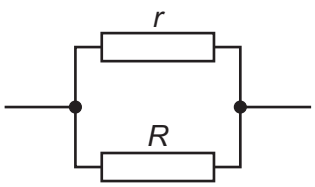
C It considers the electromotive forces and potential differences in a circuit loop and is a consequence of the conservation of charge.

D It considers the electromotive forces and potential differences in a circuit loop and is a consequence of the conservation of energy.

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35 Two resistors of resistances r and R are connected in parallel.

The value of r is less than that of R.



Which statement about the combined resistance of the two resistors is correct?

A It is between r and R.

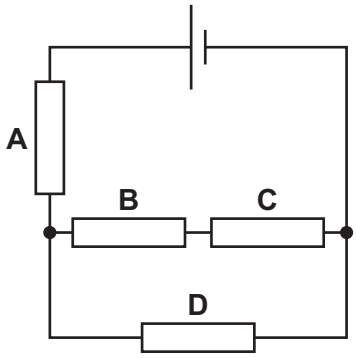
B It is equal to $(r + R)$.

C It is greater than $(r + R)$.

D It is less than r.

36 The diagram shows four identical resistors connected in a circuit.

Which resistor dissipates the most power?



37 Nuclide X with proton number Z undergoes β^+ decay to form nuclide Y.

The decay may be represented by the equation shown.

$$X \rightarrow Y + \beta^+ + W$$

What is the proton number of Y and which particle is represented by the symbol W?

	proton number of Y	particle represented by W
A	Z – 1	antineutrino
B	Z – 1	neutrino
C	Z + 1	antineutrino
D	Z + 1	neutrino

38 Which fundamental particles form a hadron?

A leptons

B nucleons

C photons

D quarks

39 The unstable nuclide $^{218}_{84}X$ decays through a sequence of emissions of α and β^- particles to form the stable nuclide $^{210}_{83}Y$.

How many α and β^- particles are emitted during this decay process?

	α-particles	β^- particles
A	1	1
B	2	1
C	2	3
D	3	2

40 Which statement about radioactive decay is correct?

A Neutrinos are always emitted during α-decay.

B The α-particles emitted from a radioactive sample have a continuous range of kinetic energies.

C The β^- particles emitted from a radioactive sample have a continuous range of kinetic energies.

D The proton number of a nucleus decreases by four when it undergoes α-decay.

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CAMBRIDGE
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Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2024**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	C	**1**
2	D	**1**
3	A	**1**
4	D	**1**
5	B	**1**
6	B	**1**
7	C	**1**
8	A	**1**
9	C	**1**
10	A	**1**
11	B	**1**
12	C	**1**
13	D	**1**
14	C	**1**
15	D	**1**
16	A	**1**
17	B	**1**
18	C	**1**
19	C	**1**
20	D	**1**
21	C	**1**
22	D	**1**
23	B	**1**
24	B	**1**
25	C	**1**
26	A	**1**
27	D	**1**
28	D	**1**

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Question	Answer	Marks
29	A	**1**
30	C	**1**
31	D	**1**
32	C	**1**
33	B	**1**
34	A	**1**
35	D	**1**
36	A	**1**
37	B	**1**
38	D	**1**
39	C	**1**
40	C	**1**

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June 2024 · Paper 13

9702_s24_qp_13

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2024**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

8319070239

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What is equal to 0.000005 J?

A 5 mJ **B** 5 MJ **C** 5 μJ **D** 5 nJ

2 The measurement of a physical quantity may be subject to random errors and to systematic errors.

Which statement is correct?

A A systematic error **cannot** be reduced by adjusting the apparatus.

B A systematic error results in a different reading each time the measurement is taken.

C Random errors are always caused by the person taking the measurement.

D Random errors can be reduced by taking the average of several measurements.

3 The Young modulus of the material of a wire is to be found. The Young modulus E is given by the equation shown.

$$E = \frac{4FL}{\pi d^2 x}$$

The wire is extended by a known force and the following measurements are made.

Which measurement has the largest effect on the uncertainty in the value of the calculated Young modulus?

	measurement	symbol	value
A	length of wire before force applied	L	2.043 ± 0.002 m
B	diameter of wire	d	0.54 ± 0.02 mm
C	force applied	F	19.62 ± 0.01 N
D	extension of wire with force applied	x	5.2 ± 0.2 mm

4 Two physical quantities combined together as a product can produce a scalar quantity or a vector quantity.

Which product of two quantities produces a scalar quantity?

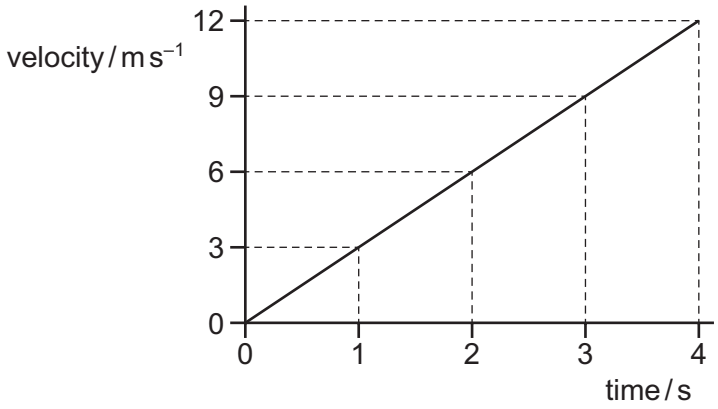
A (force) × (displacement of an object in the direction of the force)

B (mass) × (acceleration of the mass)

C (pressure) × (area on which the pressure acts)

D (velocity) × (time for which an object has that velocity)

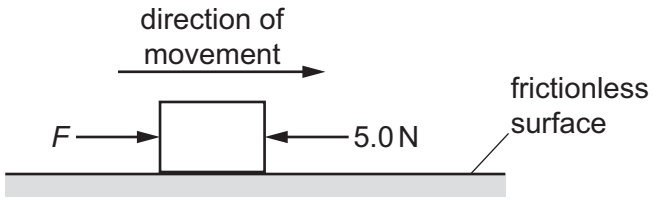
5 The diagram shows a velocity–time graph for an object moving in a straight line.



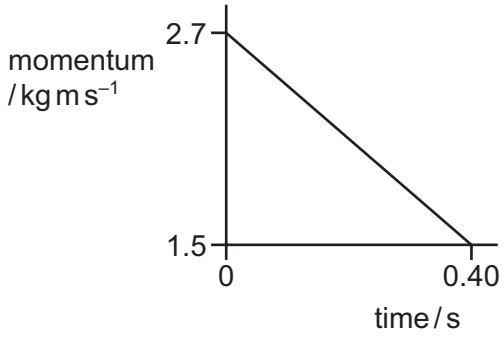
What is the displacement during the last 2 seconds of the motion?

A 6 m **B** 12 m **C** 18 m **D** 24 m

6 A block is moving along a horizontal frictionless surface. A constant force F and a constant resistive force of 5.0 N act on the block as it is moving in the direction of the force F, as shown.



The graph shows the variation with time of the momentum of the block.



What is the magnitude of force F?

A 2.0 N **B** 3.0 N **C** 5.0 N **D** 8.0 N

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7 Newton's third law describes two forces that are equal in magnitude and form a pair.

Which description of the two forces in such a pair is **not** correct?

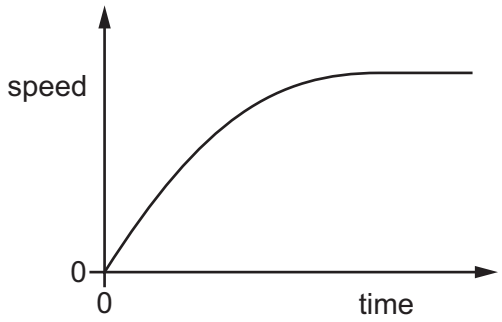
A They act in opposite directions.

B They act on different objects.

C They are the same type of force.

D They cause an object to be in equilibrium.

8 The graph shows the variation with time of the speed of a raindrop falling vertically through air.



Which statement is correct?

A The acceleration decreases to produce a steady speed.

B The acceleration increases as the speed increases.

C The air resistance decreases as the speed increases.

D The resultant force increases as the speed increases.

9 Which statement about a perfectly elastic collision between two objects is correct?

A Total kinetic energy is conserved and the relative speed of approach equals the relative speed of separation.

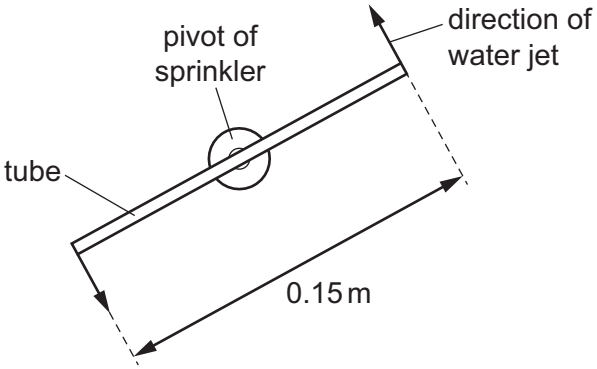
B Total kinetic energy is conserved but the relative speed of approach does **not** equal the relative speed of separation.

C Total kinetic energy is **not** conserved and the relative speed of approach does **not** equal the relative speed of separation.

D Total kinetic energy is **not** conserved but the relative speed of approach does equal the relative speed of separation.

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10 The diagram shows the view from above of a sprinkler system used to water a garden.

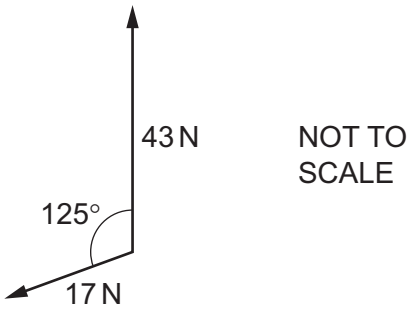


The sprinkler consists of a tube of length 0.15 m. The tube is pivoted in the middle and spins in a horizontal plane as it lets out jets of water from each end. The two water jets are in opposite directions to each other. Each water jet exerts a horizontal force of 3.0 N on the tube at right angles to the tube.

What is the magnitude of the torque on the tube from the water jets?

A 0 N m **B** 0.23 N m **C** 0.45 N m **D** 0.90 N m

11 An object is held in equilibrium by three forces. The forces all act in the same plane. The diagram shows two of the forces that act on the object.



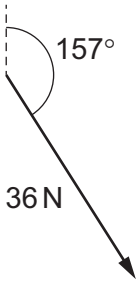
The third force is missing from the diagram.

What is the third force?

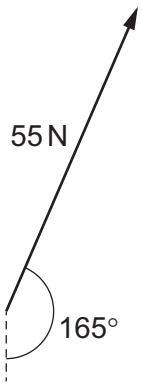
A



B



C



D



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12 The total number of forces acting on an object is two. The object is in equilibrium.

Which statements about the forces are correct?

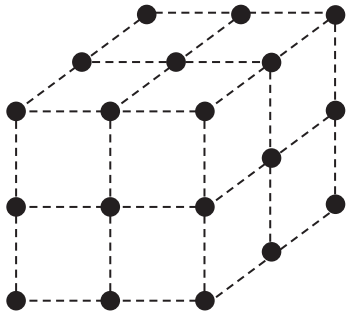
1 The two forces must have equal magnitudes.

2 The two forces must act in the same direction.

3 The two forces must act through the same point.

A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

13 The diagram shows the arrangement of atoms in a particular crystal.



Each atom is at the corner of a cube.

The mass of each atom is 3.5×10^{-25} kg. The density of the crystal is 9.2×10^{3} kg m^{-3}.

What is the shortest distance between the centres of two adjacent atoms?

A 3.8×10^{-29} m

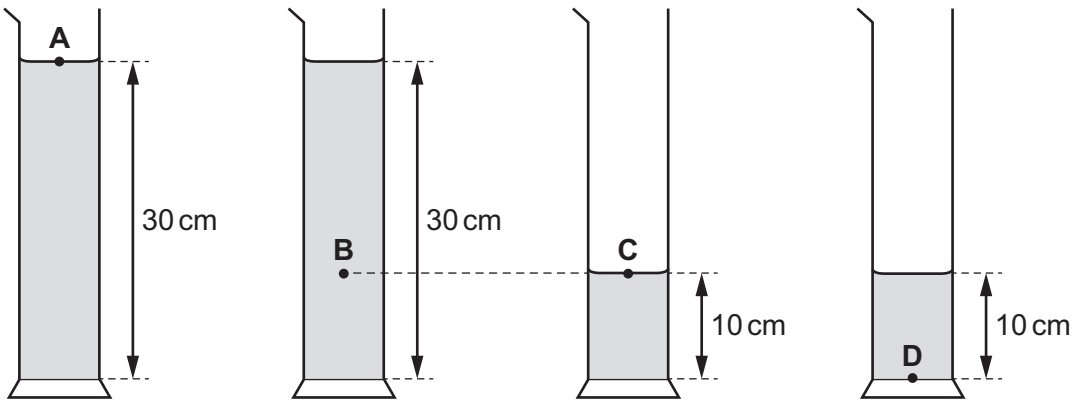
B 6.2×10^{-15} m

C 3.4×10^{-10} m

D 3.0×10^{-9} m

14 Four measuring cylinders are filled with the same liquid to the heights shown.

At which position is the pressure the greatest?



15 In many filament lamps, as much as 92 J of energy is emitted as thermal energy for every 8 J of energy emitted as light.

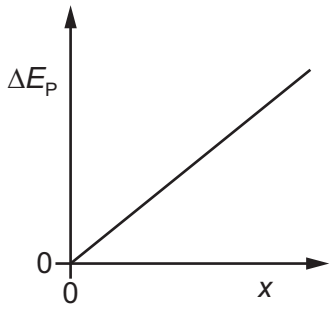
What is the efficiency of a filament lamp, as the percentage of electrical energy converted to light energy?

A 8% **B** 9% **C** 91% **D** 92%

16 What is a unit of power?

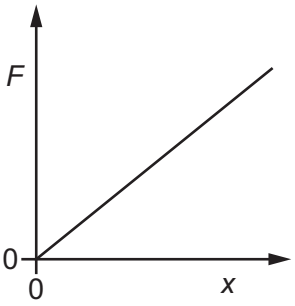
A JC^{-1} **B** Jm^{-1} **C** JN^{-1} **D** Js^{-1}

17 An object is in a uniform gravitational field. The graph shows how the change in gravitational potential energy ΔE_P of the object varies with the vertical distance x moved by the object from a fixed point.

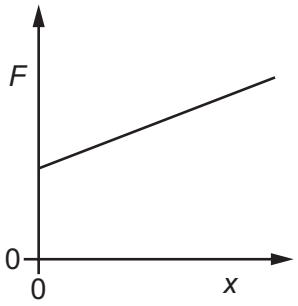


Which graph shows how the gravitational force F acting on the object varies with distance x?

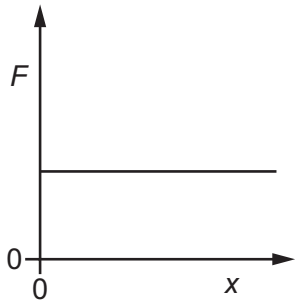
A



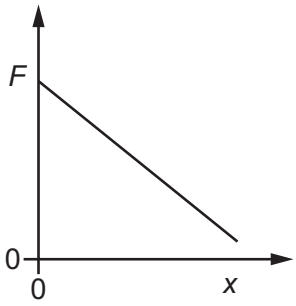
B



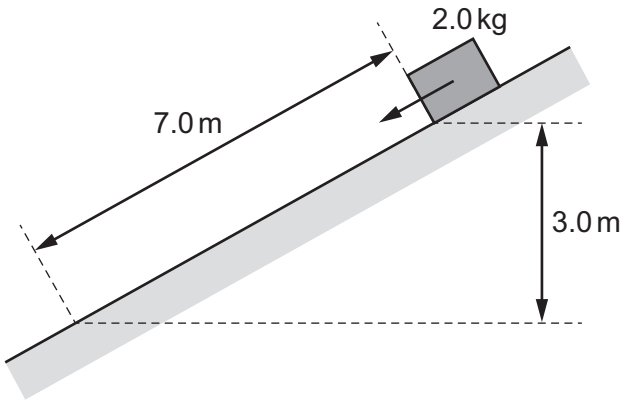
C



D



18 A block of mass 2.0 kg is released from rest on a slope. It travels 7.0 m down the slope and falls a vertical distance of 3.0 m. The block experiences a frictional force parallel to the slope of 5.0 N.



What is the speed of the block after falling this distance?

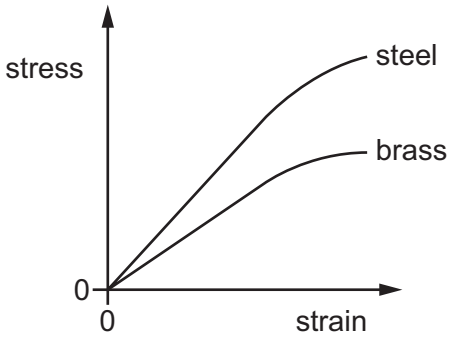
A $4.9\,m\,s^{-1}$ **B** $6.6\,m\,s^{-1}$ **C** $8.6\,m\,s^{-1}$ **D** $10\,m\,s^{-1}$

19 Two wires, one made of brass and the other of steel, are stretched in an experiment. Both wires obey Hooke's law during this experiment.

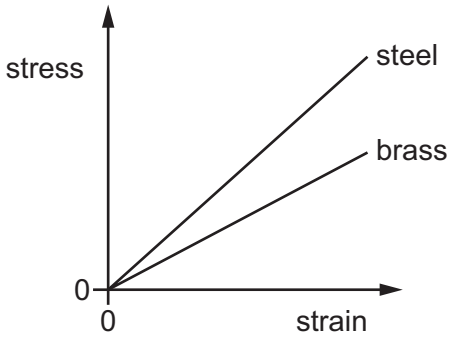
The Young modulus for brass is less than the Young modulus for steel.

Which graph shows how the stress varies with strain for both wires in this experiment?

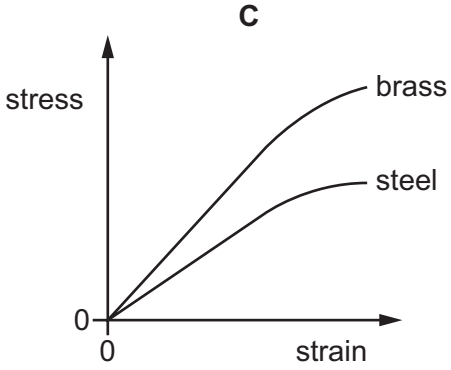
A



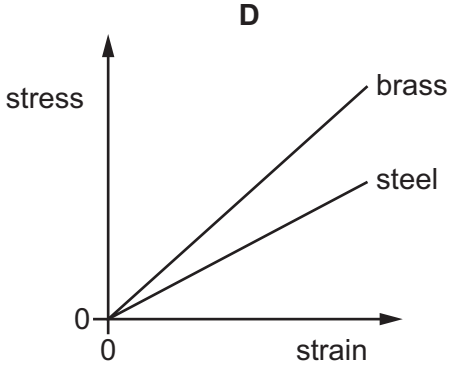
B



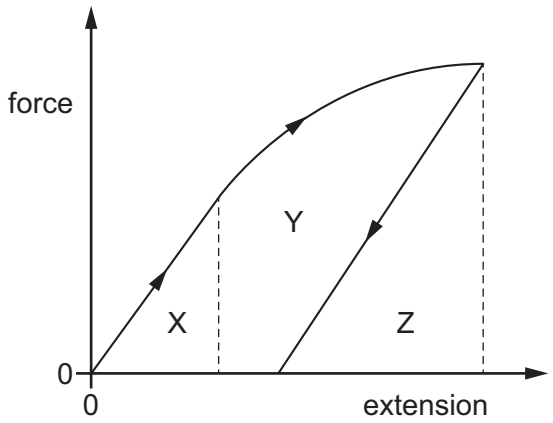
C



D



20 A sample of material is stretched by a tensile force to a point beyond its elastic limit. The tensile force is then reduced to zero. The force–extension graph is shown.



Which area represents the net work done on the sample?

A X **B** X + Y **C** Y + Z **D** Z

21 A wire is fixed at one end and is extended by a force F_1 acting on the other end. This causes the wire to have an elastic potential energy of 0.15 J.

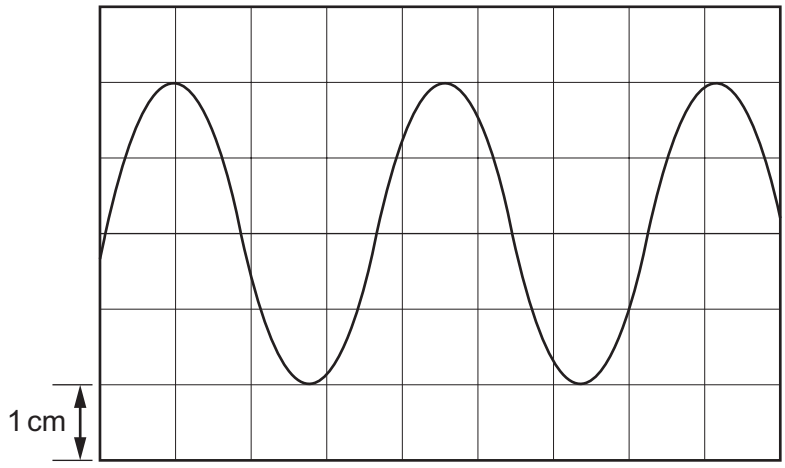
The force applied to the wire is now changed to a force F_2. This causes the wire to have a new elastic potential energy of 0.60 J.

The wire obeys Hooke's law.

What is the relationship between F_1 and F_2?

A $F_1 = 2F_2$ **B** $F_1 = 4F_2$ **C** $2F_1 = F_2$ **D** $4F_1 = F_2$

22 The diagram shows a representation of a wave on the screen of an oscilloscope.



The *y*-gain is set to 3.5 mV cm^{-1}.

What is the amplitude of the wave?

A 0.57 mV **B** 3.5 mV **C** 7.0 mV **D** 14 mV

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23 A transverse progressive wave on a string has a wavelength of 12 cm and an amplitude of 1.5 cm. The speed of the wave on the string is 48 cm s^{-1}.

What is the distance travelled by a point on the string in a time of 2.0 s?

A 12 cm **B** 24 cm **C** 48 cm **D** 96 cm

24 An ambulance siren emits a sound with a single frequency *f*.

The ambulance travels towards, passes close to, and then travels away from a stationary observer.

Which statement describes the frequency of the sound detected by the observer as the ambulance **passes** the observer?

A equal to *f* and decreasing

B equal to *f* and increasing

C greater than *f* and constant

D less than *f* and constant

25 An electromagnetic wave has a wavelength of 138 pm in a vacuum.

To which region of the electromagnetic spectrum does this wave belong?

A radio wave

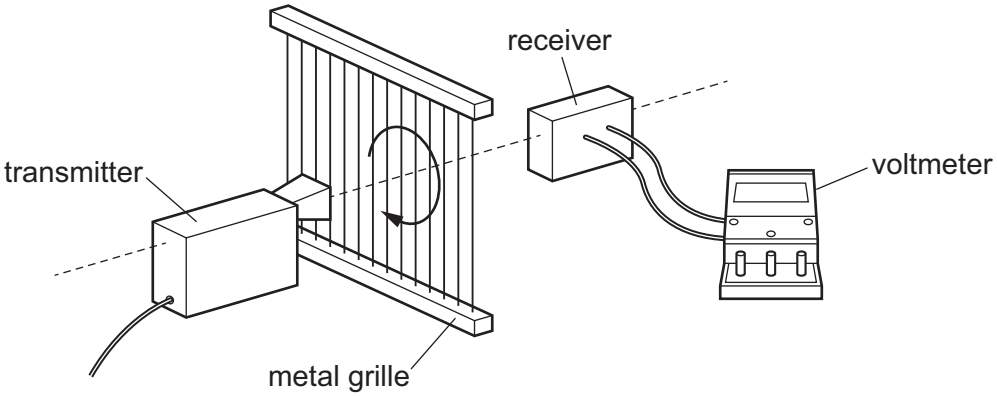
B microwave

C visible light

D X-ray

26 A student investigates the polarisation of microwaves. The microwaves from the transmitter are vertically polarised. A metal grille acts as a polarising filter when placed between the microwave transmitter and the receiver. The reading on the voltmeter is proportional to the intensity of microwaves transmitted through the grille.

When the transmission axis of the grille is vertical, the voltmeter reads 3.50 V.



The grille is then rotated through an angle θ. The voltmeter now reads 2.20 V.

What is θ?

A 37.5° **B** 39.0° **C** 51.0° **D** 52.5°

27 Two waves superpose. A resultant wave pattern is formed.

Which statement about the two waves **must** be correct?

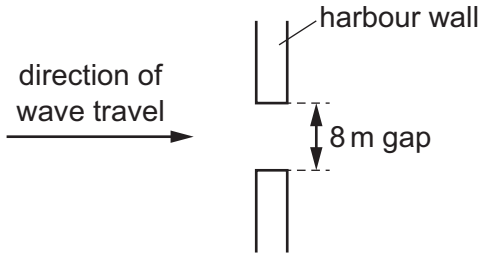
A They have the same amplitude.

B They are of the same type.

C They are transverse waves.

D They travel in opposite directions.

28 A water wave passes through a gap in a harbour wall and diffracts. The gap has a width of 8 m.



The wave travels directly towards the gap.

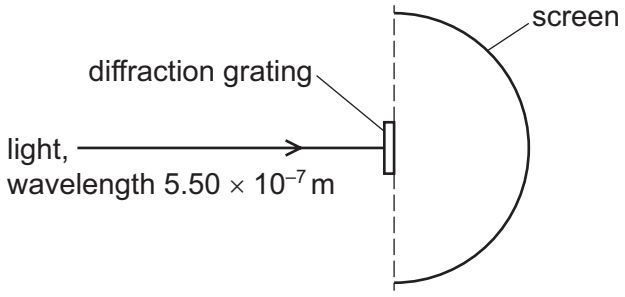
For which wavelength is the diffraction of the wave greatest?

A 1 m **B** 2 m **C** 4 m **D** 8 m

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29 Light of wavelength 5.50×10^{-7} m from a laser is incident normally on a diffraction grating.

The diffracted light is incident on a semicircular screen, as shown in the view from above.



view from above

A total of 9 bright dots are formed on the screen.

The grating is at the centre of the semicircle. The lines of the grating are vertical. The separation between adjacent lines in the grating is d.

What is a possible value of d?

A 2.25×10^{-6} m **B** 2.80×10^{-6} m **C** 4.40×10^{-6} m **D** 4.95×10^{-6} m

30 Which quantity is given by the product of charge and electric potential difference?

A current

B energy transferred

C power dissipated

D resistance

31 The current in a filament lamp is increased.

Which statement about the lamp is correct?

A The brightness of the lamp decreases.

B The potential difference across the filament decreases.

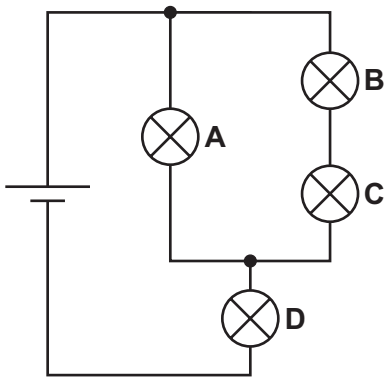
C The resistance of the filament decreases.

D The temperature of the filament increases.

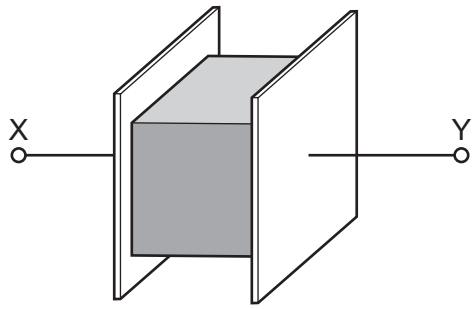
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32 A student builds the circuit shown. All the lamps are identical.

Which lamp dissipates the most power?



33 The resistance of a metal cube is measured by placing it between two parallel plates, as shown.



The cube has volume V and is made of a material with resistivity ρ. The connections to the cube have negligible resistance.

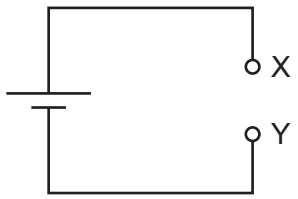
Which expression gives the electrical resistance of the metal cube between X and Y?

A $\rho V^{\frac{1}{3}}$ **B** $\rho V^{\frac{2}{3}}$ **C** $\frac{\rho}{V^{\frac{1}{3}}}$ **D** $\frac{\rho}{V^{\frac{2}{3}}}$

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34 A circuit needs to be completed by connecting two resistors between points X and Y, as shown.

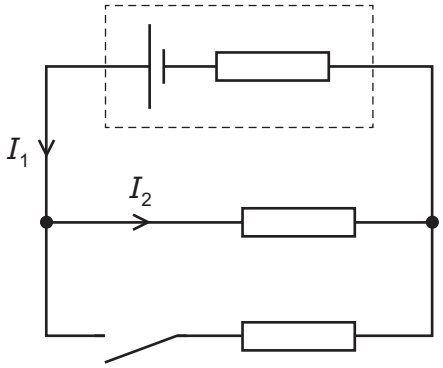
The resistors can be connected in series or in parallel with each other.



Which combination of resistors produces the least resistance between X and Y?

	resistances of resistors / Ω	type of combination
A	20 and 40	parallel
B	40 and 100	parallel
C	20 and 40	series
D	40 and 100	series

35 The diagram shows a circuit that includes a cell with internal resistance.

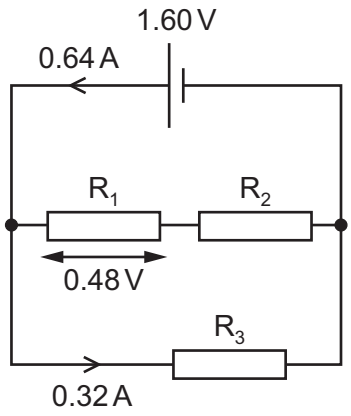


The switch is initially open.

Which row describes the effects on currents I_1 and I_2 of closing the switch?

	I_1	I_2
A	decreases	decreases
B	decreases	stays the same
C	increases	decreases
D	increases	stays the same

36 The diagram shows a circuit consisting of a cell and three resistors R_1, R_2 and R_3.



The cell has electromotive force (e.m.f.) 1.60 V and negligible internal resistance. The current in the cell is 0.64 A.

The potential difference across R_1 is 0.48 V.

The current in R_3 is 0.32 A.

What is the resistance of R_2?

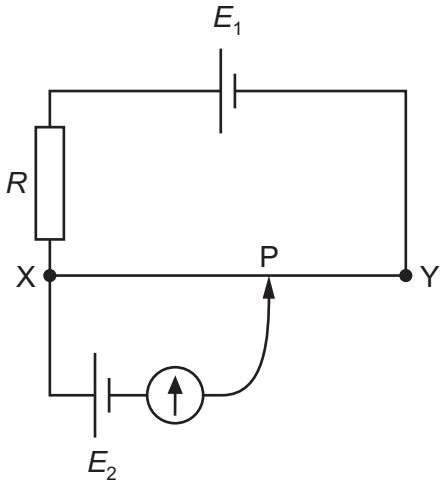
A 1.5 Ω **B** 2.5 Ω **C** 3.5 Ω **D** 5.0 Ω

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37 The diagram shows a potentiometer circuit used to compare the electromotive forces (e.m.f.), E_1 and E_2, of two cells.

XY is a uniform resistance wire. The fixed resistor has resistance R.

A sliding contact is moved along the wire XY. When the sliding contact is at position P, the galvanometer reads zero.



The circuit is changed so that the galvanometer reads zero when the sliding contact is at a new position to the left of P.

Which change could have been made to the circuit?

A The wire XY was replaced with one of lower resistance.

B E_2 was increased.

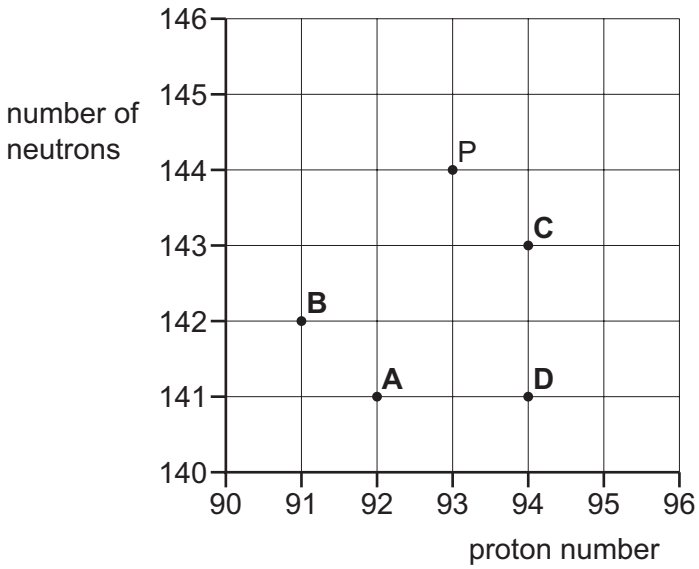
C E_1 was decreased.

D R was decreased.

38 Nucleus P has 144 neutrons and 93 protons. Nucleus P is unstable and undergoes α-decay to form nucleus Q. Nucleus Q then undergoes β^- decay to form nucleus R.

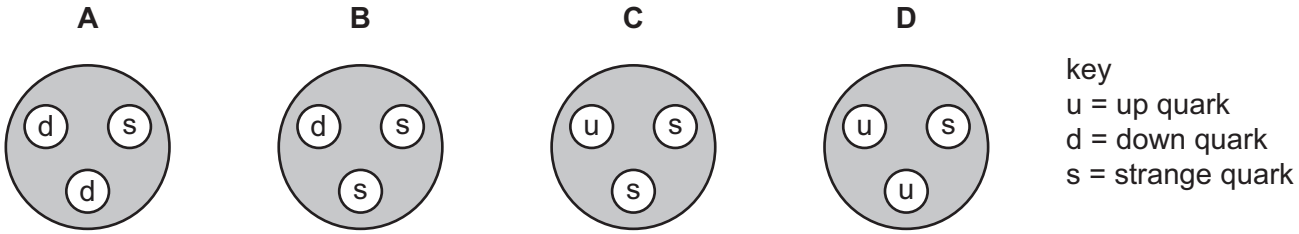
Nucleus P is represented by point P on the graph.

Which point on the graph represents nucleus R?



39 The diagrams show the quark composition of four different hadrons. One of the hadrons is a Σ^+ particle. It has a charge of $+e$, where e is the elementary charge.

Which hadron could be the Σ^+ particle?



40 Which type of particle is comprised of the most quarks?

A antiquark

B baryon

C lepton

D meson

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CAMBRIDGE
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Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2024**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	C	**1**
2	D	**1**
3	B	**1**
4	A	**1**
5	C	**1**
6	A	**1**
7	D	**1**
8	A	**1**
9	A	**1**
10	C	**1**
11	B	**1**
12	B	**1**
13	C	**1**
14	B	**1**
15	A	**1**
16	D	**1**
17	C	**1**
18	A	**1**
19	B	**1**
20	B	**1**
21	C	**1**
22	C	**1**
23	C	**1**
24	A	**1**
25	D	**1**
26	A	**1**
27	B	**1**
28	D	**1**

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Question	Answer	Marks
29	A	**1**
30	B	**1**
31	D	**1**
32	D	**1**
33	C	**1**
34	A	**1**
35	C	**1**
36	C	**1**
37	D	**1**
38	A	**1**
39	D	**1**
40	B	**1**

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November 2024 · Paper 11

9702_w24_qp_11

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **October/November 2024**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What are the SI base units for the moment of a force?

A $\mathrm{kg\,m^{-1}\,s^{2}}$ **B** $\mathrm{kg\,s^{-2}}$ **C** $\mathrm{kg\,m\,s^{-2}}$ **D** $\mathrm{kg\,m^{2}\,s^{-2}}$

2 Which statement about vector quantities is correct?

A Acceleration of free fall is a vector quantity because it has a constant magnitude.

B Temperature in °C is a vector quantity because it can be positive or negative.

C Time is a vector quantity because it can only go in the forwards direction.

D Weight is a vector quantity because it has a direction.

3 The density of the material of a rectangular block is determined by measuring the mass and linear dimensions of the block. The list shows the results obtained, together with their uncertainties.

$$\text{mass} = (25.0 \pm 0.1)\,\text{g}$$
$$\text{length} = (5.00 \pm 0.01)\,\text{cm}$$
$$\text{width} = (2.00 \pm 0.01)\,\text{cm}$$
$$\text{height} = (1.00 \pm 0.01)\,\text{cm}$$

The density is calculated to be $2.50\,\mathrm{g\,cm^{-3}}$.

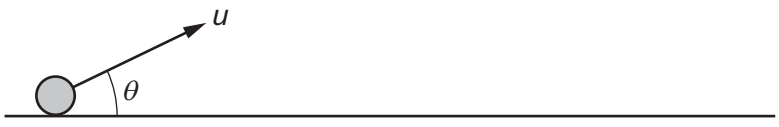
What is the uncertainty in this result?

A $\pm 0.01\,\mathrm{g\,cm^{-3}}$ **B** $\pm 0.02\,\mathrm{g\,cm^{-3}}$ **C** $\pm 0.05\,\mathrm{g\,cm^{-3}}$ **D** $\pm 0.13\,\mathrm{g\,cm^{-3}}$

4 What is a reasonable estimate of the volume of one page from this examination paper?

A $60\,\mathrm{mm^{3}}$ **B** $600\,\mathrm{mm^{3}}$ **C** $6000\,\mathrm{mm^{3}}$ **D** $60\,000\,\mathrm{mm^{3}}$

5 An object is projected from horizontal ground at a velocity of magnitude u and angle θ to the horizontal. It hits the ground at a time t after it is projected. Assume air resistance is negligible.



Which statement does **not** describe the motion of this object?

A The horizontal component of the object's velocity is constant and has the value $u\cos\theta$.

B The horizontal distance travelled by the object is $tu\cos\theta$.

C The time taken for the object to reach maximum height is $\frac{t}{2}$.

D The vertical component of the object's velocity is constant and has the value $u\sin\theta$.

6 A person, travelling on a motorway a total distance of 200 km, travels the first 90 km at an average speed of 80 km h^{-1}.

Which average speed must be obtained for the rest of the journey if the person is to reach the destination in a total time of 2 hours 0 minutes?

A 110 km h^{-1} **B** 120 km h^{-1} **C** 122 km h^{-1} **D** 126 km h^{-1}

7 A car of mass 1200 kg has momentum 18 000 kg m s^{-1}.

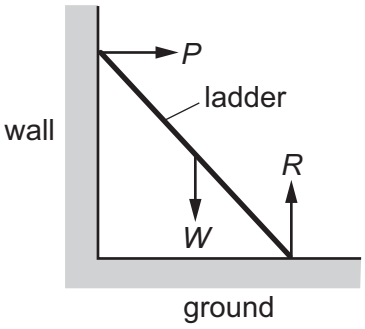
What is the kinetic energy of the car?

A 4.65 kJ **B** 6.57 kJ **C** 135 kJ **D** 270 kJ

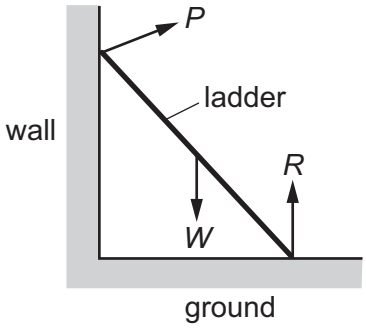
8 A ladder is positioned on icy (frictionless) ground and is leant against a rough wall. At the instant of release it begins to slide.

Which diagram shows the directions of the forces *P*, *W* and *R* acting on the ladder as it slides?

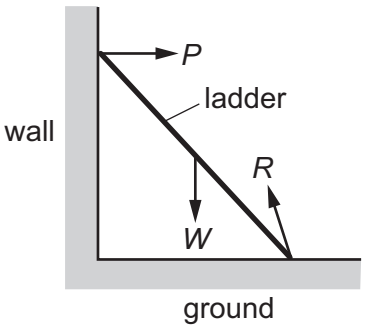
A



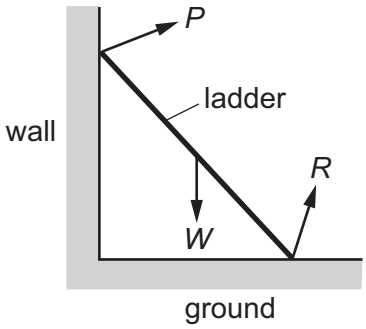
B



C



D

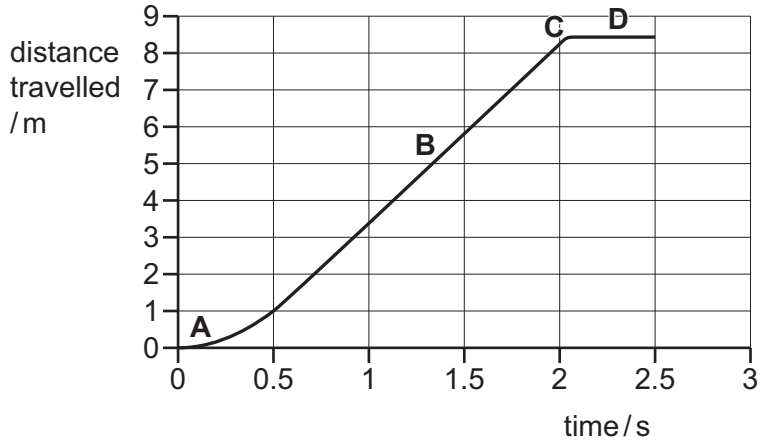


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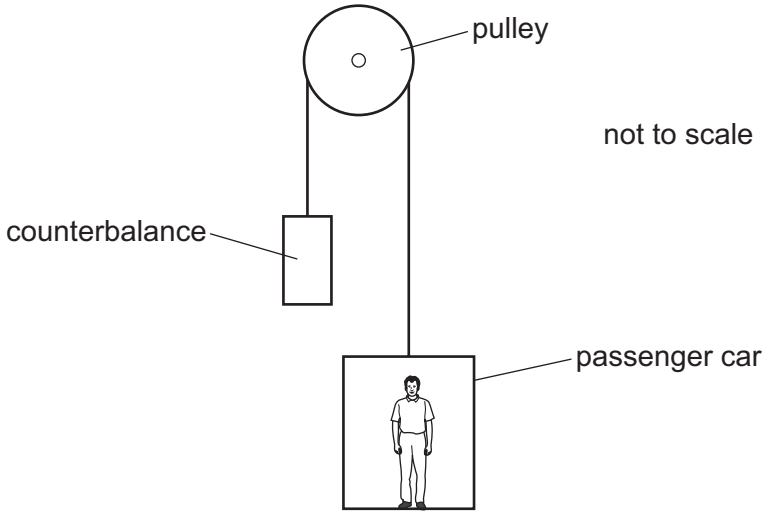
9 A toy parachute is dropped from a bridge and falls vertically through the air.

The graph shows the distance travelled by the parachute against time.

Which region of the graph shows when the parachute is at terminal velocity?



10 A lift (elevator) consists of a passenger car supported by a cable that runs over a light, frictionless pulley to a counterbalance. The counterbalance falls as the passenger car rises.



Some masses are shown in the table.

	mass / kg
passenger car	520
counterbalance	640
passenger	80

What is the magnitude of the acceleration of the car when carrying just one passenger and when the pulley is free to rotate?

A $0.032\,m\,s^{-2}$ **B** $0.32\,m\,s^{-2}$ **C** $0.61\,m\,s^{-2}$ **D** $0.65\,m\,s^{-2}$

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11 A stationary ball of mass m is hit by a bat. The ball leaves the bat with velocity v.

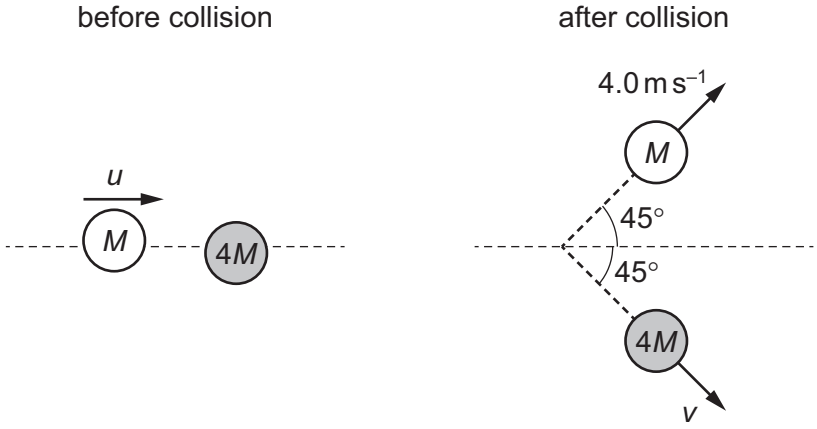
The bat is in contact with the ball for a short time Δt.

What is the average force of the bat on the ball?

A $mv\Delta t$ **B** $\frac{mv}{\Delta t}$ **C** $\frac{1}{2}mv^2\Delta t$ **D** $\frac{\frac{1}{2}mv^2}{\Delta t}$

12 A disc of mass M is moving across a horizontal frictionless surface with constant velocity u. It collides with a stationary disc of mass $4M$.

The diagram shows the view from above of the motion of the two discs before and after the collision.



What is the initial velocity u of the disc of mass M?

A $1.1\,\text{m s}^{-1}$ **B** $1.4\,\text{m s}^{-1}$ **C** $3.5\,\text{m s}^{-1}$ **D** $5.7\,\text{m s}^{-1}$

13 An object is dropped from rest on the Earth from a height of 2.0 m.

The same object is dropped from rest on the Moon from twice the height.

The acceleration of free fall on the Moon is approximately 16% of the value on the Earth.

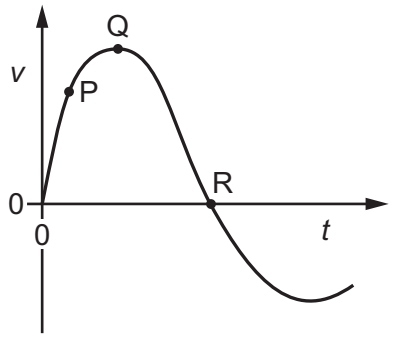
Assume that there are no resistive forces acting on the object.

What is the ratio $\frac{\text{speed of the object just before hitting the surface on the Earth}}{\text{speed of the object just before hitting the surface on the Moon}}$?

A 1.8 **B** 2.5 **C** 3.1 **D** 3.5

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14 The graph shows how velocity v varies with time t for a bungee jumper.



At which point is the bungee jumper momentarily at rest and at which point does she have zero acceleration?

	jumper at rest	jumper with zero acceleration
A	Q	P
B	Q	R
C	R	Q
D	R	R

15 A solid sphere, which is less dense than water, is held completely immersed in water a few metres below the surface. The density of the water is uniform.

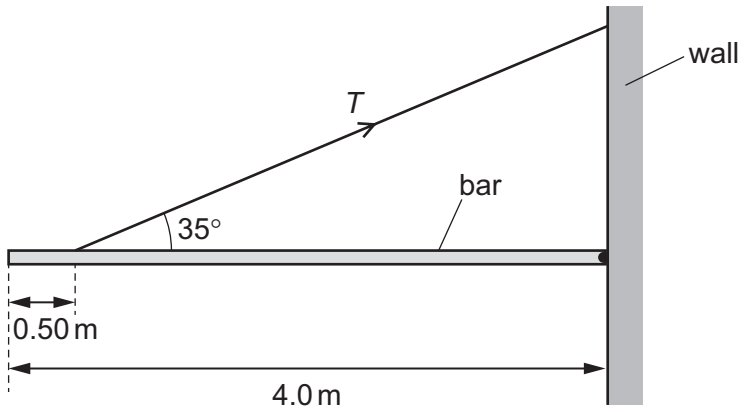
The sphere is released. Immediately after release, the sphere rises.

Which row describes the changes in the magnitudes of the upthrust on the sphere and the resultant force on the sphere as it rises?

	upthrust on the sphere	resultant force on the sphere
A	constant	decreasing
B	constant	increasing
C	decreasing	decreasing
D	decreasing	increasing

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16 A uniform bar of weight 200 N and length 4.0 m is freely hinged on a wall at one end. The bar is horizontal and is held in equilibrium by a cable attached at a distance of 0.50 m from the other end. The cable is at an angle of 35° to the horizontal.

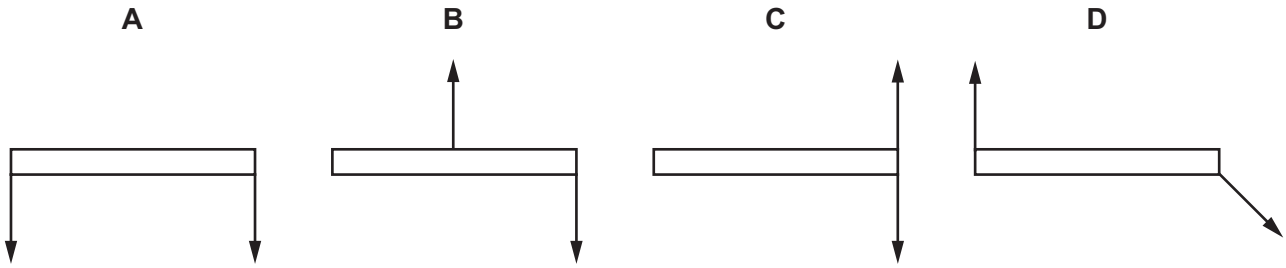


What is the tension T in the cable?

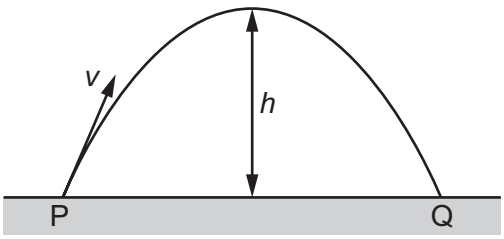
A 140 N **B** 170 N **C** 200 N **D** 400 N

17 The diagrams all show a pair of equal forces acting on a metre rule.

Which diagram shows forces that provide a couple and zero resultant force?



18 A ball of mass m is thrown up to height h in air with an initial velocity v, as shown.



Air resistance is negligible. The acceleration of free fall is g.

What is the **total** work done by the gravitational force on the ball during its flight from P to Q?

A zero **B** $\frac{1}{2}mv^2$ **C** mgh **D** $2mgh$

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19 A spring of spring constant $30\,\text{N}\,\text{m}^{-1}$ is suspended vertically from its top. The spring obeys Hooke's law. Initially the spring is not compressed and not stretched. A mass of 0.50 kg is attached to the bottom of the spring. The mass is released from rest and falls.

Frictional effects are negligible.

In the motion that follows, what is the maximum extension of the spring?

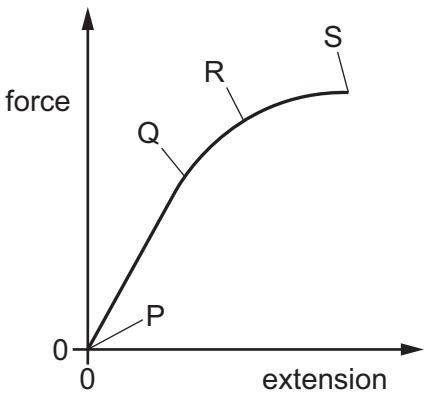
A 0.017 m **B** 0.033 m **C** 0.16 m **D** 0.33 m

20 A wire has original length L and cross-sectional area A. A tensile force F is applied to the wire which causes it to have extension x. The wire obeys Hooke's law.

What is an expression for the Young modulus of the material from which the wire is made?

A $\dfrac{\text{stress} \times x}{L}$ **B** $\dfrac{F}{A \times \text{strain}}$ **C** $\dfrac{F \times x}{A \times L}$ **D** $\dfrac{\text{strain}}{\text{stress}}$

21 A wire is stretched by a gradually increasing force. The force–extension graph for the wire is shown.



Which statement **must** be correct?

A Point Q is the elastic limit.

B Point R is the limit of proportionality.

C The area under the graph from P to S is the elastic potential energy stored in the wire.

D The area under the graph from P to S is the work done in stretching the wire.

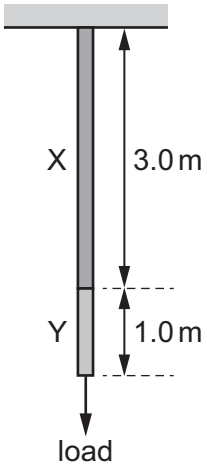
22 A spring has an unstretched length of 0.30 m and a spring constant of $400\,\text{N}\,\text{m}^{-1}$. An object is suspended from the spring and the spring is deformed within its limit of proportionality. The new length of the spring is 0.50 m.

What is the elastic potential energy stored in the spring?

A 8.0 J **B** 16 J **C** 40 J **D** 50 J

23 A wire consists of a 3.0 m length of metal X joined to a 1.0 m length of metal Y.

The cross-sectional area of the wire is uniform.



A load hung from the wire causes metal X to extend by 1.5 mm and metal Y to extend by 1.0 mm.

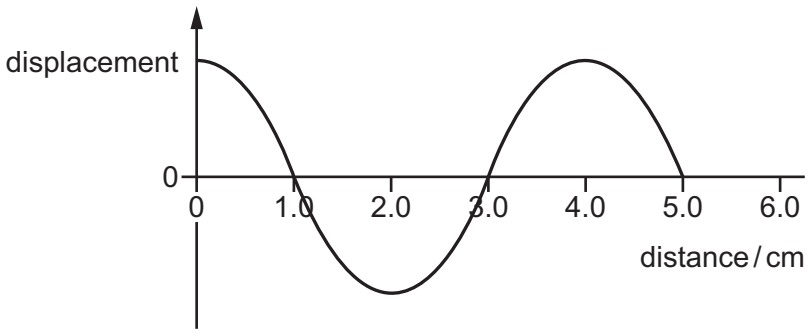
The same load is then hung from a second wire of the same cross-sectional area, consisting of a 1.0 m length of metal X and a 3.0 m length of metal Y.

Both wires are extended within their limit of proportionality.

What is the total extension of this second wire?

A 2.5 mm **B** 3.5 mm **C** 4.8 mm **D** 5.0 mm

24 The graph shows the variation of the displacement with distance for a progressive wave at one instant in time.



The period of the wave is 91 ms.

What can be determined about the wave?

A It has a velocity of 0.44 m s^{-1} and a frequency of 11 Hz.

B It has a velocity of 0.55 m s^{-1} and a wavelength of 5.0 cm.

C It is longitudinal and has a frequency of 11 Hz.

D It is transverse and has a wavelength of 4.0 cm.

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25 Which group of electromagnetic waves is arranged in order from shortest wavelength to longest wavelength?

A radio waves → visible light → gamma rays

B visible light → microwaves → infrared

C visible light → ultraviolet → X-rays

D X-rays → infrared → microwaves

26 A wave has a frequency of 5 GHz.

What is the period of the wave?

A 200 ps **B** 2 ns **C** 20 ns **D** 20 000 μs

27 Three statements about two progressive waves are listed.

1 The waves have the same frequency.

2 The waves have the same amplitude.

3 The waves are emitted with a constant phase difference.

Which statements **must** be correct for the two waves to be coherent?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

28 Waves P and Q have the same amplitude. The waves meet in phase at point X and interfere to give a resultant wave with intensity I.

The amplitude of wave P is doubled.

What is the new intensity of the resultant wave at X, in terms of I?

A $0.44I$ **B** $1.5I$ **C** $2.3I$ **D** $3.0I$

29 Radio waves can be polarised, but sound waves **cannot** be polarised.

Which statement gives the reason for this?

A Radio waves are generally of a higher frequency than sound waves.

B Radio waves are transverse waves, but sound waves are longitudinal waves.

C Radio waves can travel through a vacuum, but sound waves **cannot** travel through a vacuum.

D Radio waves travel at a much higher speed than sound waves.

30 Light of wavelength λ is incident normally on a diffraction grating with a total number of N lines in width w.

A second order maximum is observed at an angle of diffraction θ.

What is N?

A $\frac{w\lambda}{\sin\theta}$ **B** $\frac{2w\lambda}{\sin\theta}$ **C** $\frac{w\sin\theta}{2\lambda}$ **D** $\frac{w\sin\theta}{\lambda}$

31 Kirchhoff's second law is a consequence of a basic principle.

What is this principle?

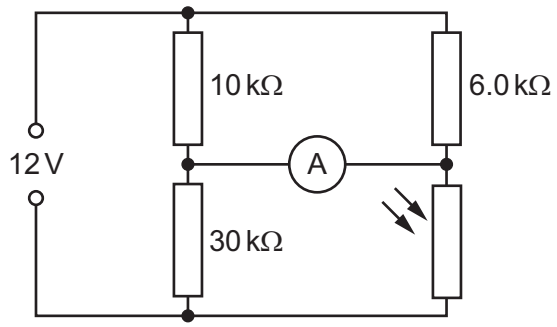
A The charge flowing in an electric circuit is conserved.

B The energy in an electric circuit is conserved.

C The sum of the electric currents entering a point in an electric circuit is equal to the sum of the electric currents leaving that point.

D The sum of the potential differences in an electric circuit is equal to the sum of the products of the current and resistance.

32 The diagram shows a circuit with a light-dependent resistor (LDR).



The ammeter reads zero current.

What is the resistance of the LDR?

A 6.0 kΩ **B** 18 kΩ **C** 26 kΩ **D** 30 kΩ

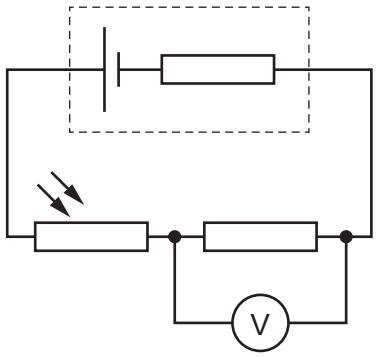
33 A torch uses three lamps connected in parallel and is powered by a cell of electromotive force (e.m.f.) 3.0 V and negligible internal resistance. Each lamp dissipates 0.60 W of power.

What is the current in the cell?

A 0.067 A **B** 0.20 A **C** 0.60 A **D** 0.83 A

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34 A cell with internal resistance is connected to a light-dependent resistor (LDR), a fixed resistor and a voltmeter, as shown.



The voltmeter reading increases.

Which quantity decreases as the voltmeter reading increases?

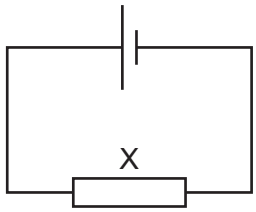
A the charge moving through the cell per unit time

B the energy transferred to the fixed resistor per unit charge

C the intensity of the light incident on the LDR

D the terminal potential difference across the cell

35 The circuit shown contains a cell with negligible internal resistance.



The energy transferred per unit charge in driving charge around the complete circuit is *E*. The potential difference (p.d.) across X is *V*.

The cell is then replaced with a different cell of the same electromotive force (e.m.f.) that has significant internal resistance.

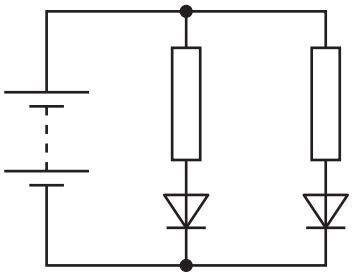
What is the effect on *E* and *V* of replacing the cell?

	effect on *E*	effect on *V*
A	decreases	decreases
B	decreases	increases
C	no change	decreases
D	no change	increases

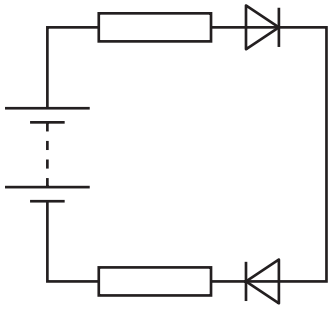
36 In the circuits shown, the batteries are identical and all have negligible internal resistance. All of the resistors have the same resistance. The diodes have zero resistance when conducting and infinite resistance when not conducting.

In which circuit is the current in the battery greatest?

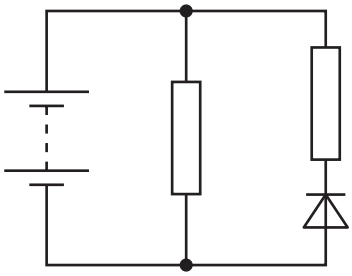
A



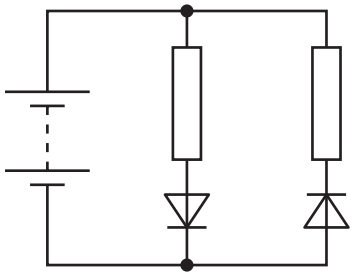
B



C

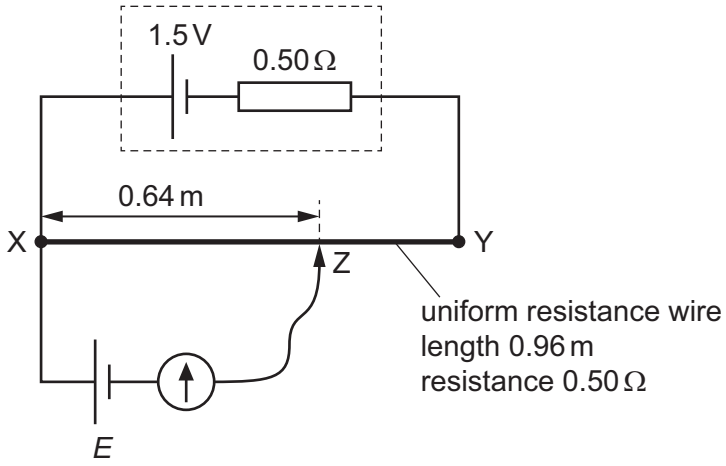


D



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37 A potentiometer circuit is used to determine the electromotive force (e.m.f.) *E* of a cell. The circuit includes a second cell of e.m.f. 1.5 V and internal resistance 0.50 Ω that is connected to a uniform resistance wire XY, as shown.



The resistance wire XY has a length of 0.96 m and a resistance of 0.50 Ω.

The movable connection Z is moved along wire XY. The galvanometer reading is zero when length XZ is 0.64 m.

What is the value of e.m.f. *E*?

A 0.50 V **B** 0.75 V **C** 1.0 V **D** 1.1 V

38 Which particle is **not** a fundamental particle?

A charm quark

B electron

C neutrino

D neutron

39 The isotope fluorine-18, $^{18}_{9}\mathrm{F}$, undergoes β^+ decay to form a stable isotope.

How many neutrons are there in a nucleus of the stable isotope?

A 7 **B** 8 **C** 9 **D** 10

40 Which statement is correct?

A A baryon is a hadron and consists of 2 quarks.

B A meson is a hadron and consists of 3 quarks.

C An electron is a fundamental particle and is a lepton.

D A neutrino is a fundamental particle and is a hadron.

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CAMBRIDGE
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PHYSICS **9702/11**

Paper 1 Multiple Choice **October/November 2024**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	D	**1**
2	D	**1**
3	C	**1**
4	C	**1**
5	D	**1**
6	D	**1**
7	C	**1**
8	B	**1**
9	B	**1**
10	B	**1**
11	B	**1**
12	D	**1**
13	A	**1**
14	C	**1**
15	A	**1**
16	C	**1**
17	B	**1**
18	A	**1**
19	D	**1**
20	B	**1**
21	D	**1**
22	A	**1**
23	B	**1**
24	A	**1**
25	D	**1**
26	A	**1**
27	C	**1**
28	C	**1**

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Question	Answer	Marks
29	B	**1**
30	C	**1**
31	B	**1**
32	B	**1**
33	C	**1**
34	D	**1**
35	C	**1**
36	A	**1**
37	A	**1**
38	D	**1**
39	D	**1**
40	C	**1**

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November 2024 · Paper 12

9702_w24_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **October/November 2024**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 A physical quantity consists of a magnitude and a unit.

Which row does **not** show a correct combination of a quantity and its unit?

	quantity	unit
A	mass	gram
B	length	metre
C	charge	ampere
D	temperature	kelvin

2 A small car travels in a town.

What is a reasonable estimate of the kinetic energy of the car?

A 5×10^4 J **B** 5×10^7 J **C** 5×10^{10} J **D** 5×10^{13} J

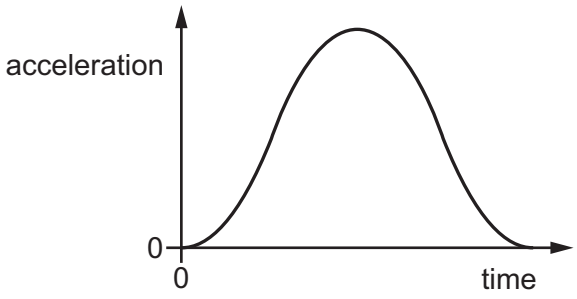
3 A solid bar has a square cross-section. Its length is measured as 50.0 ± 0.2 cm and its width is measured as 2.00 ± 0.01 cm.

These values are used to calculate the volume of the bar.

What is the percentage uncertainty in the calculated volume?

A ±0.21% **B** ±0.22% **C** ±0.90% **D** ±1.4%

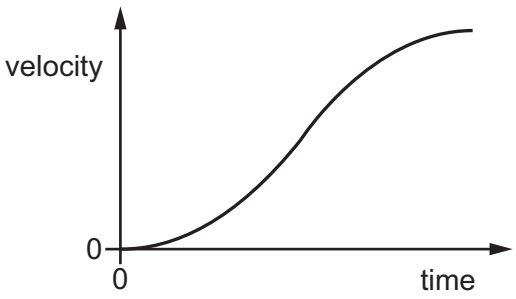
4 The graph shows how the acceleration of an object moving in a straight line varies with time.



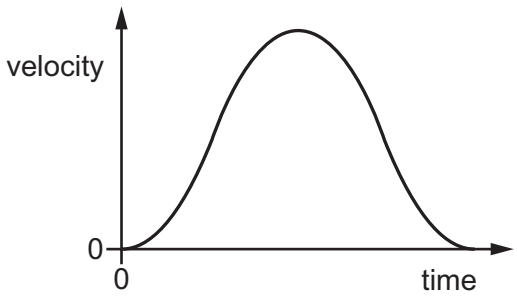
The object starts from rest.

Which graph shows the variation with time of the velocity of the object over the same time interval?

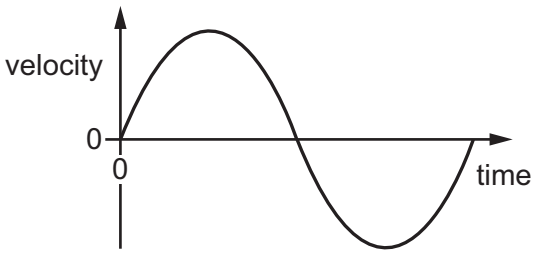
A



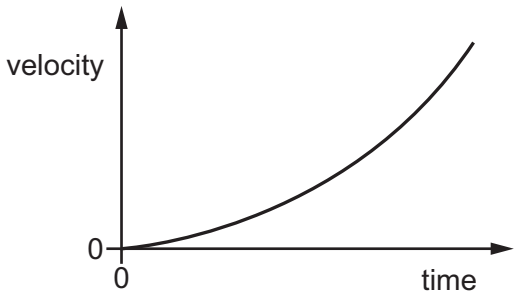
B



C



D



5 A stone falls vertically from rest. Air resistance is negligible.

What is the speed of the stone when it has fallen through a distance of 0.40 m?

A $2.0\,m\,s^{-1}$ **B** $2.8\,m\,s^{-1}$ **C** $3.9\,m\,s^{-1}$ **D** $7.8\,m\,s^{-1}$

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6 A ball is thrown horizontally off a tall building. The ground is horizontal. Air resistance is negligible.

Which statement about the motion of the ball is correct?

A The acceleration of the ball is always at right angles to the path of the ball.

B The ball follows a circular path until it hits the ground.

C The ball has constant acceleration.

D The ball's time in the air is proportional to the velocity at which the ball is thrown.

7 A moving object strikes a stationary object. The collision is inelastic. The objects move off together.

Assume that the two objects form an isolated system.

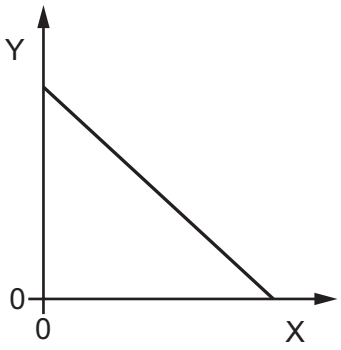
Which row shows the possible values of total momentum and total kinetic energy for the system before and after the collision?

	total momentum before collision / kg m s^{-1}	total momentum after collision / kg m s^{-1}	total kinetic energy before collision / J	total kinetic energy after collision / J
A	6	2	90	30
B	6	6	30	90
C	6	6	90	30
D	6	6	90	90

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[Turn over

8 The graph shows how a quantity Y varies with a quantity X for an object falling vertically at its terminal velocity towards the surface of the Earth.



Which quantities could X and Y represent?

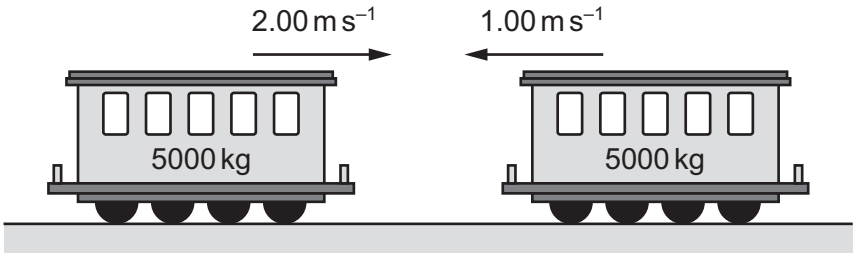
	X	Y
A	time	acceleration
B	time	height above surface
C	distance moved	kinetic energy
D	distance moved	velocity

9 A ball of mass 2.0 kg travels horizontally with a speed of 4.0 m s^{-1}. The ball collides with a wall and rebounds in the opposite direction with a speed of 2.8 m s^{-1}. The time of the collision is 150 ms.

What is the average force exerted on the wall?

A 16 N **B** 37 N **C** 53 N **D** 91 N

10 Two train carriages each of mass 5000 kg roll toward one another on a horizontal frictionless track. One is travelling at a speed of 2.00 m s^{-1} and the other at a speed of 1.00 m s^{-1}, as shown.



They collide and join together.

What is the kinetic energy lost during the collision?

A 1250 J **B** 7500 J **C** 11 250 J **D** 12 500 J

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11 Which single condition **must** apply for an object to be in equilibrium?

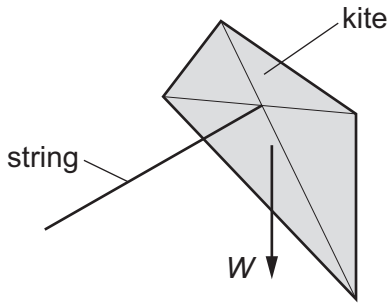
A The object has a constant non-zero acceleration.

B The object is stationary.

C There are no forces acting on the object.

D The resultant force acting on the object is zero.

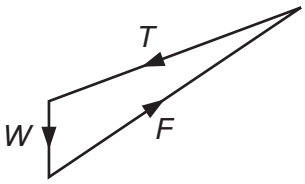
12 A kite is in equilibrium at the end of a string, as shown.



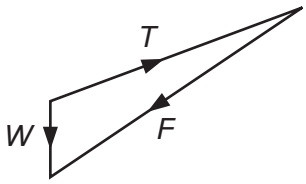
The kite has three forces acting on it: the weight *W*, the tension *T* in the string, and the force *F* from the wind.

Which vector diagram represents the forces acting on the kite?

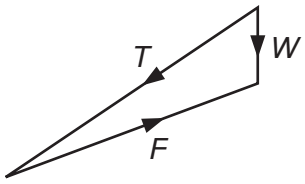
A



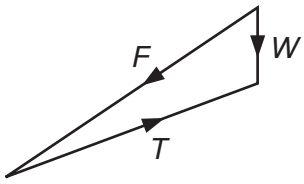
B



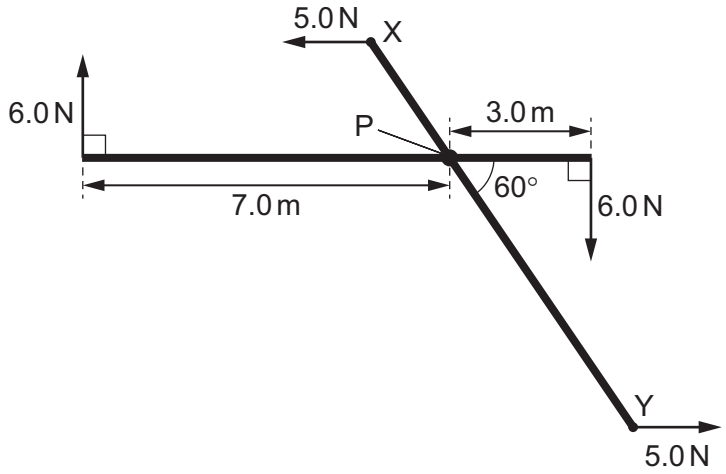
C



D



13 Four forces act about a point P, as shown.



The forces act in the same plane and produce no resultant moment about point P.

What is the length XY?

A 6.9 m **B** 12 m **C** 14 m **D** 24 m

14 A rectangular block of lead of density 1.13×10^4 kg m^{-3} has sides of length 12.0 cm, 15.0 cm and 10.0 cm.

What is the maximum pressure the block can exert when resting on a table?

A 1.13 kPa **B** 1.70 kPa **C** 11.1 kPa **D** 16.6 kPa

15 What is the definition of power?

A work done in one second

B work done in unit time

C work done per second

D work done per unit time

16 A model car travels at a constant velocity of 12 m s^{-1} in a straight horizontal line. The input power to the engine of the car is 800 W. The efficiency of the engine is 60%.

What is the total horizontal resistive force on the car?

A 27 N **B** 40 N **C** 67 N **D** 110 N

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17 Which statement represents the principle of conservation of energy?

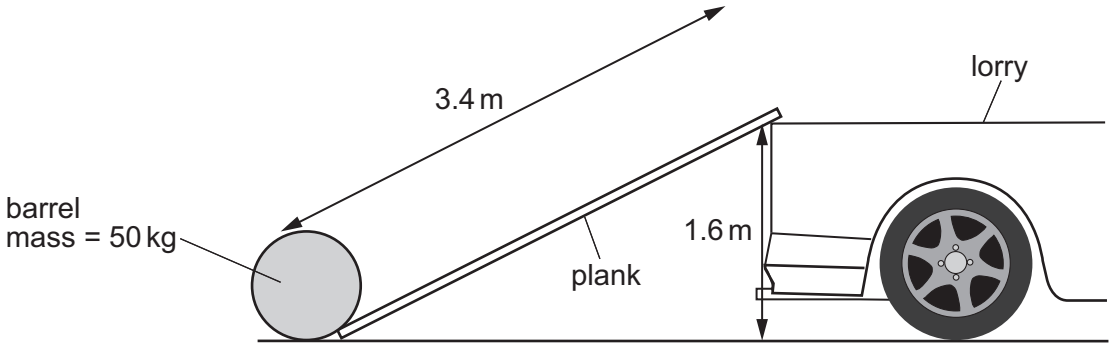
A Energy cannot be used faster than it is created.

B The supply of energy is limited, so energy must be conserved.

C The total energy in a closed system is constant.

D The total energy input to a system is equal to the useful energy output.

18 A barrel of mass 50 kg is loaded onto the back of a lorry 1.6 m high by pushing it up a frictionless plank 3.4 m long.



What is the minimum work done?

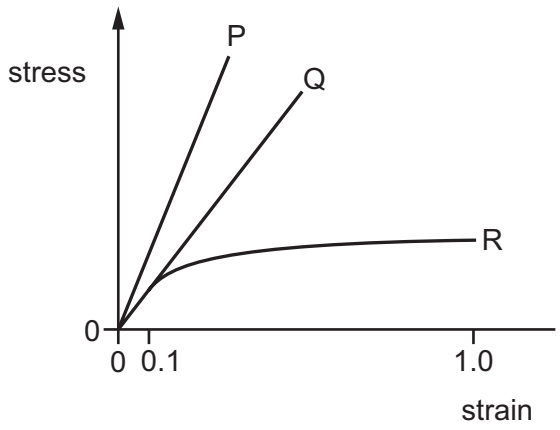
A 80 J **B** 170 J **C** 780 J **D** 1700 J

19 What is a unit for stress?

A $kg\,m^{-1}\,s^{-2}$ **B** $kg\,m^{-2}\,s^{-2}$ **C** $N\,m^{-1}$ **D** $N\,m$

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20 The graph shows the relationship between stress and strain for three wires of the same linear dimensions but made from different materials.



Which statements are correct?

1 The extension of P is approximately twice that of Q for the same stress.

2 The ratio of the Young modulus for P to that of Q is approximately two.

3 For strain less than 0.1, R obeys Hooke's law.

A 1, 2 and 3 **B** 1 and 3 only **C** 2 and 3 only **D** 2 only

21 Four solid steel rods equally support an object weighing 10 kN. Each rod is of length 2.0 m and cross-sectional area 250 mm^2. The weight of the object causes the rods to contract by 0.10 mm. The rods obey Hooke's law.

What is the Young modulus of steel?

A $2.0 \times 10^8 \, N\,m^{-2}$

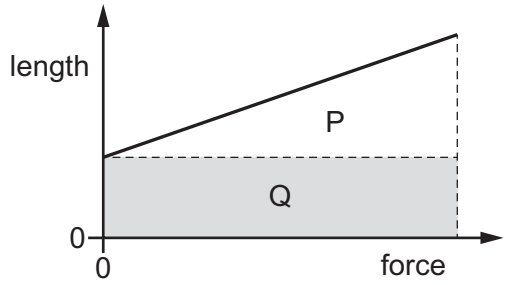
B $2.0 \times 10^{11} \, N\,m^{-2}$

C $8.0 \times 10^8 \, N\,m^{-2}$

D $8.0 \times 10^{11} \, N\,m^{-2}$

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22 The graph shows how the length of a spring varies with the force applied to it. Two areas P and Q are labelled.



Which area represents the work done in stretching the spring?

A area P

B area Q

C area P + area Q

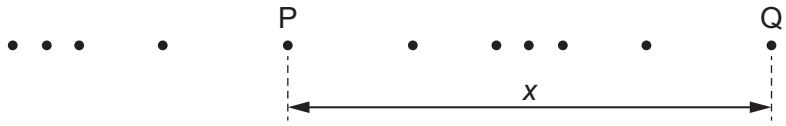
D area Q – area P

23 A horizontal beam of vertically polarised light of amplitude *A* is incident normally on a polarising filter. The transmission axis of the filter is at an angle of 50° to the vertical.

What is the amplitude of the light in the beam after it has passed through the filter?

A 0.17*A** **B** 0.41*A** **C** 0.64*A** **D** 0.80*A*

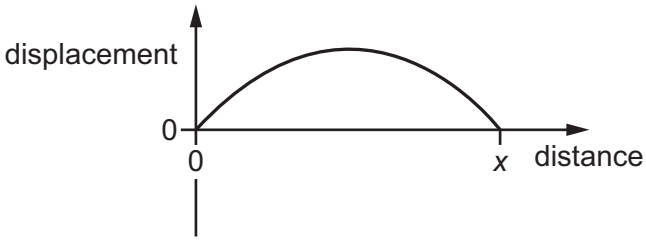
24 A progressive longitudinal sound wave moves through air. The diagram shows the positions of the air particles along part of the wave at one instant.



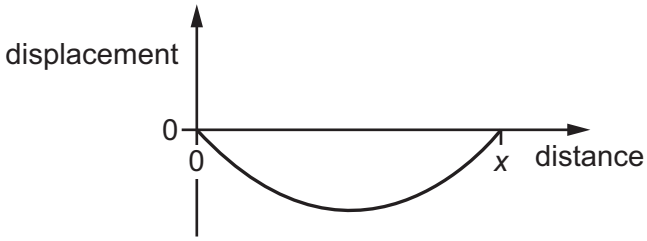
Point Q is a distance *x* from point P.

Which graph shows the variation of the displacement of the air particles with distance from P along the wave?

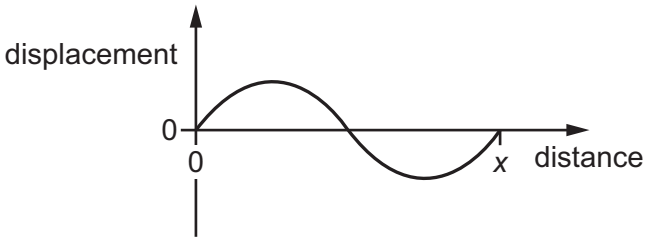
A



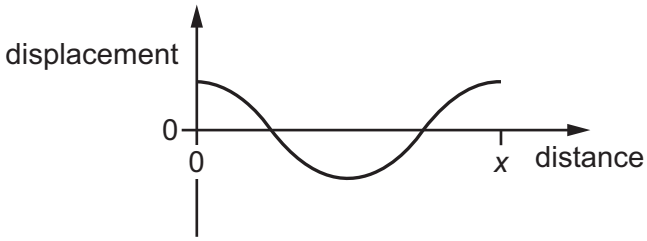
B



C



D



25 An electromagnetic wave in free space has a frequency of 3.0×10^{16} Hz.

Which row gives the principal region of this wave and an example of an electromagnetic wave with a lower frequency?

	principal region	wave with lower frequency
A	infrared	visible light
B	infrared	X-rays
C	ultraviolet	visible light
D	ultraviolet	X-rays

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26 A source of sound emits waves of a constant frequency. The source moves at a constant speed in a straight line relative to a stationary observer.

Which velocity of the source gives the smallest observed frequency?

	speed / m s^{-1}	direction
A	5	away from observer
B	10	away from observer
C	15	towards observer
D	20	towards observer

27 What could describe the time-base of a cathode-ray oscilloscope (CRO)?

A the frequency per division on the screen

B the number of divisions on the screen per unit frequency

C the number of divisions on the screen per unit time

D the time per division on the screen

28 What happens when two waves superpose at a point?

A Their amplitudes are added together.

B Their displacements are added together.

C Their frequencies are added together.

D Their velocities are added together.

29 An electromagnetic wave is diffracted as it passes through a single slit. The width of the slit is larger than the wavelength of the wave.

Which change will decrease the amount of diffraction of the wave?

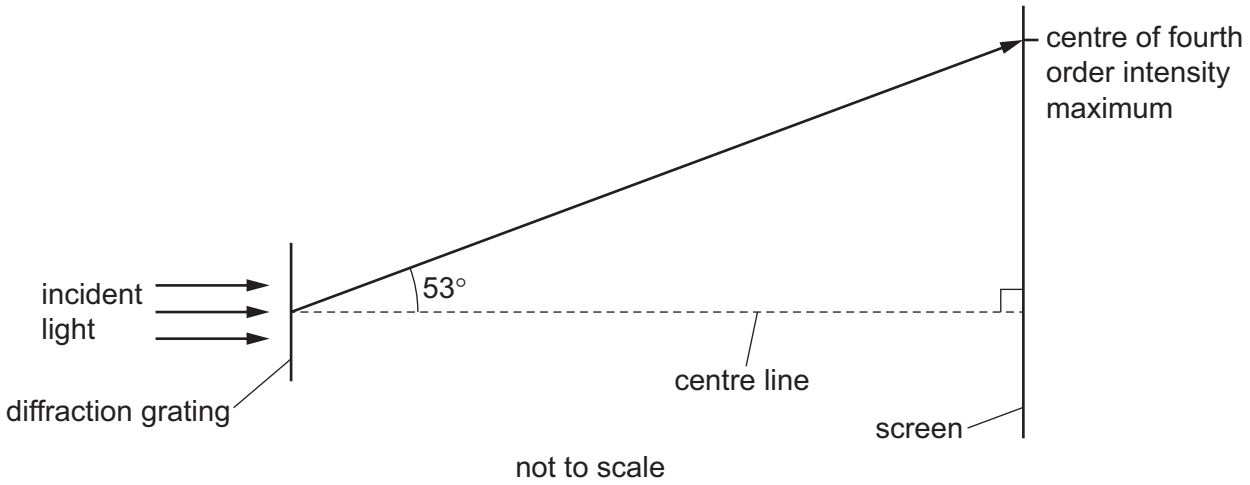
A Decrease the frequency of the wave.

B Decrease the time period of the wave.

C Decrease the width of the slit.

D Increase the wavelength of the wave.

30 The diagram shows visible light incident normally on a diffraction grating.



A pattern of intensity maxima forms on the screen. A line connecting the centre of the fourth order intensity maximum with the centre of the diffraction grating forms an angle of 53° with the centre line. The grating has a line spacing of 2.7×10^{-6} m.

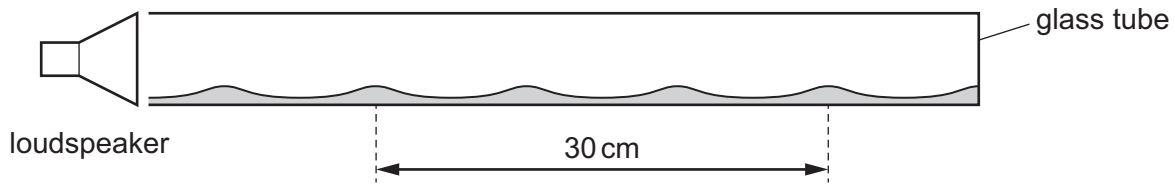
What is the wavelength of the incident light?

A 4.1×10^{-7} m **B** 5.4×10^{-7} m **C** 1.6×10^{-6} m **D** 2.2×10^{-6} m

31 A horizontal glass tube, closed at one end, has a layer of dust laid inside it on its lower side. Sound is emitted from a loudspeaker that is placed near the open end of the tube.

The frequency of the sound is varied and, at one frequency, a stationary wave is formed inside the tube so that the dust forms small heaps.

The distance between four heaps of dust is 30 cm.



The speed of sound in the air in the tube is 330 m s^{-1}.

What is the frequency of the sound emitted by the loudspeaker?

A 1650 Hz **B** 2200 Hz **C** 3300 Hz **D** 6600 Hz

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32 There is a current in a resistor for a short time interval.

Which statement about the total charge that passes through the resistor is correct?

A It can take any value.

B It is an integer multiple of the elementary charge.

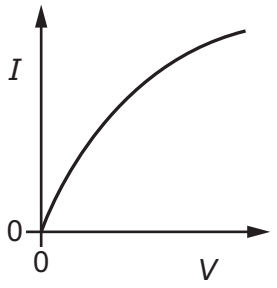
C It is the elementary charge.

D It is the rate of flow of the current.

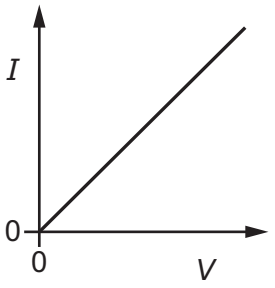
33 The potential difference V across a filament lamp is slowly raised from zero to its normal operating value.

Which graph represents the variation with V of the current I in the lamp?

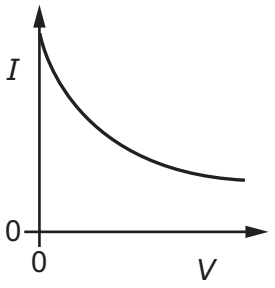
A



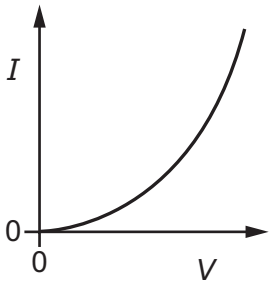
B



C



D



34 An electric current of 12 A is in a wire of length 0.35 m.

The average drift speed of the free electrons (charge carriers) in the wire is $5.0 \times 10^{-4}\,\text{m s}^{-1}$.

How many free electrons are in the wire?

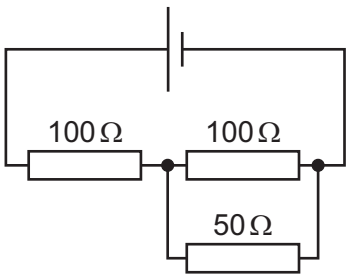
A 1.3×10^{16} **B** 5.3×10^{22} **C** 1.5×10^{23} **D** 4.3×10^{23}

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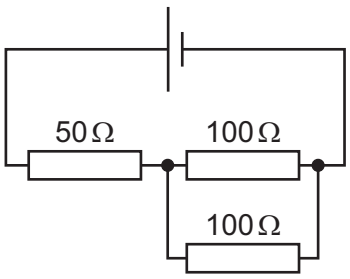
35 Three resistors are connected to a cell of negligible internal resistance.

Which circuit has a combined resistance of 100 Ω?

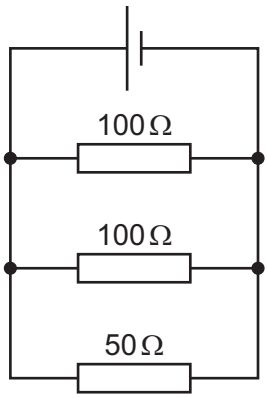
A



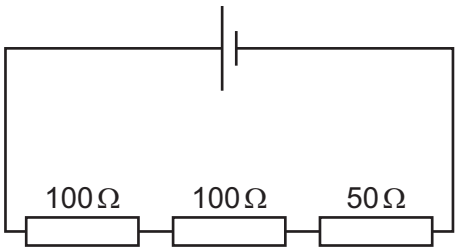
B



C



D



36 Each of Kirchhoff's laws is a statement based on the conservation of a physical quantity.

Which quantity is conserved in each law?

	Kirchhoff's first law	Kirchhoff's second law
A	charge	energy
B	energy	current
C	power	charge
D	resistance	power

37 A cell with electromotive force (e.m.f.) 1.50 V delivers a current of 0.26 A to a resistor of resistance 5.0 Ω connected between its terminals.

What is the internal resistance of the cell?

A 0.04 Ω **B** 0.20 Ω **C** 0.39 Ω **D** 0.77 Ω

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38 What is **not** a quark flavour?

A charm

B meson

C strange

D up

39 How many down quarks are in a nucleus of hydrogen-3, $^{3}_{1}\mathrm{H}$?

A 2 **B** 3 **C** 4 **D** 5

40 An isotope of boron decays to beryllium by β^{+} emission.

Which particle is emitted in addition to the β^{+} particle?

A antineutrino

B electron

C neutrino

D neutron

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CAMBRIDGE
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Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **October/November 2024**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	C	**1**
2	A	**1**
3	D	**1**
4	A	**1**
5	B	**1**
6	C	**1**
7	C	**1**
8	B	**1**
9	D	**1**
10	C	**1**
11	D	**1**
12	A	**1**
13	C	**1**
14	D	**1**
15	D	**1**
16	B	**1**
17	C	**1**
18	C	**1**
19	A	**1**
20	C	**1**
21	B	**1**
22	A	**1**
23	C	**1**
24	C	**1**
25	C	**1**
26	B	**1**
27	D	**1**
28	B	**1**

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Question	Answer	Marks
29	B	**1**
30	B	**1**
31	A	**1**
32	B	**1**
33	A	**1**
34	B	**1**
35	B	**1**
36	A	**1**
37	D	**1**
38	B	**1**
39	D	**1**
40	C	**1**

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November 2024 · Paper 13

9702_w24_qp_13

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **October/November 2024**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

IB24 11_9702_13/3RP
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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which statement about physical quantities is correct?

A A physical quantity does **not** always have a magnitude or a unit.

B A physical quantity must always have a magnitude but does **not** always have a unit.

C A physical quantity must always have a unit but does **not** always have a magnitude.

D A physical quantity must always have a magnitude and a unit.

2 What is a reasonable estimate of the mass of a solid sphere of copper that has a diameter of 60 cm?

A 0.1 kg **B** 10 kg **C** 1000 kg **D** 100 000 kg

3 The power output P of a star can be modelled with the equation

$$P = \sigma A T^4$$

where σ is a constant, A is the surface area of the star and T is the surface temperature of the star.

What are the SI base units of σ?

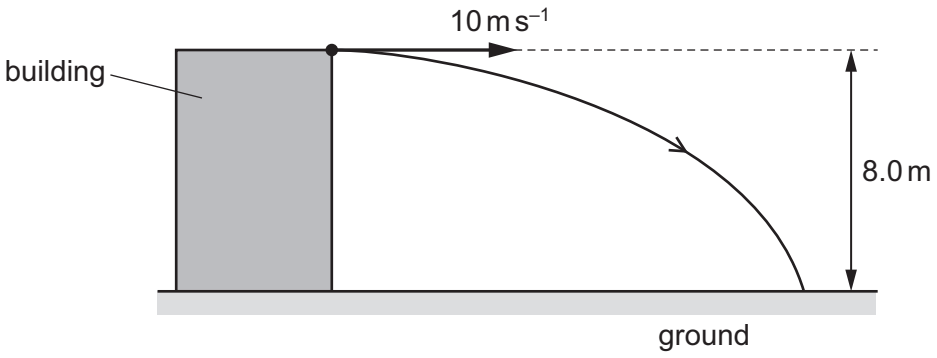
A $\text{kg s}^{-2}\text{K}^{-4}$

B $\text{kg s}^{-3}\text{K}^{-4}$

C $\text{kg m}^{-1}\text{s}^{-2}\text{K}^{-4}$

D $\text{kg m}^{-1}\text{s}^{-3}\text{K}^{-4}$

4 A boy throws a stone with a horizontal velocity of $10\,\text{m s}^{-1}$ from the top of a building. The height of the building is 8.0 m. The stone travels along a curved path until it hits the horizontal ground, as shown.

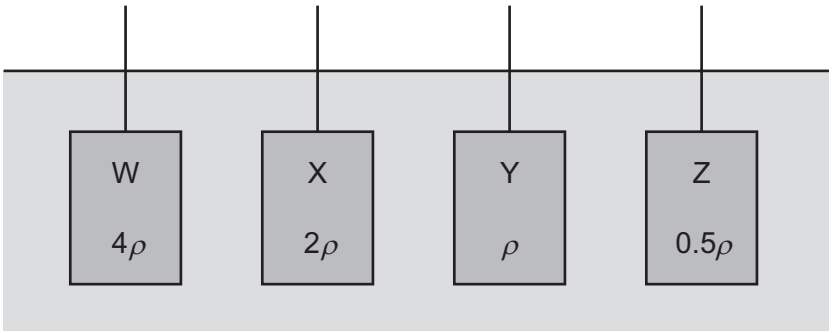


Air resistance is negligible.

How long does it take the stone to reach the ground?

A 0.61 s **B** 0.80 s **C** 1.3 s **D** 1.6 s

5 Four cuboids with identical lengths, breadths and heights are immersed in water. The cuboids are held at the same depth and in identical orientations by vertical rods, as shown.



Water has density ρ.
Cuboid W is made of material of density 4ρ.
Cuboid X is made of material of density 2ρ.
Cuboid Y is made of material of density ρ.
Cuboid Z is made of material of density 0.5ρ.

Which statement is correct?

A The upthrust of the water on each of the cuboids is the same.

B The upthrust of the water on W is twice the upthrust of the water on X.

C The upthrust of the water on X is twice the upthrust of the water on W.

D The upthrust of the water on Y is zero.

6 In which example is it **not** possible for the underlined object to be in equilibrium?

A An aeroplane climbs at a steady rate.

B An aeroplane tows a glider at a constant altitude.

C A speedboat changes direction at a constant speed.

D Two boats tow a ship into harbour.

7 Two identical balls are projected vertically upwards from ground level with the same initial velocity. Ball X is in a vacuum and ball Y is in air.

Which statement about the motion of the balls is correct?

A Ball X reaches a greater maximum height and in a longer time than ball Y.

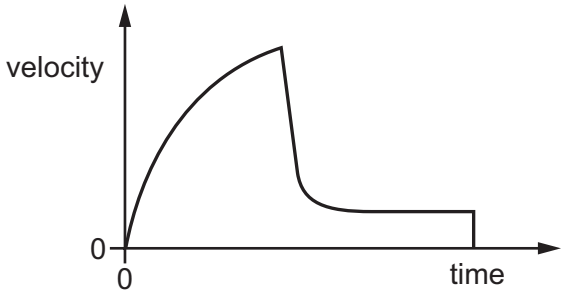
B Ball X reaches a greater maximum height and in a shorter time than ball Y.

C Ball Y reaches a greater maximum height and in a longer time than ball X.

D Ball Y reaches a greater maximum height and in a shorter time than ball X.

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8 The graph shows the variation of velocity with time for a stone that falls from a bridge into a lake and sinks to the bottom of the lake.



What can be deduced about the motion of the stone?

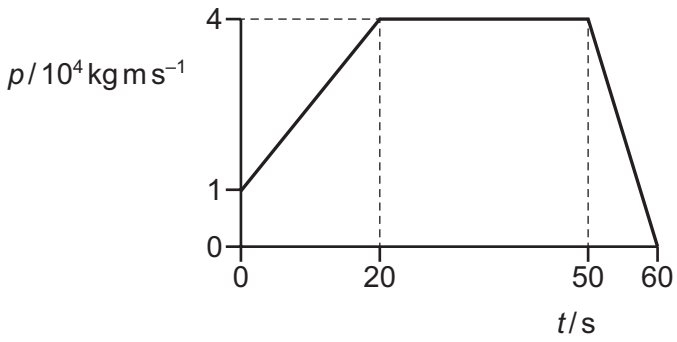
A Terminal velocity was reached in air.

B The acceleration in air was decreasing with increasing time.

C The distance travelled in water was greater than the distance travelled in air.

D The rate of change of velocity in air was constant.

9 The graph shows the variation of the momentum p with time t for a car.



What is the resultant force on the car at $t = 10\,s$?

A 0 **B** 1500 N **C** 2000 N **D** 4000 N

10 Two objects X and Y form an isolated system. X and Y collide and then separate. The mass of X is greater than the mass of Y.

Which statement about the collision is correct?

A The change of momentum of Y is greater than the change of momentum of X.

B The force on Y is greater than the force on X.

C The forces that X and Y exert on each other act for the same length of time.

D The forces that X and Y exert on each other are gravitational forces only.

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11 Which row states whether total momentum and total kinetic energy are conserved in an inelastic collision in which there are no external forces?

	total momentum	total kinetic energy
A	conserved	conserved
B	conserved	not conserved
C	not conserved	conserved
D	not conserved	not conserved

12 A tennis ball is thrown vertically upwards. The tennis ball reaches its highest point and then falls back down to the point from which it was thrown.

Air resistance is significant.

At which position on the path of the tennis ball is the resultant force on the tennis ball greatest?

A at the start just after the tennis ball is released

B when the tennis ball is halfway to its highest point on the way up

C when the tennis ball is at its highest point

D when the tennis ball is halfway from its highest point on the way down

13 Which statement correctly describes a couple?

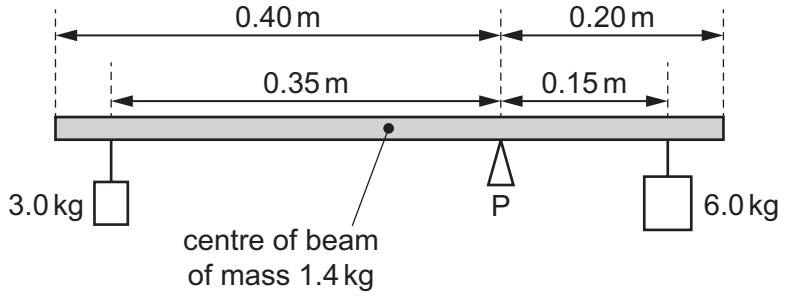
A A couple is a pair of forces that act in the same direction.

B A couple is a pair of forces that act on the centre of gravity of an object.

C A couple is a pair of forces that act to produce a resultant force.

D A couple is a pair of forces that act to produce rotation only.

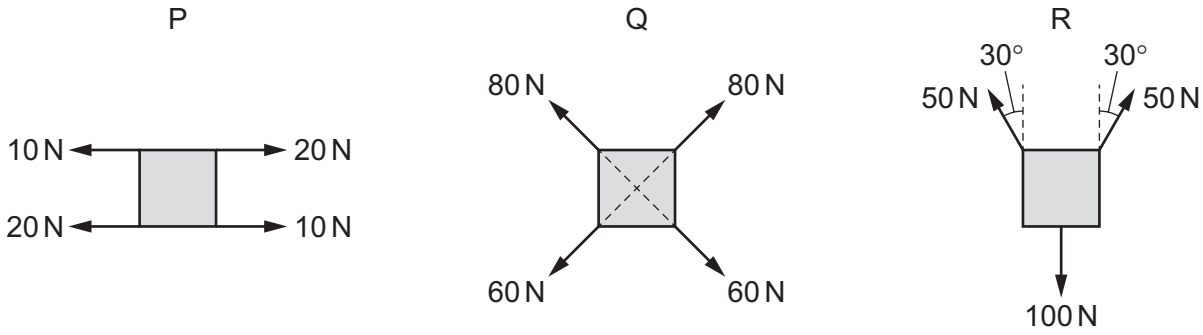
14 A uniform beam of mass 1.4 kg is pivoted at P, as shown. The beam has a length of 0.60 m and P is a distance of 0.20 m from one end. Loads of 3.0 kg and 6.0 kg are suspended at distances of 0.35 m and 0.15 m from the pivot, as shown.



What is the torque that must be applied to the beam in order to maintain it in equilibrium?

A 0.010 N m **B** 0.10 N m **C** 0.29 N m **D** 2.8 N m

15 The diagrams show three rigid objects P, Q and R being subjected to different combinations of forces.



Which objects are in equilibrium?

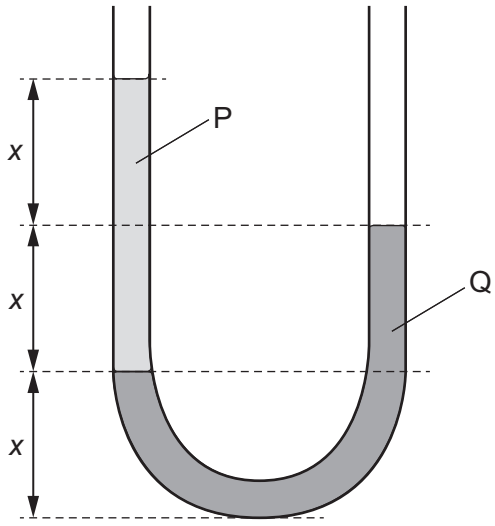
A P and Q

B P and R

C Q and R

D none of them

16 The diagram shows two liquids, labelled P and Q, that do **not** mix. The liquids are in equilibrium in an open U-tube. Three equal distances *x* are labelled.



What is the ratio $\frac{\text{density of P}}{\text{density of Q}}$?

A $\frac{1}{2}$ **B** $\frac{2}{3}$ **C** $\frac{3}{2}$ **D** 2

17 A stone of mass 0.30 kg is thrown vertically downwards with a speed of 20 m s^{-1} from a height of 12 m above the ground. It falls vertically until it hits the ground. Air resistance is negligible.

What is the kinetic energy of the stone just before it hits the ground?

A 25 J **B** 35 J **C** 60 J **D** 95 J

18 A student can run or walk up the stairs to her classroom.

Which statement describes the power required and the gravitational potential energy gained while running up the stairs compared to walking up them?

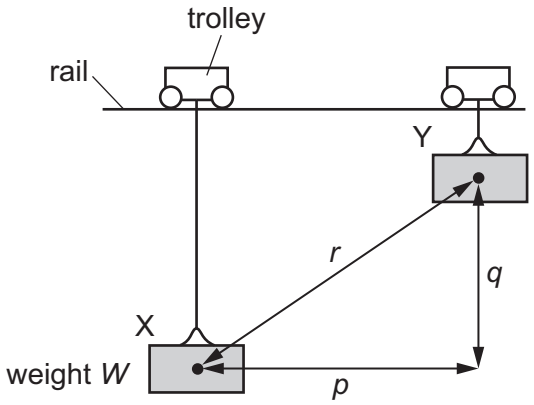
A Running provides more gravitational potential energy and uses more power.

B Running provides more gravitational potential energy and uses the same power.

C Running provides the same gravitational potential energy and uses more power.

D Running provides the same gravitational potential energy and uses the same power.

19 A weight *W* hangs from a trolley that runs along a rail. The trolley moves horizontally through a distance *p* and simultaneously raises the weight through a height *q*.



As a result, the weight moves through a distance *r* from X to Y. It starts and finishes at rest.

How much work is done on the weight during this process?

A Wp **B** $W(p + q)$ **C** Wq **D** Wr

20 The equation for kinetic energy E_K can be derived using the equations of motion.

Four equations relating to motion are listed.

1 $W = Fs$

2 $F = ma$

3 $v^2 = u^2 + 2as$

4 $P = \frac{W}{t}$

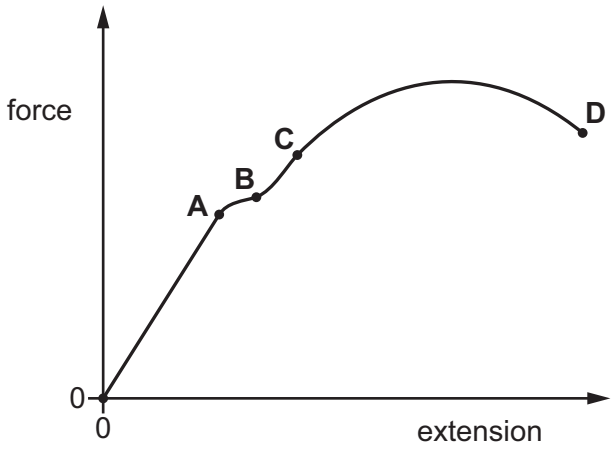
Which three equations can be used to derive the equation for E_K?

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1, 3 and 4 **D** 2, 3 and 4

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21 The force–extension graph of a metal wire is shown.

At which point on the graph does the metal wire stop obeying Hooke's law?



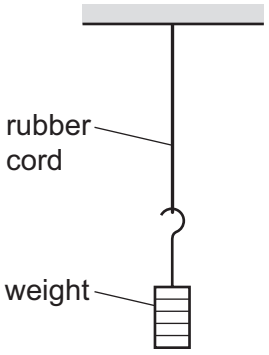
22 A uniform wire is made of a metal that has a Young modulus of 1.3×10^{11} Pa.

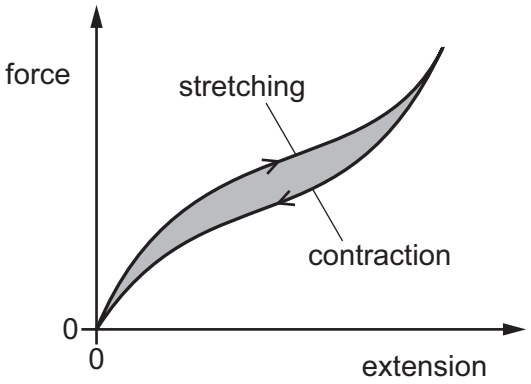
The wire is 2.4 m long and has a spring constant of $2.7 \times 10^{4}\,\text{N m}^{-1}$.

What is the volume of the wire?

A $2.6 \times 10^{-8}\,\text{m}^3$ **B** $6.3 \times 10^{-8}\,\text{m}^3$ **C** $5.0 \times 10^{-7}\,\text{m}^3$ **D** $1.2 \times 10^{-6}\,\text{m}^3$

23 A rubber cord hangs from a rigid support. A weight attached to its lower end is gradually increased from zero, and then gradually reduced to zero.





The force–extension curve for contraction is below the force–extension curve for stretching.

What does the shaded area between the curves represent?

A the elastic potential energy stored in the rubber cord

B the thermal energy dissipated in the rubber cord

C the work done by the rubber cord during contraction

D the work done on the rubber cord during stretching

24 A source of sound waves of constant frequency is travelling as shown.

In which situation would the stationary observer detect a sound with the lowest frequency?

A



B



C



D



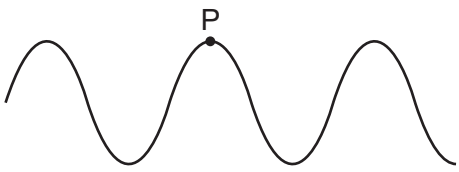
25 Each of the principal radiations of the electromagnetic spectrum has a range of wavelengths.

Which wavelength is correctly linked to its radiation?

	wavelength / m	radiation
A	10^{-9}	gamma ray
B	10^{-5}	microwave
C	10^{-8}	ultraviolet
D	10^{-14}	X-ray

26 A transverse water wave has a frequency of 15 Hz, a wavelength of 0.12 m and an amplitude of 4.0 mm.

P is a water particle that is initially at the peak of the wave, as shown.



What is the total vertical distance travelled by P in a time of 0.5 s?

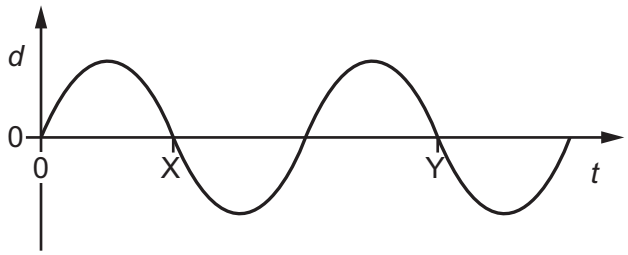
A 30 mm **B** 60 mm **C** 120 mm **D** 900 mm

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27 A wave is formed on a string.

A student plots a graph of the variation of displacement *d* with time *t* for a point on the string.

The student marks two points, X and Y, on the graph.



Which property of the wave is represented by the distance along the horizontal axis between X and Y?

A amplitude

B frequency

C period

D wavelength

28 A stationary wave is set up in a stretched string.

Which distance is equal to the wavelength of the wave?

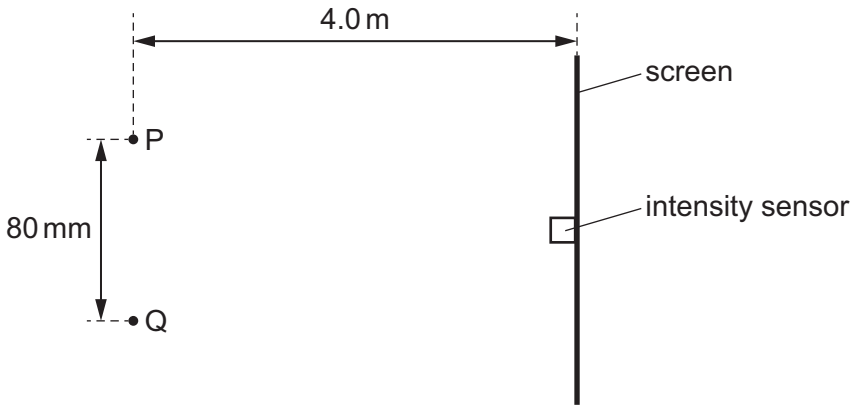
A double the distance between adjacent antinodes

B half the distance between adjacent nodes

C the distance between adjacent antinodes

D the distance between a node and an adjacent antinode

29 A source of coherent light is incident on two slits, P and Q, which are placed 80 mm apart. The light has a single frequency of 1.5×10^{12} Hz. The light from the slits meets on a screen that is a distance of 4.0 m from the slits. The screen is parallel to a line joining the slits.



not to scale

An intensity sensor is placed on the screen at the midpoint of the interference pattern such that the intensity reading is a maximum. The intensity sensor is moved along the screen.

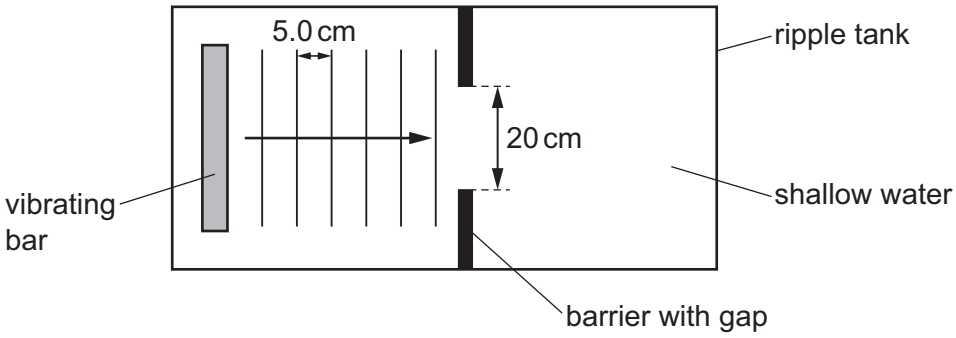
The sensor travels through two intensity minima, two intensity maxima and stops in the middle of the third intensity minimum.

Which distance does the sensor move through?

A 4.0 mm **B** 10 mm **C** 25 mm **D** 50 mm

30 A vibrating bar produces surface water waves in a ripple tank.

The wavelength of the waves is 5.0 cm and they pass through a gap of width 20 cm.



Which change will increase the amount of diffraction that is observed?

A decreasing the distance between the bar and the gap

B decreasing the frequency of the wave

C increasing the amplitude of the wave

D increasing the width of the gap

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31 What is an electric current?

A a flow of charge carriers

B a flow of energy

C an electron

D the charge on a particle

32 A wire of diameter d is connected into an electric circuit. There is a current I in the wire and the charge carriers have an average drift speed v.

The wire is replaced with a new wire of the same material but with a diameter $0.5d$.

The current is adjusted so that the charge carriers in the new wire have an average drift speed $3v$.

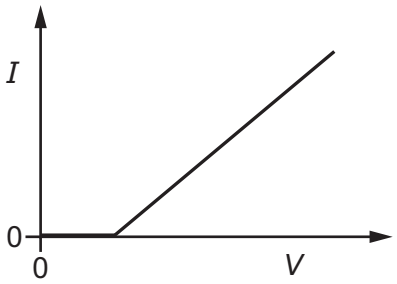
What is the current in the new wire?

A $0.75I$ **B** $1.5I$ **C** $6I$ **D** $12I$

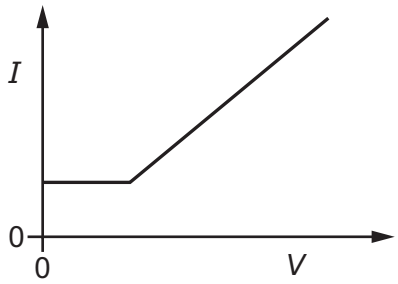
33 A fixed resistor and a diode are combined by connecting them in series. The total potential difference V across the combination is varied and the corresponding current I is measured.

Which graph could represent the variation of I with V?

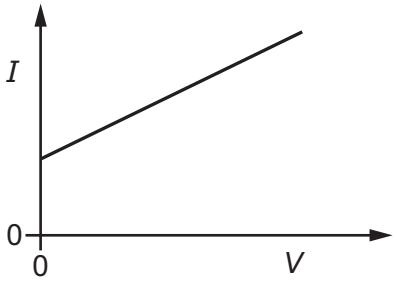
A



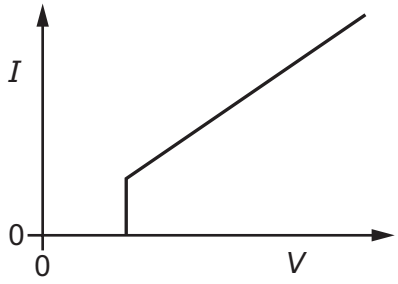
B



C



D



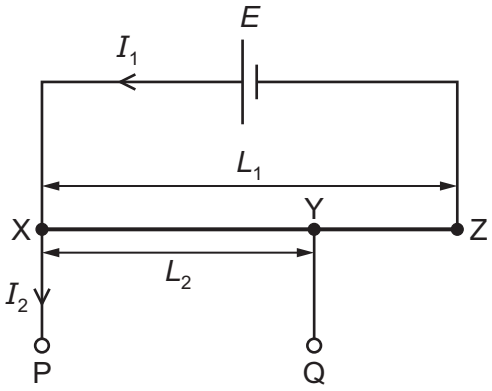
34 The potential difference (p.d.) across a fixed resistor is V. The power dissipated in the resistor is 5.0 W.

The p.d. across the resistor then changes to a new value. With the new p.d. the energy transferred to the resistor in a time of 2.25 s is 45.0 J.

What is the new p.d. across the resistor?

A $\frac{1}{2}V$ **B** $2V$ **C** $3V$ **D** $4V$

35 The diagram shows part of a circuit that uses a potentiometer wire to measure a potential difference (p.d.) in an external circuit.



The potentiometer wire XZ has length L_1. It is connected to a cell of known electromotive force (e.m.f.) E that has negligible internal resistance.

Terminals P and Q are connected to the external p.d. to be measured. The length of the potentiometer wire between points X and Y is L_2.

The ratio of the lengths L_1 and L_2 is used to determine the p.d. between P and Q in terms of E.

Which condition must be met in order to determine this p.d.?

A The current I_1 must be zero.

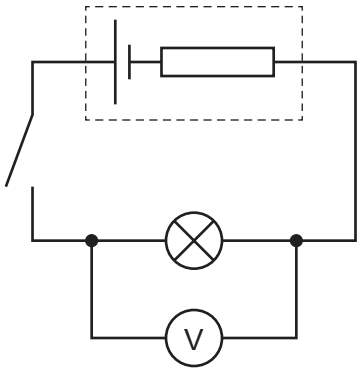
B The current I_2 must be zero.

C The p.d. across YZ must be zero.

D The resistance of the external circuit must be zero.

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36 The diagram shows a circuit.



Which statement about the circuit is **not** correct?

A Electromotive force is the energy transferred per unit charge.

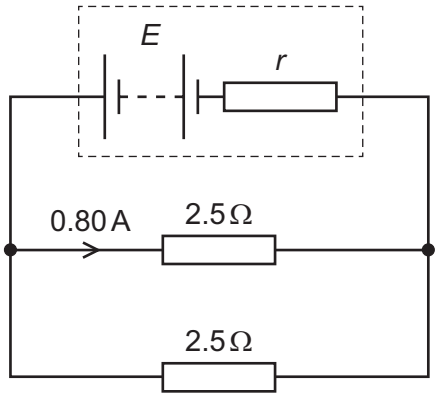
B Energy is transferred from chemical potential energy in the cell to other forms when the switch is closed.

C The electromotive force of the cell is greater than the terminal potential difference when the switch is closed.

D When the switch is open, the voltmeter measures the electromotive force of the cell.

37 A circuit contains a battery of electromotive force E and internal resistance r connected to two resistors each of resistance $2.5\,\Omega$.

The resistors are connected in parallel as shown.



The current in one of the resistors is 0.80 A.

Which expression, where E is in volts and r is in ohms, gives the internal resistance r?

A $\frac{E - 4.0}{1.6}$ **B** $\frac{E - 4.0}{0.80}$ **C** $\frac{E - 2.0}{1.6}$ **D** $\frac{E - 2.0}{0.80}$

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38 What is the rest mass of a beta-particle?

A 0

B 9.11×10^{-31} kg

C 1.66×10^{-27} kg

D 1.67×10^{-27} kg

39 What is the name of the group (class) of particles containing mesons, and the name of the group (class) of particles containing baryons?

	group (class) of particles containing mesons	group (class) of particles containing baryons
A	hadrons	hadrons
B	hadrons	leptons
C	leptons	hadrons
D	leptons	leptons

40 A magnesium nucleus $^{23}_{12}\text{Mg}$ decays by emitting two particles.

The resulting nucleus is sodium $^{23}_{11}\text{Na}$.

Which two particles are emitted?

A α-particle, antineutrino

B β^+ particle, antineutrino

C β^- particle, neutrino

D β^+ particle, neutrino

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CAMBRIDGE
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PHYSICS **9702/13**

Paper 1 Multiple Choice **October/November 2024**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	D	**1**
2	C	**1**
3	B	**1**
4	C	**1**
5	A	**1**
6	C	**1**
7	A	**1**
8	B	**1**
9	B	**1**
10	C	**1**
11	B	**1**
12	A	**1**
13	D	**1**
14	D	**1**
15	D	**1**
16	A	**1**
17	D	**1**
18	C	**1**
19	C	**1**
20	A	**1**
21	A	**1**
22	D	**1**
23	B	**1**
24	D	**1**
25	C	**1**
26	C	**1**
27	C	**1**
28	A	**1**

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Question	Answer	Marks
29	C	**1**
30	B	**1**
31	A	**1**
32	A	**1**
33	A	**1**
34	B	**1**
35	B	**1**
36	D	**1**
37	C	**1**
38	B	**1**
39	A	**1**
40	D	**1**

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March 2025 · Paper 12

9702_m25_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **February/March 2025**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

4146915581

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which quantity is a scalar quantity?

A force

B momentum

C velocity

D work

2 What is the effect of a systematic error on the measurement of a physical quantity?

A It limits the precision of the measured value.

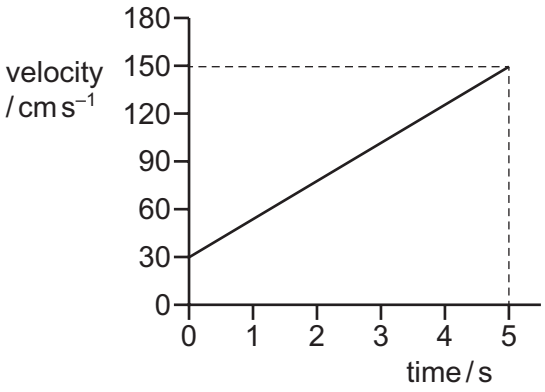
B It limits the range of values obtained in repeated measurements.

C It results in repeated measurements having different values from each other.

D It results in the measured value being different from the correct value.

3 A car is accelerated by a constant resultant force of 300 N for 5.0 s.

The variation with time of the velocity, in $cm\,s^{-1}$, of the car is shown.



What is the mass of the car?

A 13 kg **B** 1000 kg **C** 1300 kg **D** 10 000 kg

4 An aircraft, initially stationary on a runway, takes off with a speed of $85\,km\,h^{-1}$ in a distance of no more than 1.20 km.

What is the minimum constant acceleration necessary for the aircraft?

A $0.23\,m\,s^{-2}$ **B** $0.46\,m\,s^{-2}$ **C** $3.0\,m\,s^{-2}$ **D** $6.0\,m\,s^{-2}$

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5 An object is fired upwards from horizontal ground. The object has an initial velocity of $20\,m\,s^{-1}$ at an angle of 45° to the horizontal. Air resistance is negligible.

Which statement describes the speed of the object after it is fired until immediately before it reaches the ground again?

A Its speed decreases to a value greater than zero, then increases to $20\,m\,s^{-1}$.

B Its speed decreases to a value greater than zero, then increases to a value greater than $20\,m\,s^{-1}$.

C Its speed decreases to zero, then increases to $20\,m\,s^{-1}$.

D Its speed decreases to zero, then increases to a value less than $20\,m\,s^{-1}$.

6 What is a statement of the principle of conservation of momentum for a system?

A The total momentum and the total kinetic energy are always conserved.

B The total momentum is conserved only in elastic collisions.

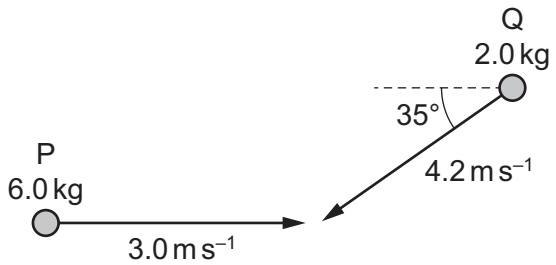
C The total momentum is conserved provided that no external forces act.

D The total momentum of each object in the system is the product of its mass and velocity.

7 Objects P and Q form an isolated system.

Object P has mass 6.0 kg and is moving at a speed of $3.0\,m\,s^{-1}$.

Object Q has mass 2.0 kg and is moving at a speed of $4.2\,m\,s^{-1}$ at an angle of 35° to the path of P.



Objects P and Q collide and stick together.

What is the magnitude of the component of the final momentum of the combined objects in the original direction of P?

A $9.6\,kg\,m\,s^{-1}$ **B** $11\,kg\,m\,s^{-1}$ **C** $13\,kg\,m\,s^{-1}$ **D** $25\,kg\,m\,s^{-1}$

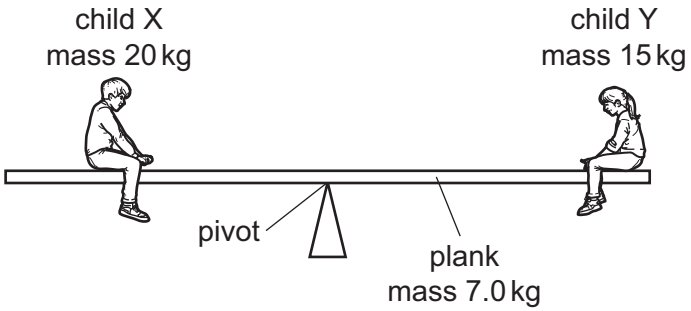
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8 An astronaut of mass m in a spacecraft experiences a gravitational force $F = mg$ when stationary on the launchpad.

What is the gravitational force on the astronaut when the spacecraft is launched vertically upwards with an acceleration of $0.2g$?

A $1.2mg$ **B** mg **C** $0.8mg$ **D** 0

9 The diagram shows a child X of mass 20 kg and a child Y of mass 15 kg seated on a uniform plank.



The plank has a mass of 7.0 kg and has a pivot at its midpoint. The plank is horizontal and in equilibrium.

Which statement about the weight of the plank is correct?

A The weight of the plank can be considered to be acting at its midpoint.

B The weight of the plank is causing an anticlockwise moment.

C The weight of the plank is causing a clockwise moment.

D The weight of the plank equals the force on the plank from the pivot.

10 An object is fully submerged in a liquid.

A student determines the ratio $\frac{\text{upthrust acting on the object}}{\text{weight of the object}}$.

Which single change would double the value of this ratio?

A Use a different liquid that has twice the density and the same volume as the original liquid.

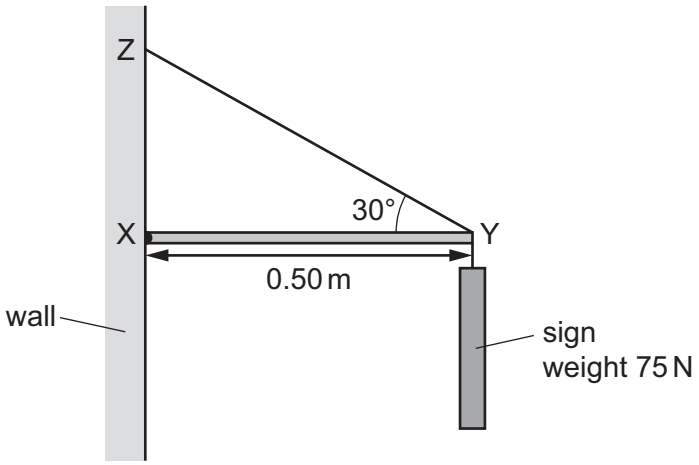
B Use a different object that has half the volume and the same density as the original object.

C Use a different object that has twice the density and the same volume as the original object.

D Use a different object that has twice the volume and the same density as the original object.

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11 A shop sign weighing 75 N hangs from a frame attached to a vertical wall.



The frame consists of a horizontal rod XY and a rod YZ that is at an angle of 30° to the horizontal. Rod XY is attached to the wall by a hinge at X and has length 0.50 m. Assume that the weights of the rods are negligible.

What is the horizontal force exerted by the wall on rod XY?

A 0 N **B** 43 N **C** 130 N **D** 150 N

12 A student takes measurements to calculate the density of a liquid in a beaker.

The height of the liquid in the beaker is 0.20 m ± 2%.

The internal diameter of the beaker is 0.05 m ± 3%.

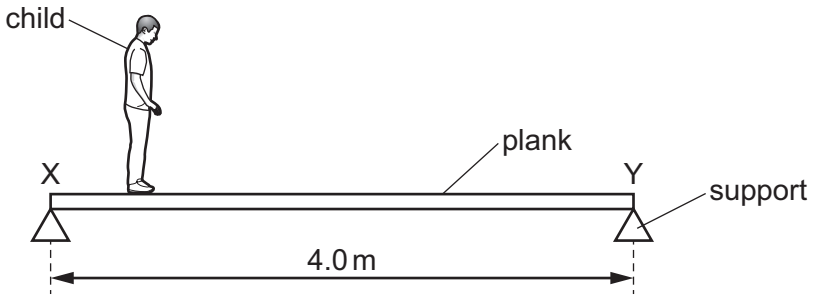
The mass of the liquid is 0.36 kg ± 10%.

What is the percentage uncertainty in the calculated density of the liquid?

A 2% **B** 5% **C** 15% **D** 18%

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13 The diagram shows a uniform plank XY of length 4.0 m and weight 300 N.



The plank rests on fixed supports at its ends X and Y.

A child of weight 600 N stands in different positions on the plank.

The support at end X exerts a force *F* vertically upwards on the plank.

What is the magnitude of *F* when the child stands at X and when the child stands at Y?

	F / N when child is at X	*F* / N when child is at Y
A	600	0
B	600	150
C	750	0
D	750	150

14 Which relationship is used in the derivation of the equation shown?

$$\text{power} = \text{force} \times \text{velocity}$$

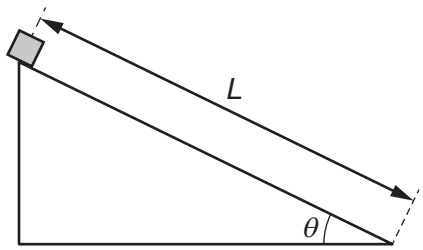
A displacement = velocity × time

B force = mass × acceleration

C momentum = mass × velocity

D velocity = acceleration × time

15 A block is released from rest at the top of a slope inclined at an angle to the horizontal. The slope has length L as shown in the diagram.

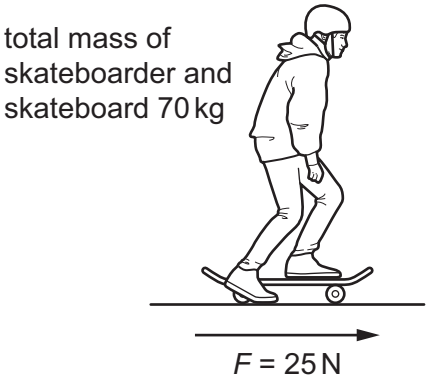


There are no resistive forces acting on the block.

What is the speed of the block at the bottom of the slope?

A $4.43\sqrt{L\cos\theta}$ **B** $4.43\sqrt{L\sin\theta}$ **C** $19.6L\cos\theta$ **D** $19.6L\sin\theta$

16 A skateboarder and her skateboard have a total mass of 70 kg. She pushes on the ground with her foot to create a forward force F of 25 N on herself and the skateboard, as shown in the diagram.



The skateboarder and skateboard travel forwards a distance of 0.50 m before the skateboarder lifts her foot from the ground.

What is the work done by F on the skateboarder and skateboard?

A 13 J **B** 50 J **C** 340 J **D** 360 J

17 A turbine at a hydroelectric power station is situated at a vertical distance of 30 m below the level of the surface of a large lake. The water passes through the turbine at a rate of 340 m^3 per minute.

The overall efficiency of the turbine and generator system is 90%. The density of water is 1000 $kg\,m^{-3}$.

What is the useful power output of the power station?

A 0.15 MW **B** 1.5 MW **C** 1.7 MW **D** 90 MW

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18 A projectile is launched at 45° to the horizontal with initial kinetic energy E.

Assuming air resistance to be negligible, what will be the kinetic energy of the projectile when it reaches its highest point?

A 0.50E **B** 0.71E **C** 0.87E **D** E

19 A wire is extended by a tensile force so that its deformation is elastic.

What is meant by elastic deformation?

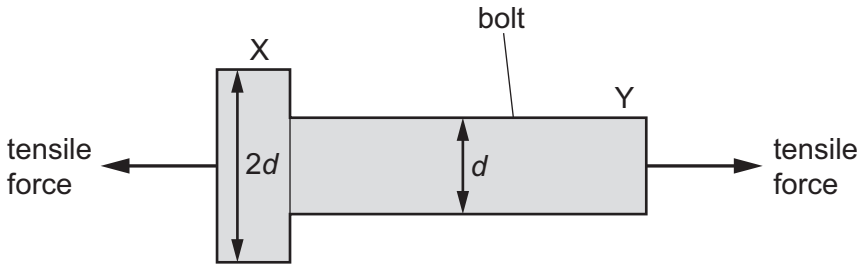
A The extension of the wire is proportional to the tensile force.

B The extension of the wire is **not** proportional to the tensile force.

C When the tensile force is removed, the wire does **not** return to its original length.

D When the tensile force is removed, the wire returns to its original length.

20 A bolt is subjected to a tensile force, as shown.

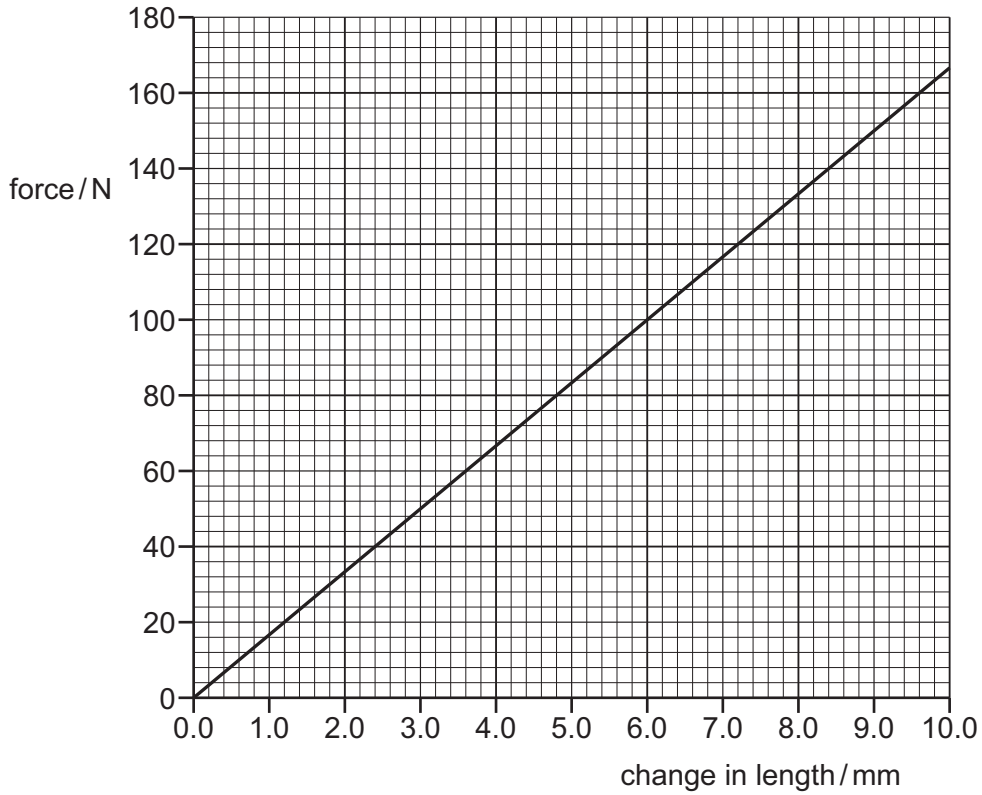


The bolt has a circular cross-section. At end X, the diameter is 2d. At end Y, the diameter is d.

What is the ratio $\frac{\text{stress at Y}}{\text{stress at X}}$?

A 0.25 **B** 0.50 **C** 2.0 **D** 4.0

21 The graph shows the relationship between force acting on a compression spring and change in length of the spring.



One of these springs is placed in each corner of a horizontal square plate. The axis of each spring is in a vertical direction. These four springs support a total load of 160 N.

What is the total elastic potential energy stored in the four springs?

A 0.048 J **B** 0.19 J **C** 0.38 J **D** 0.77 J

22 Which row correctly identifies the properties of all electromagnetic waves?

	transverse wave	longitudinal wave	can travel in free space
A	✓	✗	✓
B	✓	✗	✗
C	✗	✓	✓
D	✗	✓	✗

key

✓ = property of an electromagnetic wave

✗ = **not** a property of an electromagnetic wave

23 What is the approximate range of wavelengths in free space for infrared radiation?

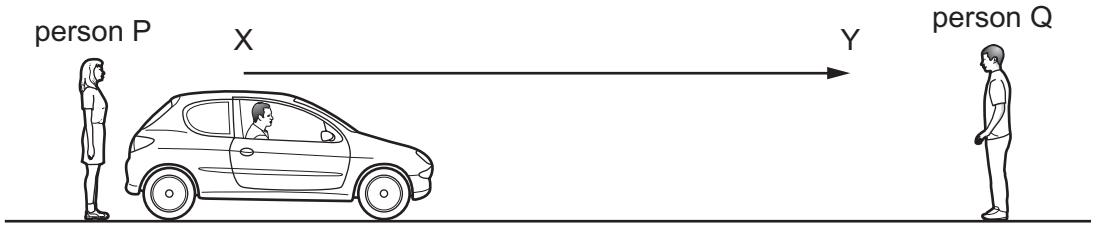
A 100 nm to 400 nm

B 300 μm to 30 cm

C 400 nm to 700 nm

D 800 nm to 1000 μm

24 The diagram shows a car travelling at a constant speed in a straight line between person P and person Q from point X to point Y.



The car sounds its horn continuously as it travels. The horn emits sound of constant frequency.

Which statements about what person P and person Q hear during the motion of the car are correct?

1 Person P hears a sound of increasing frequency.

2 Person Q hears a sound of decreasing frequency.

3 Person Q always hears a sound of higher frequency than person P.

A 1, 2 and 3

B 1 and 2 only

C 3 only

D none of them

25 A progressive wave of frequency 300 Hz is travelling with a speed of 600 m s^{-1}.

What is the phase difference between two points on the wave that are a distance of 0.50 m apart?

A 45° **B** 90° **C** 180° **D** 360°

26 A polarised beam of light with intensity I is incident normally on a polarising filter.

The transmitted light has intensity I.

The filter is rotated about the normal axis through an angle θ.

The transmitted light has intensity $0.75I$.

What is the angle θ?

A 30° **B** 42° **C** 49° **D** 60°

27 Light waves are emitted from two sources.

What is a necessary condition for observable interference fringes to be produced?

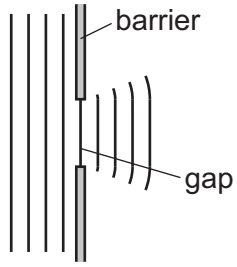
A The waves must be polarised.

B The waves must **not** be polarised.

C The waves must be coherent.

D The waves must have equal amplitudes.

28 The diagram shows a water wave in a shallow tank. The wave is diffracted through a gap in a barrier and spreads. The wavelength of the wave is much smaller than the width of the gap.



The wavelength of the wave and the width of the gap are both changed by a small amount.

Which combination of changes **must** increase the amount of spreading due to diffraction?

	wavelength	width of gap
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

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29 Light of wavelength 567 nm is incident normally on a diffraction grating. The grating has 400 lines per mm. A number of diffraction maxima are observed on the far side of the grating.

What is the angle between the second-order maximum and the third-order maximum?

A 13.1° **B** 13.9° **C** 15.9° **D** 27.0°

30 Two cylindrical conductors, X and Y, are made from the same material. The conductors have equal lengths, but Y has a smaller diameter than X.

X and Y are connected in series to a cell.

Which row compares the number of charge carriers per unit time passing through X and through Y and compares the average drift speed of the charge carriers in X and in Y?

	number of charge carriers per unit time	average drift speed of charge carriers
A	Y greater than X	Y greater than X
B	Y same as X	Y same as X
C	Y greater than X	Y same as X
D	Y same as X	Y greater than X

31 A copper wire is 6.4 m long and has a resistance of 0.92 Ω.

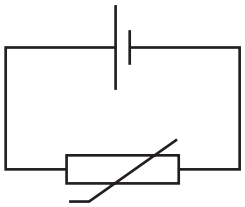
The resistivity of copper is $1.8 \times 10^{-8}\,\Omega\,\text{m}$.

What is the diameter of the wire?

A $5.7 \times 10^{-5}\,\text{m}$ **B** $1.0 \times 10^{-4}\,\text{m}$ **C** $4.0 \times 10^{-4}\,\text{m}$ **D** $7.1 \times 10^{-4}\,\text{m}$

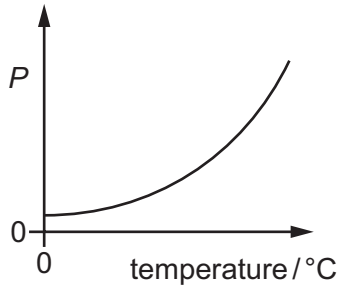
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32 A thermistor is connected to a cell with negligible internal resistance.

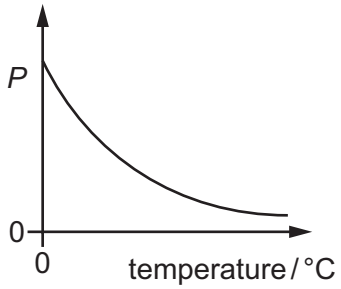


Which graph shows the variation with temperature of power, *P*, dissipated in the thermistor?

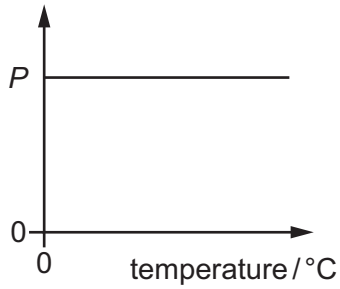
A



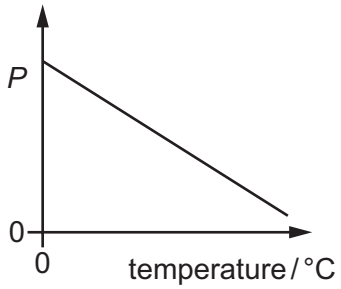
B



C



D



33 A metal electrical conductor has a resistance of 5.6 kΩ. A potential difference (p.d.) of 9.0 V is applied across its ends.

How many electrons pass a point in the conductor in one minute?

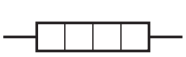
A 6.0×10^{20} **B** 1.0×10^{19} **C** 6.0×10^{17} **D** 1.0×10^{16}

34 Which circuit symbol does **not** represent an electric component that is designed to emit sound waves?

A



B



C

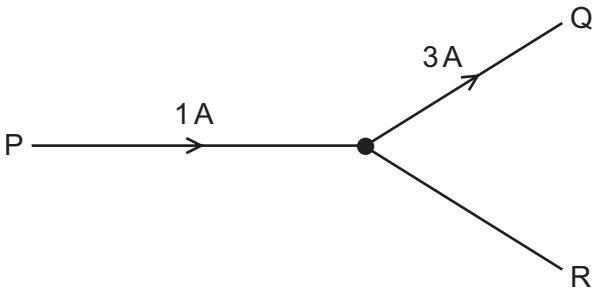


D



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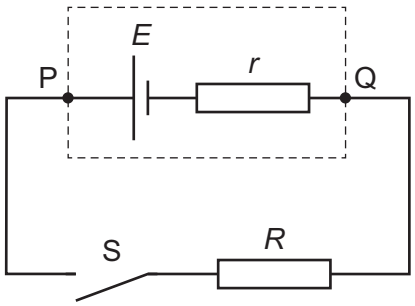
35 The diagram shows a junction in a circuit where three wires, P, Q and R, meet. The currents in P and Q are 1 A and 3 A respectively, in the directions shown.



How much charge passes a given point in wire R in a time of 5 s?

A 0.4 C **B** 2 C **C** 10 C **D** 20 C

36 A cell of electromotive force (e.m.f.) E and internal resistance r is connected in series with a switch S and an external resistor of resistance R.



The potential difference (p.d.) between P and Q is V.

Which statement is correct when S is changed from open to closed?

A V increases because there is a p.d. across R.

B V decreases because there is a p.d. across r.

C V remains the same because the decrease of p.d. across r is balanced by the increase of p.d. across R.

D V remains the same because the sum of the p.d.s across r and R is still equal to E.

37 What is a general description of a baryon?

A It consists of three quarks that **must** all be the same flavour.

B It consists of three quarks that do **not** need to be the same flavour.

C It consists of two quarks that **must** both be the same flavour.

D It consists of two quarks that do **not** need to be the same flavour.

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38 A stationary nucleus has nucleon number A.

The nucleus decays by emitting a proton with speed v to form a new nucleus with speed u. The new nucleus and the proton move away from one another in opposite directions.

Which equation gives v in terms of A and u?

A $v = (\frac{A}{4} - 1)u$

B $v = (A - 1)u$

C $v = Au$

D $v = (A + 1)u$

39 What is the change to the quark composition of a nucleus that takes place during β^+ decay?

A down to antiup

B down to up

C up to antidown

D up to down

40 What is the charge, in terms of the elementary charge e, on a charm quark?

A $-\frac{2}{3}e$ **B** $-\frac{1}{3}e$ **C** $+\frac{1}{3}e$ **D** $+\frac{2}{3}e$

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CAMBRIDGE
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PHYSICS **9702/12**

Paper 1 Multiple Choice **February/March 2025**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	D	**1**
2	D	**1**
3	C	**1**
4	A	**1**
5	A	**1**
6	C	**1**
7	B	**1**
8	B	**1**
9	A	**1**
10	A	**1**
11	C	**1**
12	D	**1**
13	D	**1**
14	A	**1**
15	B	**1**
16	A	**1**
17	B	**1**
18	A	**1**
19	D	**1**
20	D	**1**
21	B	**1**
22	A	**1**
23	D	**1**
24	C	**1**
25	B	**1**
26	A	**1**
27	C	**1**
28	C	**1**

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Question	Answer	Marks
29	C	**1**
30	D	**1**
31	C	**1**
32	A	**1**
33	C	**1**
34	B	**1**
35	C	**1**
36	B	**1**
37	B	**1**
38	B	**1**
39	D	**1**
40	D	**1**

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June 2025 · Paper 11

9702_s25_qp_11

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2025**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 Which quantity is a vector?

A pressure

B temperature

C weight

D work

2 Four students, **A**, **B**, **C** and **D**, have completed an experiment to determine the acceleration of free fall, *g*. Each student repeated the experiment three times. The determined values of *g* are shown in the table.

Which set of results has a high precision and a low accuracy?

	acceleration of free fall / $m\,s^{-2}$		
	experiment 1	experiment 2	experiment 3
A	7.2	9.4	8.3
B	9.5	9.8	10.2
C	9.8	9.8	9.9
D	10.1	10.2	10.1

3 The diameter of a ball is measured as $(5.26 \pm 0.02)\,cm$.

What is the absolute uncertainty in the volume of the ball?

A $0.29\,cm^3$ **B** $0.87\,cm^3$ **C** $1.1\,cm^3$ **D** $7.0\,cm^3$

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4 Two vectors, X and Y, are shown.



What are the directions of X + Y and X – Y?

	X + Y	X – Y
A	→	↓
B	→	↑
C	←	↓
D	←	↑

5 A ball is projected vertically downwards with an initial velocity of $20\,m\,s^{-1}$. Air resistance is negligible.

What is the displacement from its initial position of the ball after a time of 1.5 s?

A 11 m **B** 19 m **C** 37 m **D** 41 m

6 A surveyor's device emits a pulse of light. The light is reflected from a wall 150 m away.

What is the total time taken for the pulse to travel from the device to the wall and then back to the device?

A 0.05 ns **B** 0.10 ns **C** 0.50 μs **D** 1.0 μs

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7 Which equation of uniformly accelerated motion can be derived using only the gradient of a velocity–time graph?

A $s = \frac{1}{2}(u + v)t$

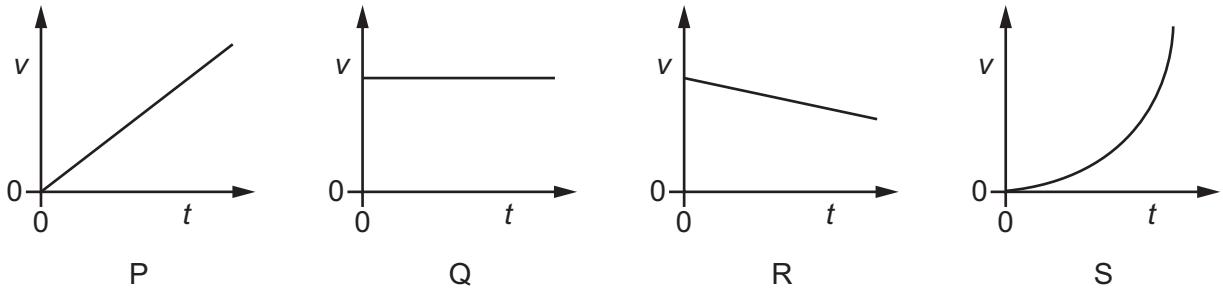
B $s = ut + \frac{1}{2}at^2$

C $v = u + at$

D $v^2 = u^2 + 2as$

8 An object is projected horizontally from a table at time $t = 0$. The object falls in a uniform gravitational field. Air resistance is negligible.

Graphs P, Q, R and S are velocity–time graphs.



Which graphs represent the horizontal and vertical components of the velocity of the object?

	horizontal	vertical
A	Q	P
B	Q	S
C	R	P
D	R	S

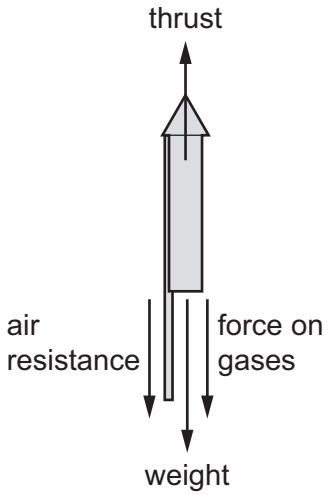
9 What is always conserved in elastic collisions?

	total kinetic energy	total velocity
A	yes	yes
B	yes	no
C	no	yes
D	no	no

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10 A firework travels vertically upwards in air. Gases are pushed vertically downwards from the firework.

Several forces that act on the firework and gases are shown.



NOT TO SCALE

Which forces are a Newton's third law pair?

A air resistance and force on gases

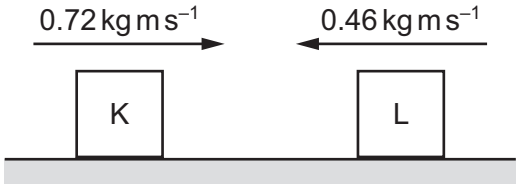
B air resistance and thrust

C force on gases and thrust

D thrust and weight

11 Two blocks K and L slide towards each other along a horizontal frictionless surface.

The diagram shows the momentum of the two blocks just before they collide.



During the collision, the blocks are in contact with each other for a time of 0.084 s.

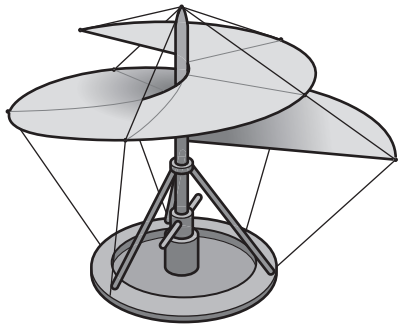
After the collision, the blocks separate and block L moves back along its original path with a momentum of 0.12 kg m s^{-1}.

What is the magnitude of the average force exerted on block L by block K during the collision?

A 3.1 N **B** 4.0 N **C** 6.9 N **D** 7.1 N

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12 Leonardo da Vinci proposed a flying machine that would work like a screw to lift the pilot into the air. The 'screw' is rotated by the pilot.



The machine and the pilot together have a total mass of 120 kg.

Which useful output power must the pilot provide to move vertically upwards at a constant speed of $2.5\,m\,s^{-1}$?

A 48 W **B** 300 W **C** 470 W **D** 2900 W

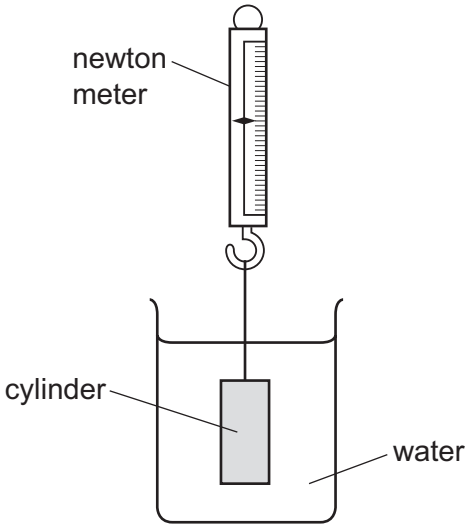
13 Which row describes a pair of forces that forms a couple?

	direction of the forces	magnitude of the forces
A	opposite	different
B	opposite	equal
C	same	different
D	same	equal

14 A uniform cylinder of weight 25.0 N is suspended from a newton meter.

The cylinder is fully submerged in water, as shown.

The reading on the newton meter is 10.0 N.

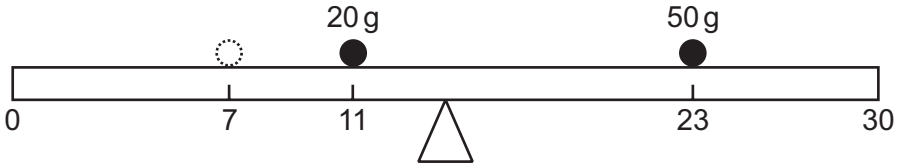


The water is replaced by a liquid with a density 10% greater than the density of water. The cylinder remains fully submerged.

What is the new reading on the newton meter?

A 8.5 N **B** 9.0 N **C** 11.0 N **D** 11.5 N

15 A student balances a 30 cm ruler on a fulcrum set at the 15 cm mark. She then places a 50 g mass on the 23 cm mark and a 20 g mass on the 11 cm mark, as shown.



Which mass should she place on the 7 cm mark to restore the balance?

A 30 g **B** 40 g **C** 47 g **D** 133 g

16 A submarine is at a depth of 130 m below the surface of the sea.

The pressure on the submarine due to the sea water is p.

The submarine sinks to a depth of 260 m below the surface.

Assume that the density of sea water is constant.

What is the **difference** between the pressures on the submarine due to the sea water at the two depths?

A 0 **B** 0.5p **C** p **D** 2p

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17 An object is falling in a uniform gravitational field.

Which two quantities are sufficient to calculate the change in gravitational potential energy?

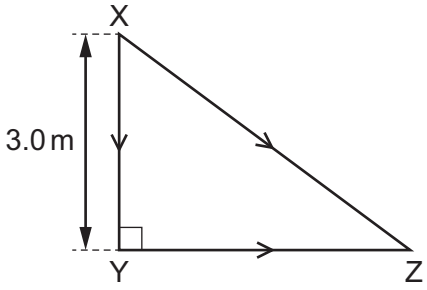
A mass and acceleration of free fall

B mass and change in vertical displacement

C weight and acceleration of free fall

D weight and change in vertical displacement

18 The points X, Y and Z are on a rough, horizontal surface.



A box P is pushed across the surface from X to Y and then from Y to Z.

The distance from X to Y is 3.0 m. The work done against the frictional force in moving the box from X to Y is 150 J.

The work done against the frictional force in moving the box from Y to Z is 200 J.

An identical box Q is pushed in a straight line from X to Z.

The magnitude of the frictional force between the boxes and the surface is constant.

How much extra work is done against the frictional force in moving P than Q?

A 100 J **B** 250 J **C** 350 J **D** 600 J

19 The momentum of a car of mass m increases from p_1 to p_2.

What is the increase in the kinetic energy of the car?

A $\frac{(p_2^2 - p_1^2)}{2m}$ **B** $\frac{(p_2 - p_1)^2}{2m}$ **C** $\frac{p_2 - p_1}{2m}$ **D** $\frac{p_1 - p_2}{2m}$

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20 A parachutist is falling at constant (terminal) velocity.

Which statement is **not** correct?

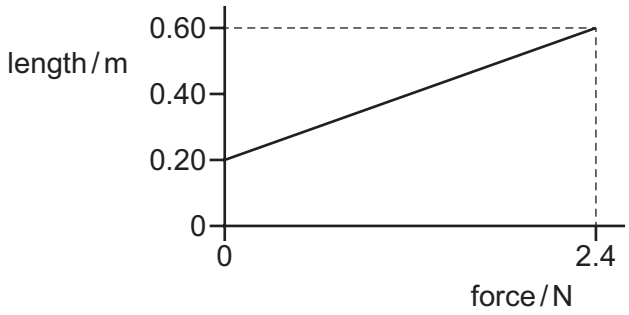
A Gravitational potential energy is converted into kinetic energy of the air.

B Gravitational potential energy is converted into kinetic energy of the parachutist.

C Gravitational potential energy is converted into thermal energy of the air.

D Gravitational potential energy is converted into thermal energy of the parachutist.

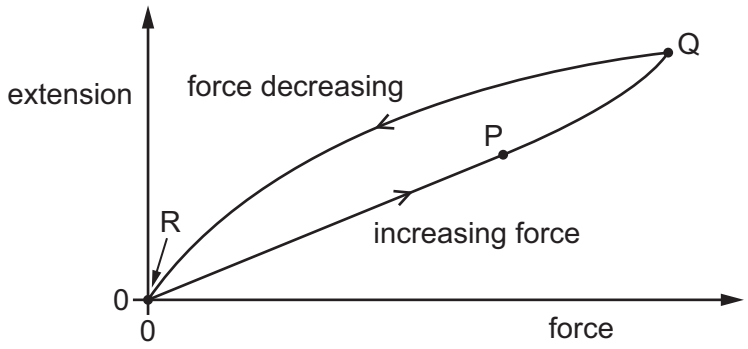
21 A student investigates a spring. The variation of the length of the spring with the force applied to the spring is shown.



What is the spring constant of the spring?

A $0.17\,N\,m^{-1}$ **B** $0.25\,N\,m^{-1}$ **C** $4.0\,N\,m^{-1}$ **D** $6.0\,N\,m^{-1}$

22 The graph shows the variation with force of the extension of a wire.



The force is gradually increased to a maximum at Q and then gradually decreased to zero at R.

Which statement is correct?

A Along the line PQ, the wire obeys Hooke's law.

B Along the line RP, the spring constant is equal to $\frac{\text{extension}}{\text{force}}$.

C The wire has elastic deformation at point Q.

D The work done in stretching the wire to P is equal to (force × extension) at point P.

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23 A uniform metal wire of length L and diameter d has spring constant k.

What is the Young modulus of the metal?

A $\frac{\pi d^2}{4kL}$ **B** $\frac{\pi d^2}{kL}$ **C** $\frac{kL}{\pi d^2}$ **D** $\frac{4kL}{\pi d^2}$

24 An aircraft produces a sound at a frequency of 30.0 Hz.

The speed of sound in air is 330 m s^{-1}.

The aircraft is directly in front of the stationary observer and travels in a straight line towards or away from the observer.

The observer hears the sound from the aircraft at a frequency of 20.0 Hz.

What is the speed and direction of the aircraft?

	speed / m s^{-1}	direction
A	110	away from observer
B	110	towards observer
C	165	away from observer
D	165	towards observer

25 Which statement about longitudinal and transverse wave motion for a progressive wave is correct?

A All transverse and longitudinal waves require a medium for propagation.

B All transverse waves travel at the same speed, but longitudinal waves can travel at different speeds.

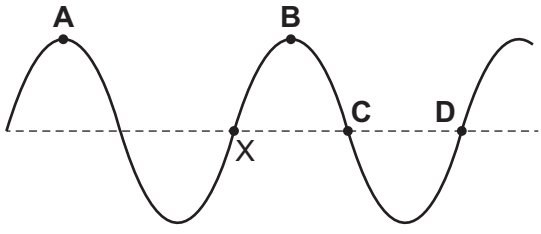
C In both transverse waves and longitudinal waves, a particle with zero displacement has a maximum speed.

D In a longitudinal wave there is a net movement of particles in the direction of travel of the wave, but there is no net movement of particles in a transverse wave.

26 The diagram shows a progressive transverse wave on a stretched string at one instant in time.

One point on the string is labelled X.

Which point on the string is 270° out of phase with X?



27 Which wavelength of electromagnetic radiation in free space could be green light?

A $4.9 \times 10^{-9}\,\text{m}$ **B** $6.6 \times 10^{-9}\,\text{m}$ **C** $5.5 \times 10^{-7}\,\text{m}$ **D** $8.6 \times 10^{-7}\,\text{m}$

28 Which group contains only waves that can be polarised?

A infrared waves, radio waves, sound waves

B visible light waves, microwaves, radio waves

C visible light waves, radio waves, sound waves

D microwaves, visible light waves, sound waves

29 One wave has an amplitude of $2A$. A second wave has an amplitude of $\frac{A}{2}$. The waves are otherwise identical.

The two waves travel in opposite directions and overlap.

What is the ratio $\frac{\text{maximum amplitude of combined wave}}{\text{minimum amplitude of combined wave}}$?

A $\frac{3}{2}$ **B** $\frac{5}{3}$ **C** $\frac{5}{2}$ **D** 4

30 Light of wavelength 680 nm is incident normally on a diffraction grating with 450 lines mm^{-1}.

What is the angle of diffraction of the second-order maximum in the diffraction pattern that is produced?

A 1.8° **B** 3.5° **C** 18° **D** 38°

31 A hollow tube is closed at one end and open at the other.

A stationary sound wave of the lowest possible frequency, 820 Hz, is produced in the tube.

The speed of sound in air is $330\,\text{m}\,\text{s}^{-1}$.

What is the length of the tube?

A 10 cm **B** 20 cm **C** 40 cm **D** 160 cm

32 Which expression gives the definition of resistance?

A current divided by potential difference

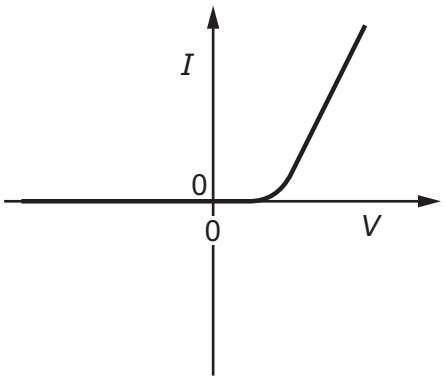
B current multiplied by potential difference

C potential difference divided by current

D resistivity multiplied by length

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33 The graph shows the I–V characteristic of an electrical component.



What is the component?

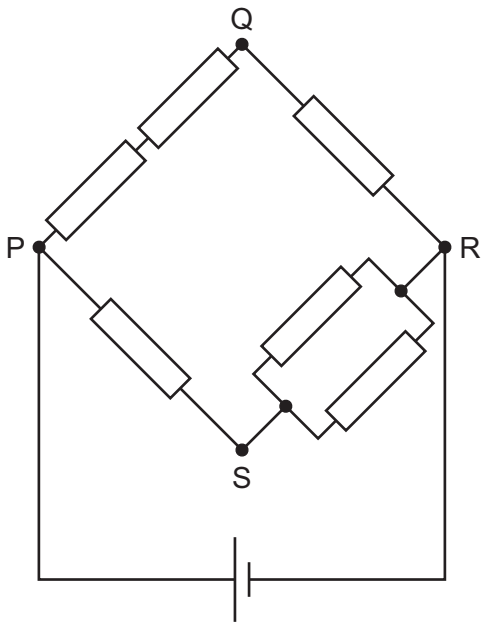
A a filament lamp

B a metallic conductor at constant temperature

C a resistor

D a semiconductor diode

34 The diagram shows six identical resistors connected in a circuit.

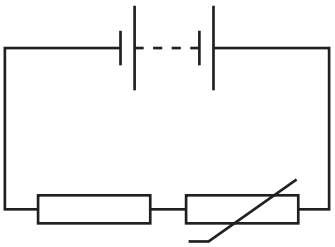


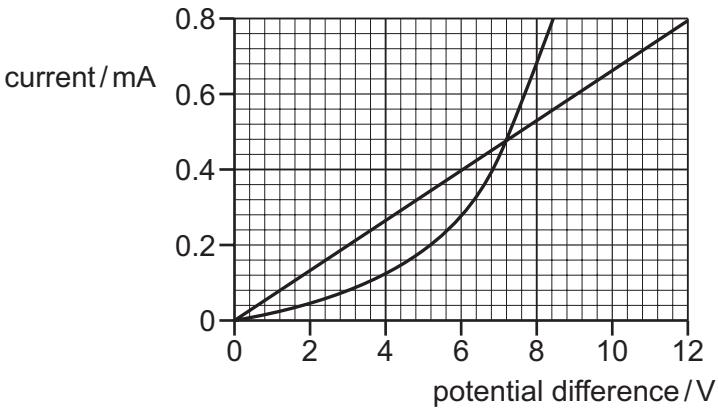
In which branch of the circuit is the most power dissipated?

A PQ **B** QR **C** RS **D** SP

35 The diagram shows a circuit containing a thermistor, a fixed resistor and a battery.

The graph shows the $I-V$ characteristics of both components.



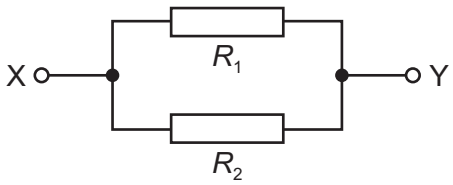


The battery has an e.m.f. of 12 V and negligible internal resistance.

What is the current in the fixed resistor?

A 0.36 mA **B** 0.40 mA **C** 0.48 mA **D** 0.80 mA

36 Two resistors of resistances R_1 and R_2 are connected in parallel.



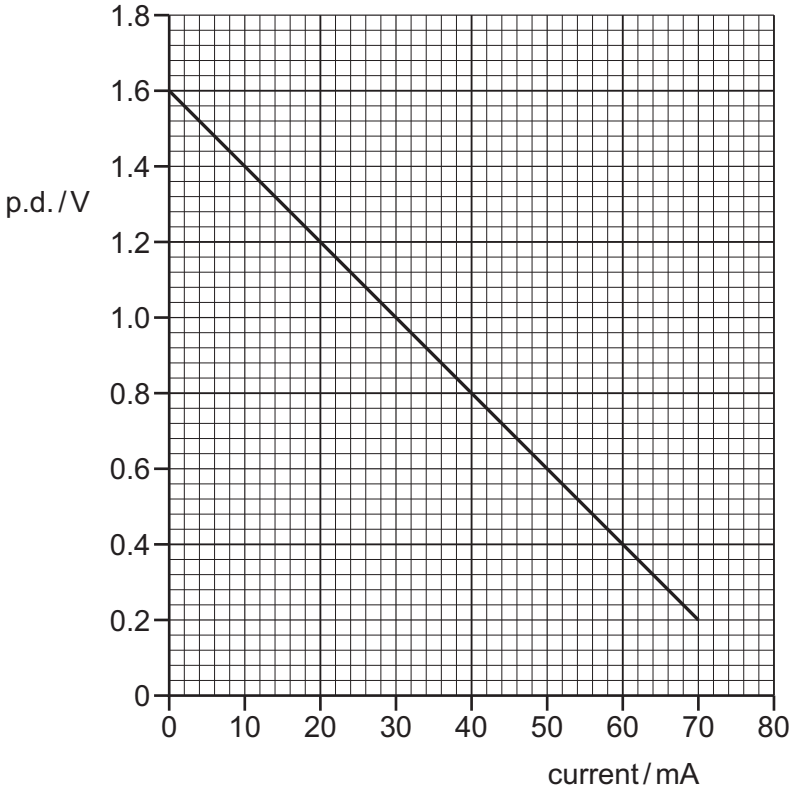
What is the combined resistance between X and Y?

A $R_1 + R_2$ **B** $\dfrac{R_1 R_2}{R_1 + R_2}$ **C** $\dfrac{R_1 + R_2}{R_1 R_2}$ **D** $\dfrac{R_1}{R_2}$

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37 A cell is connected in series with an ammeter and a variable resistor. A voltmeter is used to measure the p.d. across the variable resistor. The resistance of the variable resistor is varied, and the p.d. and the current are recorded.

The graph shows the variation with current of the p.d. across the variable resistor.



What is the internal resistance of the cell?

A 0.050 Ω **B** 1.6 Ω **C** 20 Ω **D** 23 Ω

38 Which list contains only fundamental particles?

A antineutrinos, baryons, neutrons, electrons

B mesons, electrons, neutrinos, protons

C positrons, quarks, hadrons, protons

D quarks, positrons, neutrinos, leptons

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39 Which statement explains why alpha-particles have discrete energies, but beta-particles have a continuous range of energies?

A Beta-particles have a much smaller mass than alpha-particles which means beta-particles can be emitted with a larger range of velocities.

B Only alpha-particles experience repulsion from the nucleus which is dependent on the number of protons in the nucleus.

C Only beta-particles are emitted with another lepton and some energy is transferred to the other lepton.

D Only the energy of alpha-particles is discrete because the composition of alpha-particles is always the same.

40 A hadron consists of antiquarks that are all identical.

What is a possible value, in terms of the elementary charge *e*, for the charge on the hadron?

A 0*e* **B** +1*e* **C** +2*e* **D** +3*e*

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CAMBRIDGE
International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **May/June 2025**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	C	**1**
2	D	**1**
3	B	**1**
4	B	**1**
5	D	**1**
6	D	**1**
7	C	**1**
8	A	**1**
9	B	**1**
10	C	**1**
11	C	**1**
12	D	**1**
13	B	**1**
14	A	**1**
15	B	**1**
16	C	**1**
17	D	**1**
18	A	**1**
19	A	**1**
20	B	**1**
21	D	**1**
22	C	**1**
23	D	**1**
24	C	**1**
25	C	**1**
26	A	**1**
27	C	**1**
28	B	**1**

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Question	Answer	Marks
29	B	**1**
30	D	**1**
31	A	**1**
32	C	**1**
33	D	**1**
34	D	**1**
35	A	**1**
36	B	**1**
37	C	**1**
38	D	**1**
39	C	**1**
40	B	**1**

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June 2025 · Paper 12

9702_s25_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2025**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What must all physical quantities have?

A a direction and a magnitude

B a direction and a unit

C a magnitude and a prefix

D a magnitude and a unit

2 What is $0.25\,\text{kN}\,\text{mm}^{-2}$ expressed in $\text{N}\,\text{m}^{-2}$?

A $0.00025\,\text{N}\,\text{m}^{-2}$

B $0.25\,\text{N}\,\text{m}^{-2}$

C $250\,000\,\text{N}\,\text{m}^{-2}$

D $250\,000\,000\,\text{N}\,\text{m}^{-2}$

3 A student calculates the density of a solid steel cube in an experiment.

The measured mass is $975\,\text{g} \pm 10\,\text{g}$ and the measured length of side is $50\,\text{mm} \pm 1\,\text{mm}$.

What is the density of the steel?

A $7.8\,\text{g}\,\text{cm}^{-3} \pm 3.0\%$

B $7.8\,\text{g}\,\text{cm}^{-3} \pm 7.0\%$

C $7.8\,\text{g}\,\text{cm}^{-3} \pm 11\%$

D $7.8\,\text{g}\,\text{cm}^{-3} \pm 13\%$

4 The time period T of a pendulum is given by

$$T = 2\pi\left(\frac{L}{g}\right)^{n}$$

where L is the length of the pendulum and g is the acceleration of free fall.

The equation is homogeneous.

What is the value of n?

A -2 **B** $-\frac{1}{2}$ **C** $\frac{1}{2}$ **D** 2

5 Radio waves can be used to measure the distance between Earth and the planet Jupiter.

A pulse of radio waves is emitted from the surface of Earth. The pulse reflects from the surface of Jupiter and is detected again on Earth.

The time between emitting and receiving the pulse is 3960 s.

What is the distance between Earth and Jupiter?

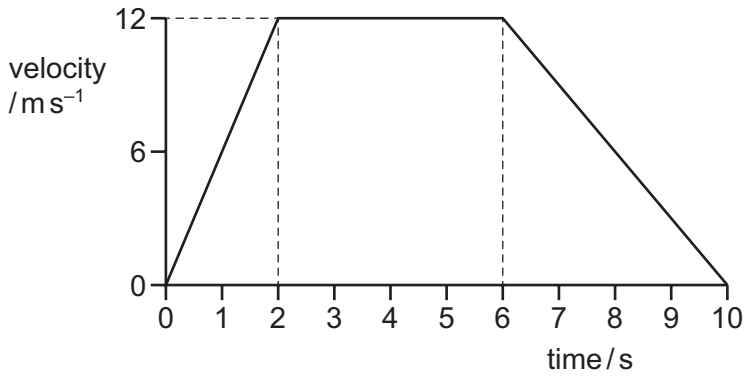
A 5.94×10^{8} km

B 1.19×10^{9} km

C 5.94×10^{11} km

D 1.19×10^{12} km

6 The graph shows the variation with time of the velocity of a car.



Which statement is correct?

A The car accelerates for 2 s, then stops for 4 s and then reverses.

B The car accelerates at 12 m s^{-2} for 2 s.

C The car travels a distance of 36 m in the first 4 s.

D The car travels a distance of 48 m in the last 4 s.

7 A solid object of mass 1.0 kg falls vertically downwards in a vacuum.

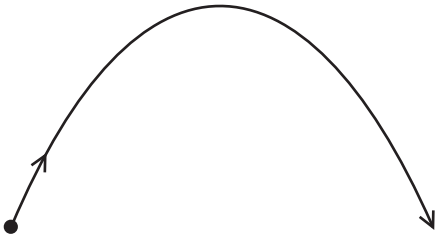
When the speed of the object is 60 m s^{-1}, an additional constant force of 50 N suddenly starts to act vertically upwards on the object.

What is the speed of the object 2.0 s after the additional force starts to act?

A 20 m s^{-1} **B** 40 m s^{-1} **C** 80 m s^{-1} **D** 100 m s^{-1}

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8 A stone is thrown upwards and follows a curved path.



Air resistance is negligible.

Why does the path have this shape?

A The stone has a constant horizontal acceleration and constant vertical velocity.

B The stone has a constant horizontal velocity and constant vertical acceleration.

C The stone has a constant upward acceleration followed by a constant downward acceleration.

D The stone has a constant upward velocity followed by a constant downward velocity.

9 A rocket engine ejects 90 kg of exhaust gas per second at a velocity of 190 $m\,s^{-1}$ relative to the rocket.

What is the force acting on the rocket due to the ejected gas?

A 2.1 kN **B** 17 kN **C** 18 kN **D** 162 kN

10 Which statement does **not** describe an elastic collision between two objects?

A The relative speed of approach of the two objects equals the relative speed of separation.

B The total kinetic energy of the objects is conserved.

C The total kinetic energy of the objects is reduced.

D The total linear momentum of the objects is conserved.

11 A cyclist is riding at a constant speed on a level road.

According to Newton's third law of motion, what is equal and opposite to the backward push of the back wheel on the road?

A the force exerted by the cyclist on the pedals

B the forward push of the road on the back wheel

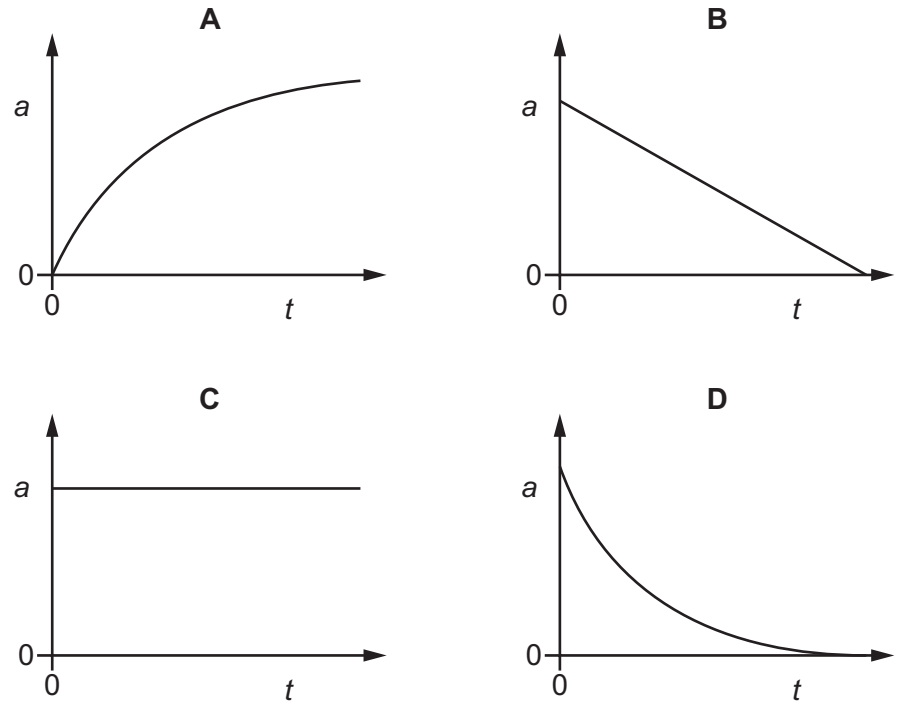
C the tension in the cycle chain

D the total air resistance and friction force

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12 A stone is released from rest and falls a long distance in air.

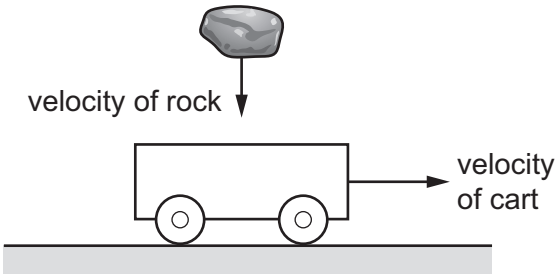
Which graph could show the variation with time *t* of the acceleration *a* of the stone?



13 An empty cart is moving along a horizontal track at a constant velocity.

Resistive forces acting on the cart are negligible.

A heavy rock is dropped vertically into the cart.



The cart continues to move horizontally with the rock inside.

How does the momentum and kinetic energy of the cart with the rock inside compare with the momentum and kinetic energy of the empty cart?

A The cart with the rock inside has a smaller momentum and a smaller kinetic energy.

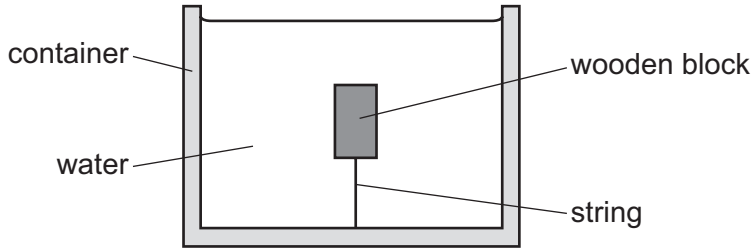
B The cart with the rock inside has a smaller momentum and the same kinetic energy.

C The cart with the rock inside has the same momentum and a smaller kinetic energy.

D The cart with the rock inside has the same momentum and the same kinetic energy.

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14 A wooden block is held stationary in a container of water using a string that is attached to both the wooden block and the bottom of the container.



The wooden block has mass m and volume V. The water has density ρ. The acceleration of free fall is g.

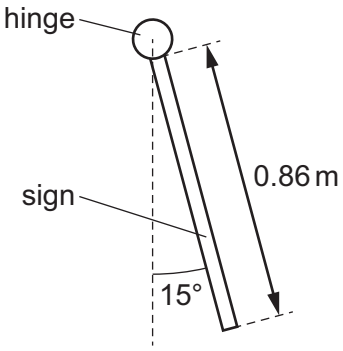
What is the magnitude of the force acting on the block due to the tension in the string?

A ρgV **B** $mg + \rho gV$ **C** mg **D** $\rho gV - mg$

15 A square shop sign of uniform density has mass 2.4 kg and sides of length 0.86 m.

The sign is supported by a hinge along its top edge.

There is friction in the hinge so that the sign hangs from it in equilibrium at an angle of 15° to the vertical, as shown.

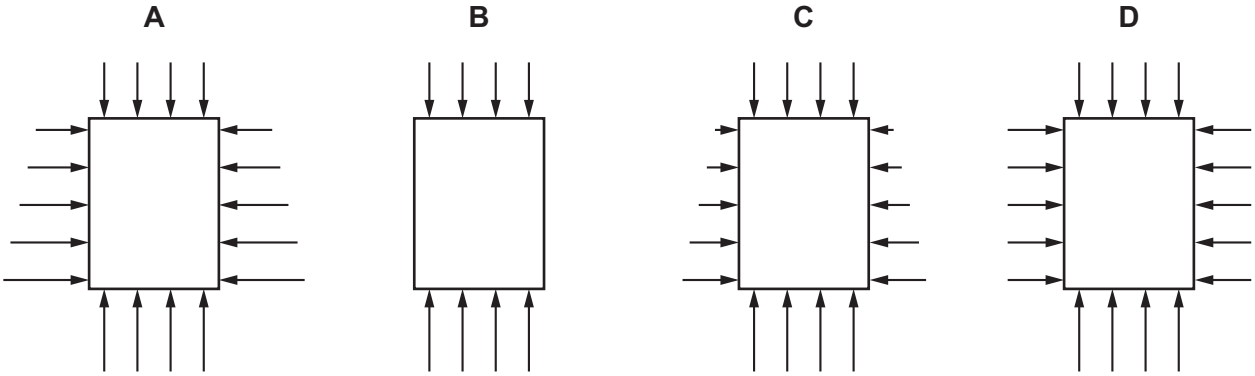


What is the moment about the hinge of the weight of the sign?

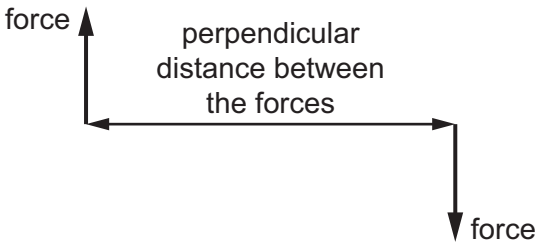
A 2.6 N m **B** 4.0 N m **C** 5.2 N m **D** 9.8 N m

16 A block is submerged vertically in a liquid. The four diagrams show the block viewed from the side.

Which diagram shows, to scale, the forces exerted on equal areas of the block by the liquid?



17 The diagram shows a couple.



How is the torque of the couple calculated?

A $\frac{1}{2}$ × perpendicular distance between the forces × magnitude of one of the forces

B perpendicular distance between the forces × magnitude of one of the forces

C perpendicular distance between the forces × magnitude of the sum of the forces

D 2 × perpendicular distance between the forces × magnitude of one of the forces

18 A ball of mass 1.2 kg travels horizontally at a speed of 3.0 m s^{-1}.

The ball hits a cushion and comes to rest over a horizontal distance of 0.020 m.

What is the work done by the cushion on the ball to bring the ball to rest?

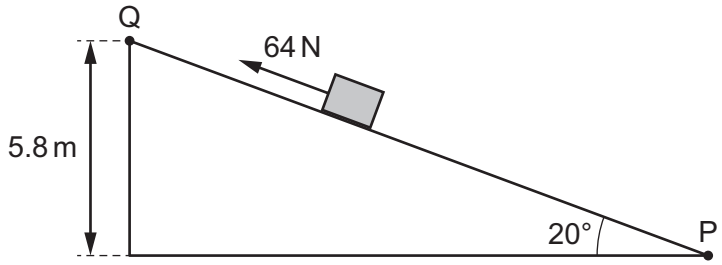
A 0.24 J **B** 1.8 J **C** 5.4 J **D** 11 J

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19 A box of mass 4.9 kg is pushed at a constant velocity from point P at the bottom of an inclined plane to point Q at the top.

The box is pushed by a force of 64 N acting parallel to the slope.

The slope is inclined at an angle of 20° to the horizontal and the box moves through a vertical height of 5.8 m.



What is the work done against the frictional force acting on the block between P and Q?

A 280 J **B** 810 J **C** 960 J **D** 1100 J

20 Which expression gives the efficiency of a system?

A $\dfrac{\text{total energy input}}{\text{useful energy output}}$

B $\dfrac{\text{useful energy output + wasted energy output}}{\text{total energy input}}$

C $\dfrac{\text{useful energy output}}{\text{total energy input}}$

D $\dfrac{\text{wasted energy output}}{\text{total energy input}}$

21 A student has a copper wire and a steel wire with equal lengths and cross-sectional areas.

The student hangs identical loads on the two wires.

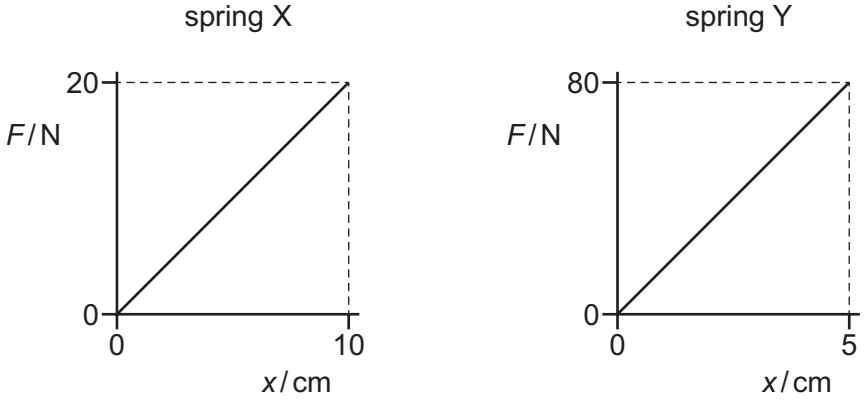
The extensions of the two wires are different.

The student calculates the stress, strain and Young modulus of each wire.

Which row identifies with a tick (✓) the calculated values that are equal for both wires?

	stress	strain	Young modulus
A	✓		
B		✓	
C		✓	✓
D	✓		✓

22 Two springs X and Y stretch elastically. The graphs show the variation with extension x of the force F applied to each spring.



Which statement is correct?

A When each spring is given the same extension, the energy stored in Y is 4 times the energy stored in X.

B When each spring is given the same extension, the energy stored in Y is 8 times the energy stored in X.

C When the same force is applied to each spring, the energy stored in Y is 4 times the energy stored in X.

D When the same force is applied to each spring, the energy stored in Y is 8 times the energy stored in X.

23 Which phrase describes the strain at the elastic limit on a stress–strain graph?

A the maximum strain below which Hooke's law is obeyed

B the maximum strain below which the deformation is plastic

C the minimum strain above which Hooke's law is obeyed

D the minimum strain above which the deformation is plastic

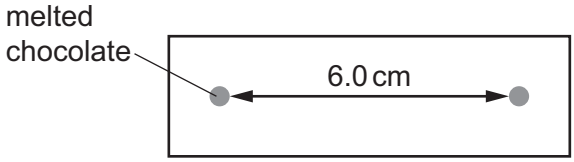
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24 A teacher removes the turntable from a microwave oven and places a bar of chocolate in the oven. She then switches the oven on for a short time.

A stationary wave is formed in the oven.

When the chocolate is removed, the teacher observes that there are two small sections of melted chocolate 6.0 cm apart with unmelted chocolate in between.

Each section of melted chocolate is located at an antinode.



Assume that the speed of the microwaves is $3.0 \times 10^8\,\text{m s}^{-1}$.

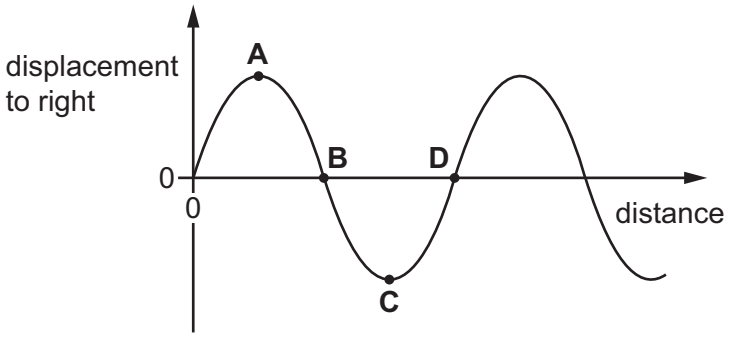
What is the frequency of the microwaves emitted by the oven?

A 25 MHz **B** 50 MHz **C** 2.5 GHz **D** 5.0 GHz

25 A sound wave travels from the left to the right.

The graph shows the variation of the displacement to the right of particles in the sound wave with distance, at one instant.

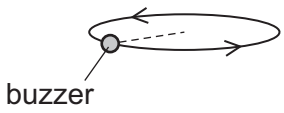
Which letter represents the centre of a compression?



26 A buzzer emitting sound of frequency 846 Hz is attached to a string and rotated in a horizontal circle. The linear speed of the buzzer is 25.0 m s^{-1}.



observer



The speed of sound is 340 m s^{-1}.

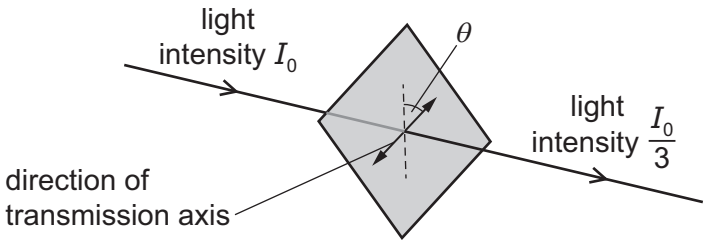
What is the maximum frequency heard by the observer?

A 783 Hz **B** 788 Hz **C** 908 Hz **D** 913 Hz

27 Vertically polarised light of intensity I_0 is incident normally on a polarising filter.

The transmission axis of the filter is at an angle θ to the plane of polarisation of the light.

The intensity of the light after passing through the filter is $\frac{I_0}{3}$.



What is θ?

A 8.4° **B** 55° **C** 71° **D** 84°

28 Which statement compares the behaviour of transverse and longitudinal waves?

A Only longitudinal waves can be diffracted.

B Only longitudinal waves can travel in free space.

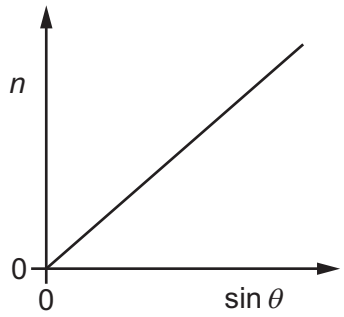
C Only transverse waves can be coherent.

D Only transverse waves can be polarised.

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29 A student uses a diffraction grating to determine the wavelength of visible light from a source.

The diffraction grating has 300 lines per mm. The student measures the angle θ of each order n of the intensity maxima. A graph of n against $\sin\theta$ is plotted. The line of best fit for the plotted points is shown and has gradient G.



Which expression represents the wavelength, in m, of the visible light in terms of G?

A $3 \times 10^{5} G$ **B** $3.3 \times 10^{-6} G$ **C** $\dfrac{3.3 \times 10^{-6}}{G}$ **D** $\dfrac{300}{G}$

30 Two waves meet.

What is **not** a necessary condition for the waves to produce a stationary wave?

A They must be of the same type.

B They must have the same period.

C They must have the same wavelength.

D They must travel in the same direction.

31 Light of a single wavelength is incident normally on a double slit. Interference fringes are observed on a screen.

The distance from the double slit to the screen is 0.60 m and the fringe separation is 1.8 mm.

The distance from the double slit to the screen increases by 0.90 m.

What is the new fringe separation?

A 1.2 mm **B** 2.7 mm **C** 4.5 mm **D** 5.4 mm

32 A wire is made from a metal of constant resistivity. There is a constant current in the wire.

Which statement about the potential difference across the wire is correct?

A It is directly proportional to the length of the wire.

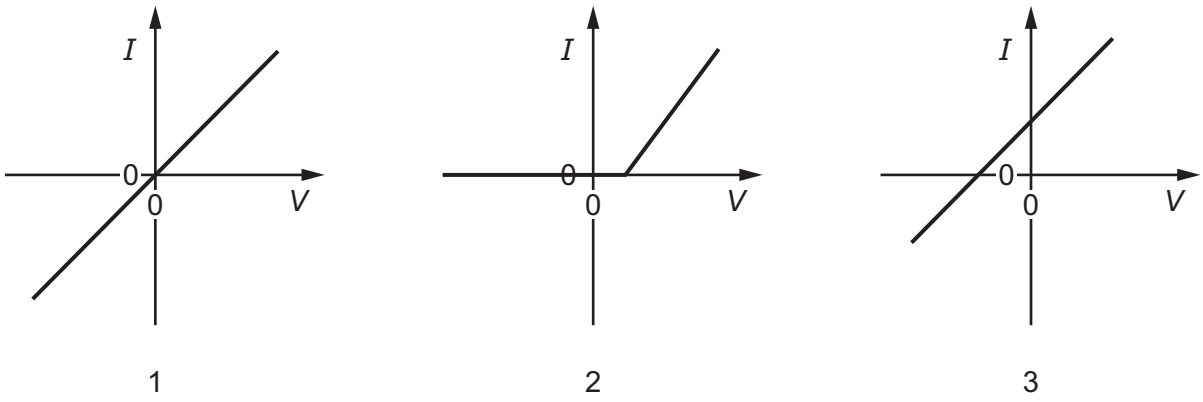
B It is inversely proportional to the length of the wire.

C It is directly proportional to the diameter of the wire.

D It is inversely proportional to the diameter of the wire.

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33 The I–V characteristics for three electrical components are shown.



1 2 3

Which components obey Ohm's law?

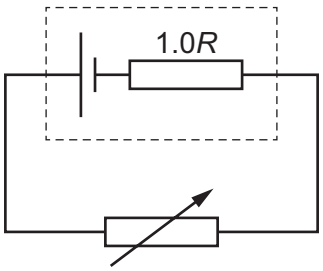
A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 1 only

34 A resistor of resistance 200 Ω is connected to a supply that has a p.d. of 4.00 V.

How many electrons enter the resistor in 4.00 s?

A 3.13×10^{16} **B** 1.25×10^{17} **C** 5.00×10^{17} **D** 1.25×10^{21}

35 The diagram shows a cell of internal resistance $1.0R$ connected to a variable resistor.



When the variable resistor has an initial resistance of $2.0R$, the current in the cell is I_1 and the terminal potential difference (p.d.) across the cell is V_1.

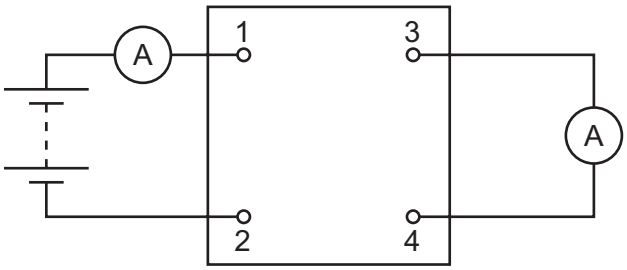
The variable resistor is now adjusted to a new resistance of $4.0R$.

What is the new current in the cell and the new terminal p.d. across the cell?

	current	terminal p.d.
A	$0.50I_1$	$1.0V_1$
B	$0.50I_1$	$1.2V_1$
C	$0.60I_1$	$1.0V_1$
D	$0.60I_1$	$1.2V_1$

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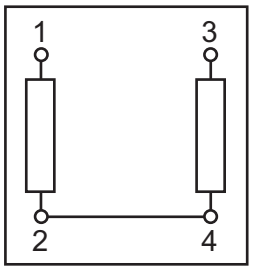
36 The diagram shows a four-terminal box connected to a battery and two ammeters.



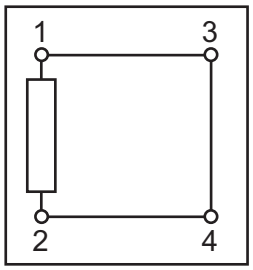
The currents in the two ammeters are identical.

Which circuit, within the box, gives this result?

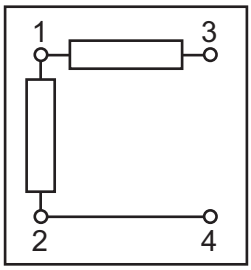
A



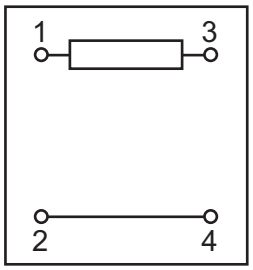
B



C



D



37 Each of Kirchhoff's two laws presumes that some quantity is conserved.

Which row states Kirchhoff's **first** law and names the quantity that is conserved?

	statement	quantity
A	The algebraic sum of currents at a junction is zero.	charge
B	The algebraic sum of currents at a junction is zero.	energy
C	The e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop.	charge
D	The e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop.	energy

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[Turn over

38 Two neutral atoms are isotopes of the same element.

Which statement about the atoms is correct?

A They have a different number of neutrons and a different number of electrons.

B They have a different number of neutrons and the same number of protons.

C They have the same number of neutrons and a different number of electrons.

D They have the same number of neutrons and the same number of protons.

39 What is the composition of a meson?

A 1 quark and 1 antiquark

B 1 quark and 2 antiquarks

C 2 quarks and 1 antiquark

D 3 quarks

40 A nucleus of carbon-11 contains 6 protons and 5 neutrons.

The nucleus of carbon-11 decays by β^+ emission.

What is the total number of up and down quarks in the product of the decay of this nucleus?

	up quarks	down quarks
A	15	18
B	16	17
C	17	16
D	18	15

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PHYSICS **9702/12**

Paper 1 Multiple Choice **May/June 2025**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	D	**1**
2	D	**1**
3	B	**1**
4	C	**1**
5	A	**1**
6	C	**1**
7	A	**1**
8	B	**1**
9	B	**1**
10	C	**1**
11	B	**1**
12	D	**1**
13	C	**1**
14	D	**1**
15	A	**1**
16	A	**1**
17	B	**1**
18	C	**1**
19	B	**1**
20	C	**1**
21	A	**1**
22	B	**1**
23	D	**1**
24	C	**1**
25	B	**1**
26	D	**1**
27	B	**1**
28	D	**1**

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Question	Answer	Marks
29	C	**1**
30	D	**1**
31	C	**1**
32	A	**1**
33	D	**1**
34	C	**1**
35	D	**1**
36	D	**1**
37	A	**1**
38	B	**1**
39	A	**1**
40	B	**1**

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June 2025 · Paper 13

9702_s25_qp_13

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2025**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What represents a vector quantity?

A 1 N upwards

B 2 kg decrease

C 3 K cooler

D 4 s later

2 What is a reasonable estimate of the current in an electric kettle that is in use?

A 8 μA **B** 8 mA **C** 8 A **D** 8 kA

3 Which statement about errors in measurements is correct?

A An accurate set of measurements always has a small random error.

B A precise set of measurements always has a small systematic error.

C A random error can be reduced by taking an average of several measurements.

D A systematic error creates a random set of measurements spread out about the true value.

4 A sample of material has cross-sectional area A and length L. The temperatures at the two sides of the sample are T_1 and T_2. Thermal energy Q is transferred through the sample in time t.

These quantities are related by

$$\frac{Q}{t} = \frac{k \times A \times (T_1 - T_2)}{L}$$

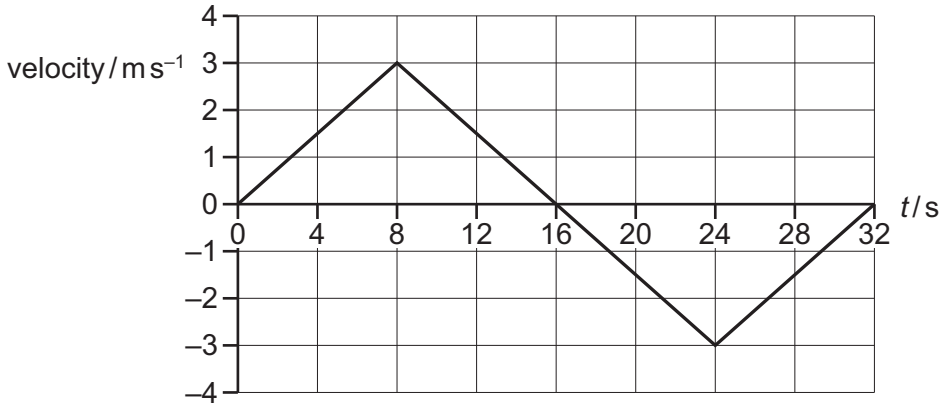
where k is a constant.

What are the SI base units of k?

A $\text{kg m s}^{-3}\,^{\circ}\text{C}^{-1}$ **B** $\text{kg m s}^{-3}\,\text{K}^{-1}$ **C** $\text{kg m s}^{-1}\,^{\circ}\text{C}^{-1}$ **D** $\text{kg m s}^{-1}\,\text{K}^{-1}$

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5 The graph shows the variation of velocity with time t of an object moving in a straight line.



At $t = 0$, the displacement of the object is zero.

What is the displacement of the object at $t = 20$ s?

A –3 m **B** 21 m **C** 24 m **D** 27 m

6 A science museum designs an experiment to show the fall of a feather in a vertical glass vacuum tube.

The time of fall from rest in the vacuum is to be close to 0.5 s.

Which length of tube is required?

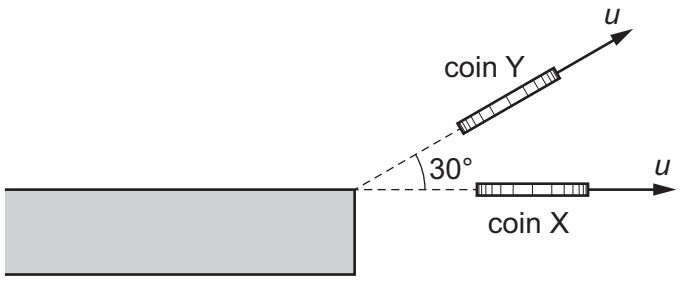
A 1.2 m **B** 2.5 m **C** 4.9 m **D** 9.8 m

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7 Two coins are projected from a horizontal table at the same initial speed u.

Coin X is projected horizontally.

Coin Y is projected upwards at an angle of 30° to the horizontal.



Both coins hit the horizontal ground without bouncing.

Assume the air resistance on each coin is negligible.

Which statement about the motion of the coins is correct?

A Both coins hit the ground with the same speed.

B Both coins travel the same vertical distance.

C Coin Y hits the ground before coin X.

D Coin Y has a smaller vertical acceleration.

8 Two spheres are released from rest at equal heights above the ground.

Both spheres reach terminal velocity.

One sphere has a larger density than the other sphere.

Both spheres have equal volumes.

Which statement is correct while both spheres are at terminal velocity?

A The drag forces on the spheres are equal.

B The resultant forces on the spheres are equal.

C The velocities of the spheres are equal.

D The weights of the spheres are equal.

9 Which statement **defines** force?

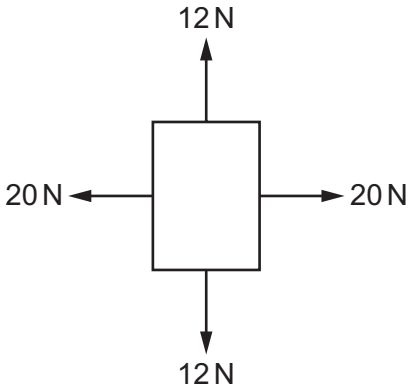
A When a force acts on a body that is free to move, the force is the product of the mass of the body and its acceleration.

B When a force acts on a body that is free to move, the force is the rate of change of momentum of the body.

C When a force acts on a body that is free to move, the force is the work done by the force divided by the distance moved by the body.

D When a force acts on a lever and causes a moment, the force is the moment divided by the perpendicular distance of the force from the pivot.

10 Four forces, all in the same plane, act on an object.



What could describe the motion of the object?

A It is accelerating.

B It is moving in a straight line.

C It is moving with decreasing speed.

D It is moving with increasing momentum.

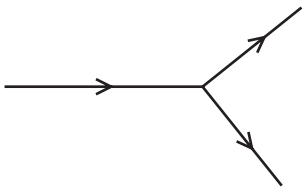
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11 A nucleus collides with a stationary nucleus in a vacuum. The diagrams show the paths of the nuclei before and after the collision.

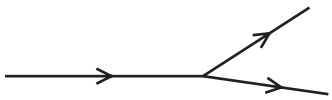
No other particles are involved in the collision.

Which diagram is **not** possible?

A



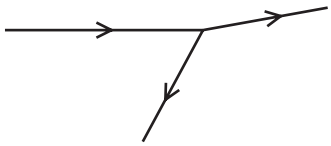
B



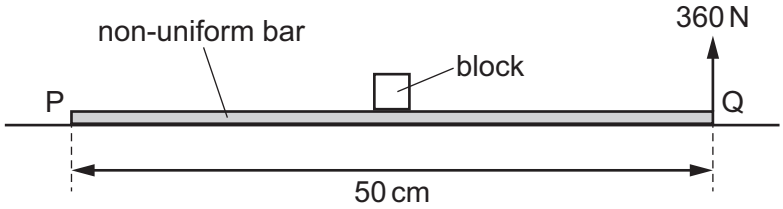
C



D



12 A non-uniform bar PQ has length 50 cm and weight 200 N.



A block of weight 400 N is attached to the top of the bar at its centre. The bar rests on horizontal ground.

A vertical force of 360 N is now exerted upwards on the bar at end Q and is just sufficient to lift end Q from the ground.

What is the distance from the centre of gravity of the bar to end P?

A 0.050 m **B** 0.10 m **C** 0.25 m **D** 0.40 m

13 What is the definition of density?

A mass of one cubic metre

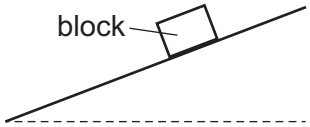
B mass of a unit volume

C mass per cubic metre

D mass per unit volume

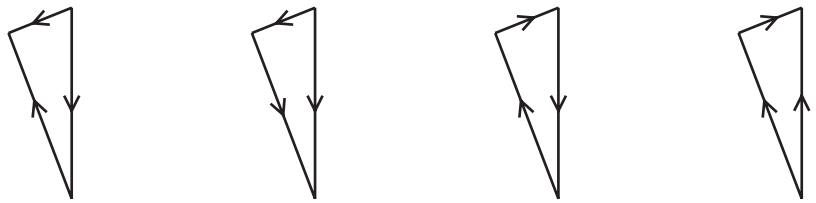
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14 A block is in equilibrium on a slope.



Which vector triangle represents the three forces acting on the block?

A **B** **C** **D**



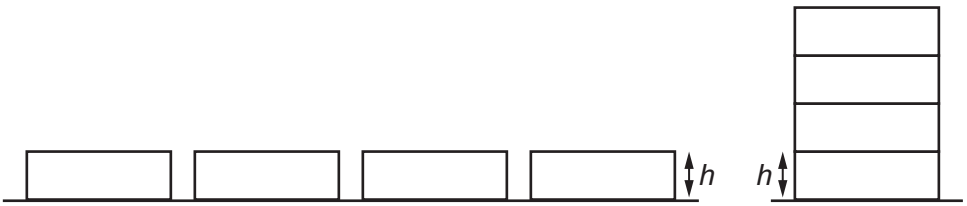
15 A measuring cylinder contains 80 cm^3 of a liquid.

The pressure due to the liquid at the 60 cm^3 mark is 1200 Pa.

What is the pressure due to the liquid at the base of the measuring cylinder?

A 1200 Pa **B** 1600 Pa **C** 3600 Pa **D** 4800 Pa

16 Four identical uniform blocks are spread on a table. Each block has mass *m* and thickness *h*.



The acceleration of free fall is *g*.

How much work is done on the blocks in stacking them on top of one another?

A 3*mgh* **B** 6*mgh* **C** 8*mgh* **D** 10*mgh*

17 An electric motor uses 1.7 kW of power when operating normally.

The efficiency of the motor is 53%.

What is the useful output power of the motor?

A 0.032 kW **B** 0.90 kW **C** 3.2 kW **D** 90 kW

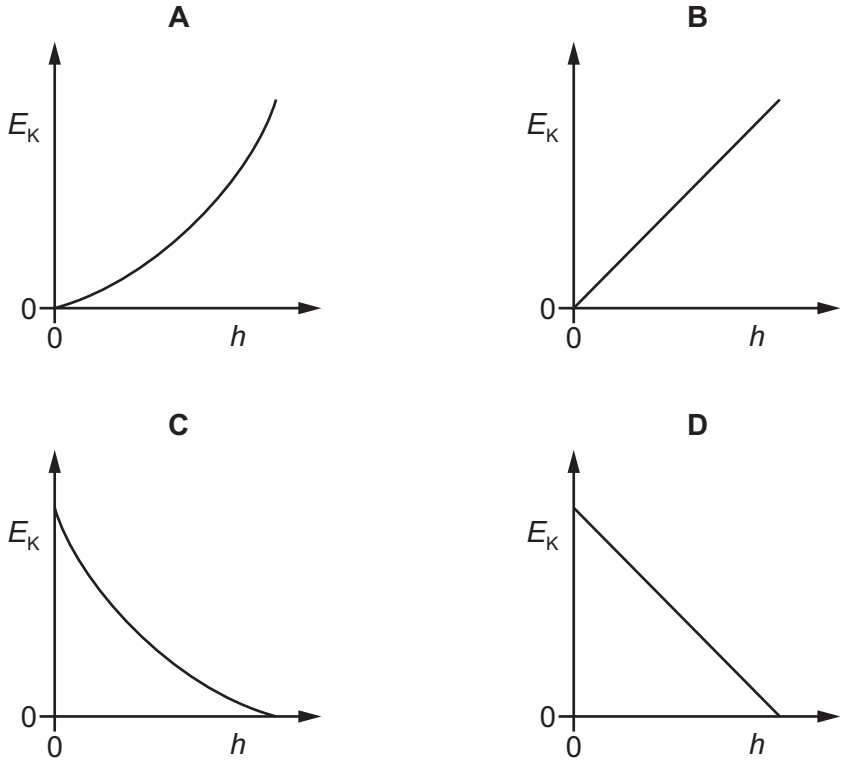
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18 A ball is released from rest and falls vertically to the ground.

The kinetic energy E_K of the ball varies as the height h of the ball above the ground changes.

Air resistance is negligible.

Which graph shows the variation of E_K with h?



19 An object travelling with a speed of $10\,m\,s^{-1}$ has kinetic energy of 1500 J.

The speed of the object is increased to $40\,m\,s^{-1}$.

What is the new kinetic energy of the object?

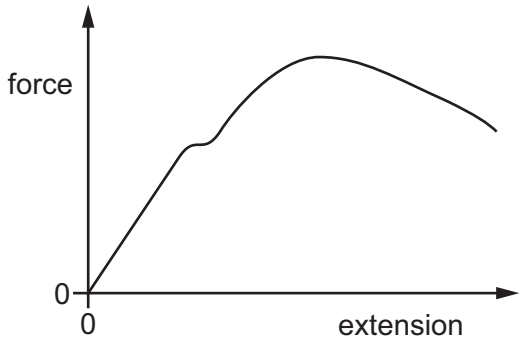
A 4500 J **B** 6000 J **C** 24 000 J **D** 1 350 000 J

20 What are the units of stress, strain and the Young modulus?

	stress	strain	Young modulus
A	newton	metre	pascal
B	newton	no unit	newton
C	pascal	metre	newton
D	pascal	no unit	pascal

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21 The force–extension graph for a metal wire is shown.



Which quantity is represented by the area under the graph?

A power transferred to the wire

B temperature increase in the wire

C time taken for the wire to extend

D work done on the wire

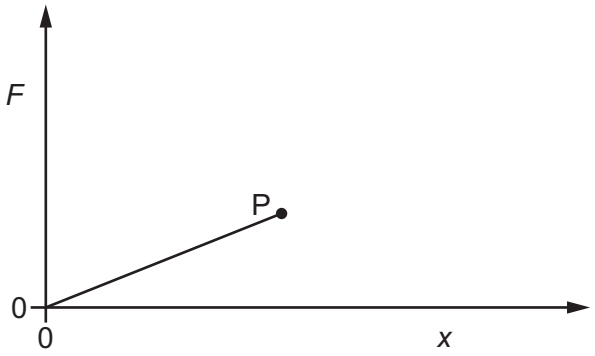
22 A wire of length L_0 is attached at one end to a fixed point. A tensile force is applied to the other end so that the wire extends and has a new length L_1.

What is the strain of the wire?

A $\left(\frac{L_1}{L_0}\right) - 1$ **B** $L_1 - L_0$ **C** $\left(\frac{L_1}{L_0}\right) + 1$ **D** $L_1 + L_0$

23 A wire of length *L* is stretched to determine its limit of proportionality.

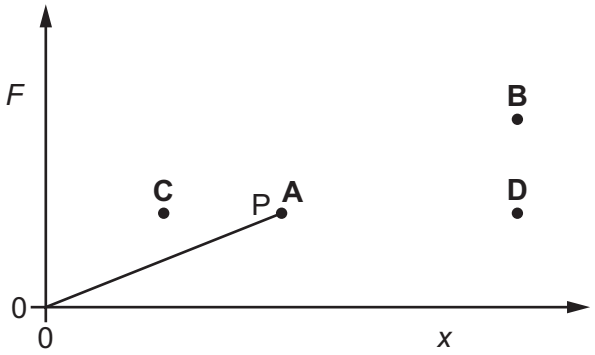
The graph shows the variation of the extension *x* with the force *F* applied to the wire.



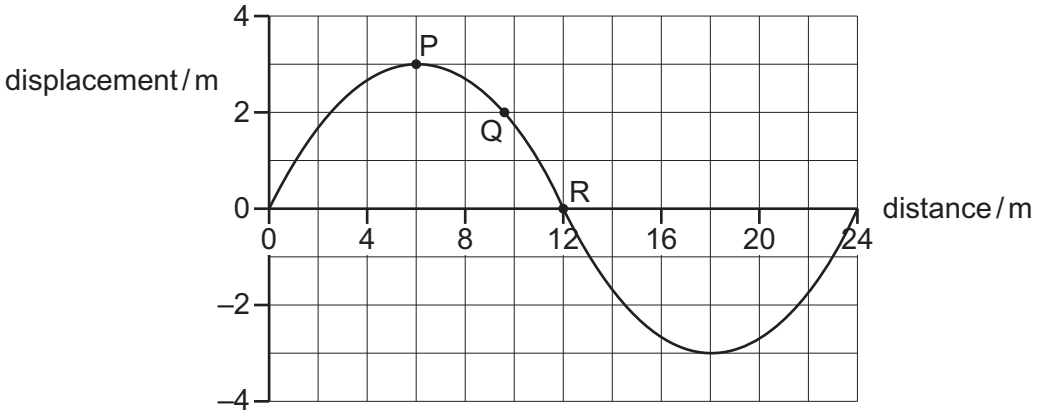
The limit of proportionality is shown by point P on the graph.

The experiment is repeated with another wire of length 2*L* but of the same material and same diameter as the first wire.

Which point on the graph shows the limit of proportionality for the new wire?



24 The graph shows the variation with distance along the wave of the displacement of water particles at a particular instant in time for a transverse water wave.



P, Q and R show the positions of three water particles in the wave.

Which particle has the greatest speed at the instant shown?

A all have the same speed

B particle P

C particle Q

D particle R

25 Which phenomenon is only associated with transverse waves?

A diffraction

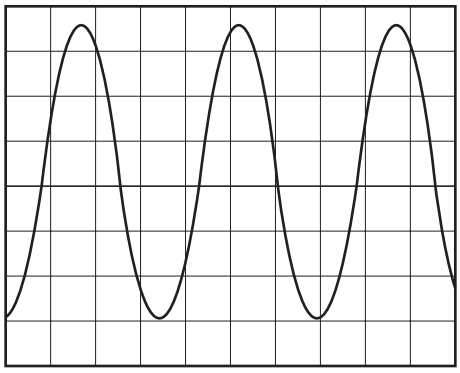
B interference

C polarisation

D reflection

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26 A wave is displayed on an oscilloscope.



The oscilloscope settings are:

time–base: 300 μs div^{-1}

y-gain: 40 μV div^{-1}.

What is the frequency of the wave?

A 130 Hz **B** 950 Hz **C** 1100 Hz **D** 7100 Hz

27 An electromagnetic wave travelling in free space is **not** visible to the human eye.

What is a possible wavelength of the wave?

A 4.5×10^{-10} km

B 6.2×10^{-7} m

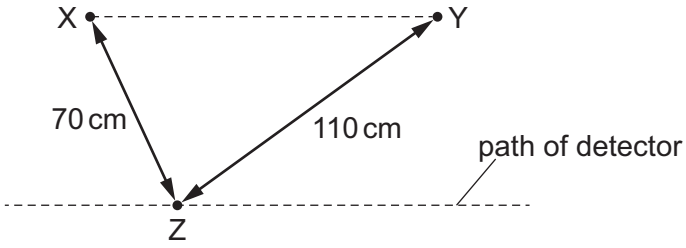
C 5.4×10^{-6} cm

D 4.2×10^{-4} mm

28 Electromagnetic waves of equal wavelengths are emitted from two sources, X and Y. The waves are emitted from X and Y with a phase difference of 180°.

A detector moves along a path that is parallel to the line XY and detects a pattern of intensity maxima and minima.

The diagram shows the arrangement of the sources and the path of the detector.



An intensity maximum is detected at point Z. Length XZ is 70 cm and length YZ is 110 cm.

What is a possible wavelength of the waves?

A 10 cm **B** 16 cm **C** 20 cm **D** 40 cm

29 Two waves of the same type overlap.

When does the principle of superposition apply?

A always

B only when the waves have the same amplitude

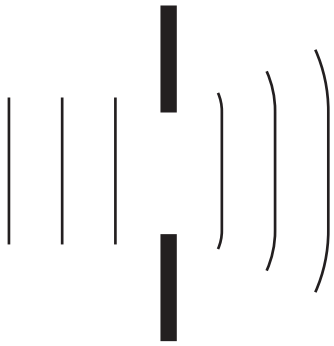
C only when the waves travel in opposite directions

D only when the waves have the same frequency

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30 A ripple tank contains water at a constant depth.

A water wave of constant frequency travels towards a gap in a barrier placed in the ripple tank.



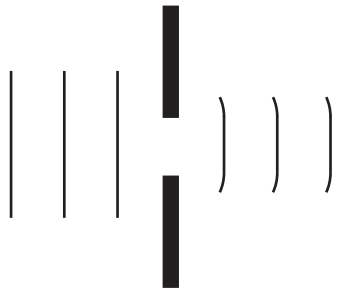
The gap is made smaller.

Which diagram represents the wave before and after the barrier?

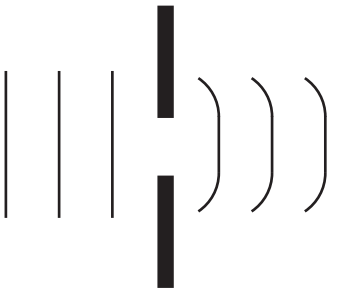
A



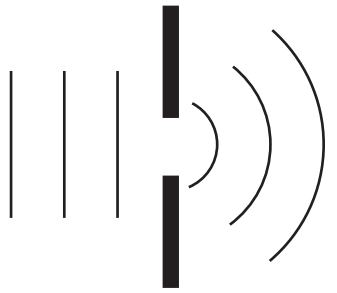
B



C



D



31 Light of a single frequency from two coherent sources interferes to produce a pattern of bright and dark fringes on a screen.

Which change results in a larger fringe separation?

A increasing the distance between the sources and the screen

B increasing the distance between the two sources

C increasing the frequency of the light

D increasing the intensity of the light

32 The current I in a metallic conductor of cross-sectional area A is given by

$$I = Anve$$

where e is the elementary charge and v is the mean drift velocity of the conduction electrons.

What is represented by the letter n in the equation?

A density of conduction electrons

B number of conduction electrons per unit time

C number of conduction electrons per unit volume

D volume of conduction electrons

33 What **cannot** be the charge on a charge carrier?

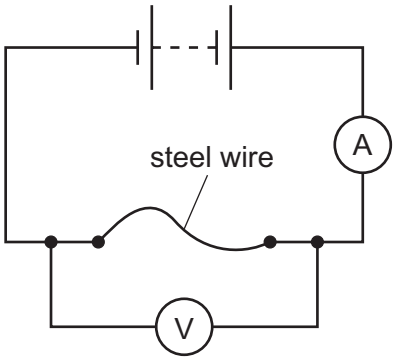
A $-4.8 \times 10^{-19}\,\text{C}$

B $-3.2 \times 10^{-19}\,\text{C}$

C $-2.4 \times 10^{-19}\,\text{C}$

D $+3.2 \times 10^{-19}\,\text{C}$

34 A steel wire with a length of 2.20 m is connected to a battery as shown.



The reading on the voltmeter is 1.50 V and the reading on the ammeter is 4.90 A.

The resistivity of steel is $6.90 \times 10^{-7}\,\Omega\,\text{m}$.

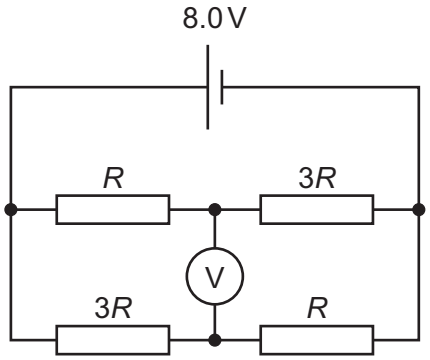
What is the diameter of the steel wire?

A $1.76 \times 10^{-4}\,\text{m}$ **B** $3.52 \times 10^{-4}\,\text{m}$ **C** $1.26 \times 10^{-3}\,\text{m}$ **D** $2.51 \times 10^{-3}\,\text{m}$

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35 Two resistors of resistance R and two resistors of resistance $3R$ are connected to a cell of e.m.f. 8.0 V as shown.

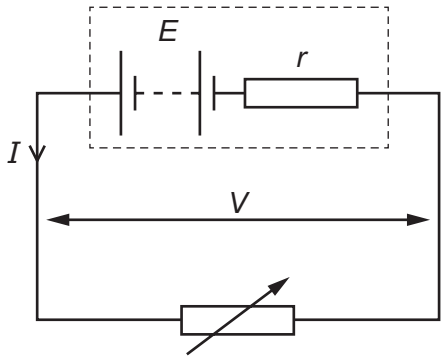
The cell has negligible internal resistance.



What is the reading on the voltmeter?

A 0.0 V **B** 2.0 V **C** 4.0 V **D** 6.0 V

36 A battery has an e.m.f. E and internal resistance r. The battery delivers a current I to a variable resistor and the p.d. across its terminals is V.



The variable resistor is adjusted so that I increases.

Why does V decrease?

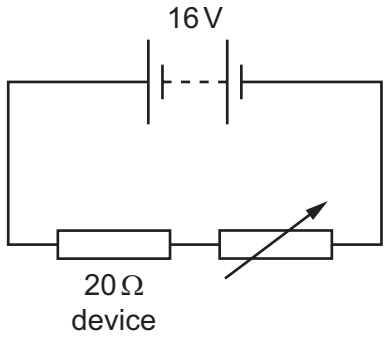
A The e.m.f. E decreases.

B The internal resistance r increases.

C The p.d. across r increases.

D The resistance of the variable resistor increases.

37 An electrical device of fixed resistance 20 Ω is connected in series with a variable resistor and a battery of e.m.f. 16 V and negligible internal resistance.



The power dissipated in the electrical device is 4.0 W.

What is the resistance of the variable resistor?

A 16 Ω **B** 36 Ω **C** 44 Ω **D** 64 Ω

38 A nucleus decays by emitting a β^+ particle.

Which particle **must** also be emitted from the nucleus?

A antineutrino

B β^- particle

C α-particle

D neutrino

39 What are the charges on an antidown quark and on an antistrange quark?

	antidown quark	antistrange quark
A	$+\frac{1}{3}e$	$+\frac{1}{3}e$
B	$+\frac{1}{3}e$	$-\frac{1}{3}e$
C	$-\frac{1}{3}e$	$+\frac{1}{3}e$
D	$-\frac{1}{3}e$	$-\frac{1}{3}e$

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40 The nucleus of a radioactive isotope of an element emits an α-particle. The daughter nucleus then emits a β^- particle and then the daughter nucleus of that reaction emits another β^- particle.

Which statement describes the final nuclide that is formed?

A It is a different isotope of the original element.

B It is a nuclide of a different element of higher proton number.

C It is a nuclide of the same element but with different proton number.

D It is identical to the original nuclide.

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CAMBRIDGE
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PHYSICS **9702/13**

Paper 1 Multiple Choice **May/June 2025**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	A	**1**
2	C	**1**
3	C	**1**
4	B	**1**
5	B	**1**
6	A	**1**
7	A	**1**
8	B	**1**
9	B	**1**
10	B	**1**
11	C	**1**
12	D	**1**
13	D	**1**
14	C	**1**
15	D	**1**
16	B	**1**
17	B	**1**
18	D	**1**
19	C	**1**
20	D	**1**
21	D	**1**
22	A	**1**
23	D	**1**
24	D	**1**
25	C	**1**
26	B	**1**
27	C	**1**
28	B	**1**

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Question	Answer	Marks
29	A	**1**
30	D	**1**
31	A	**1**
32	C	**1**
33	C	**1**
34	D	**1**
35	C	**1**
36	C	**1**
37	A	**1**
38	D	**1**
39	A	**1**
40	A	**1**

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November 2025 · Paper 11

9702_w25_qp_11

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/11**

Paper 1 Multiple Choice **October/November 2025**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 The table shows some physical quantities.

Which row correctly identifies the quantities as scalars or vectors?

	acceleration	charge	kinetic energy	wavelength
A	scalar	vector	vector	scalar
B	vector	vector	scalar	scalar
C	scalar	scalar	scalar	vector
D	vector	scalar	scalar	scalar

2 Two quantities are measured.

$$L = 6.8 \pm 0.1\,\text{cm}$$

$$T = 2.42 \pm 0.08\,\text{s}$$

L and T are related to X by the equation shown.

$$X = \frac{4\pi^2 L}{T^2}$$

What is the calculated value and uncertainty of X?

A $45.8 \pm 0.2\,\text{cm s}^{-2}$

B $45.8 \pm 0.3\,\text{cm s}^{-2}$

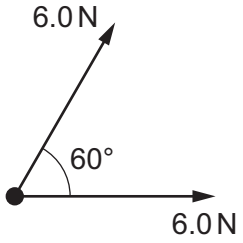
C $46 \pm 2\,\text{cm s}^{-2}$

D $46 \pm 4\,\text{cm s}^{-2}$

3 What are the SI base units of the watt?

A J s^{-1} **B** $\text{kg m}^2\text{ s}^{-3}$ **C** $\text{kg m}^2\text{ s}^{-1}$ **D** N m s^{-1}

4 The diagram shows two forces of 6.0 N acting on an object. The angle between the lines of action of the two forces is 60°.



What is the magnitude of the resultant force?

A 6.0 N **B** 7.9 N **C** 10 N **D** 12 N

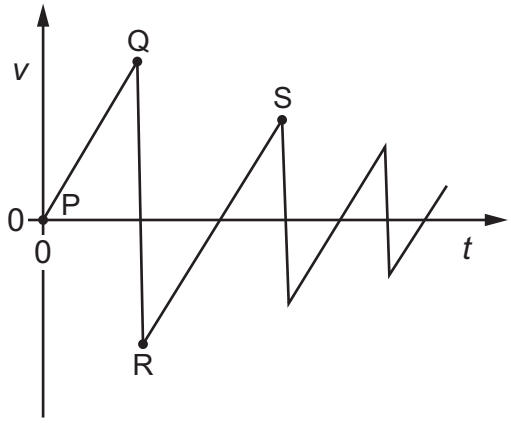
5 A goods train passes through a station at a constant speed of $10\,m\,s^{-1}$ at time $t = 0$. An express train is at rest at the station. The express train leaves the station with a uniform acceleration of $0.5\,m\,s^{-2}$ just as the goods train goes past. Both trains move in the same direction on straight, parallel tracks.

At which time t does the express train overtake the goods train?

A 6 s **B** 10 s **C** 20 s **D** 40 s

6 A ball is held above the ground and released. It falls to the ground and bounces several times.

The graph shows the variation with time t of the velocity v of the ball.



Four points on the graph are labelled P, Q, R and S.

Which statement is **not** correct?

A The area under line PQ represents the initial height of the ball.

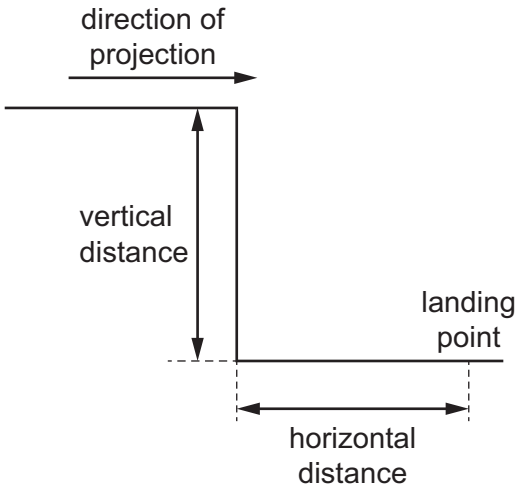
B The collisions of the ball with the ground are inelastic.

C The gradient of the line RS represents the acceleration due to free fall.

D The maximum upwards velocity of the ball is reached at point P.

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7 An object is projected horizontally. The object falls a vertical distance y and travels a horizontal distance x before landing on the ground.



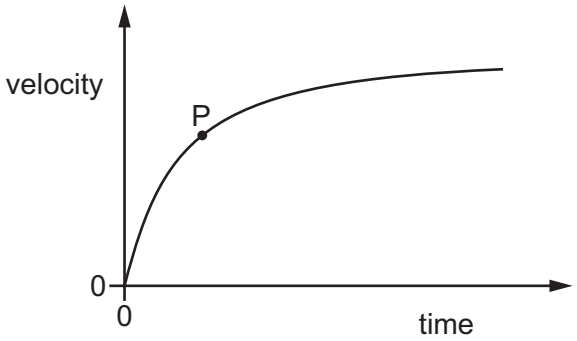
A second object is projected horizontally with the same initial velocity and falls a vertical distance $4y$ before landing on the ground.

Assume that air resistance is negligible.

Which horizontal distance does the second object travel?

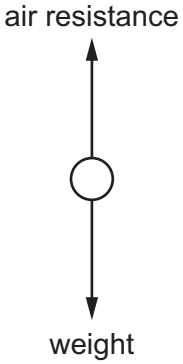
A x **B** $2x$ **C** $4x$ **D** $16x$

8 A sphere falls from rest through the air. The graph shows the variation with time of the sphere's velocity.



Which diagram shows the forces acting on the sphere when it is at the velocity corresponding to point P on the graph?

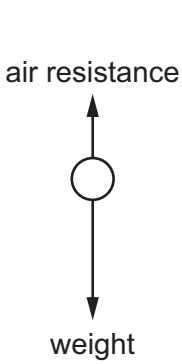
A



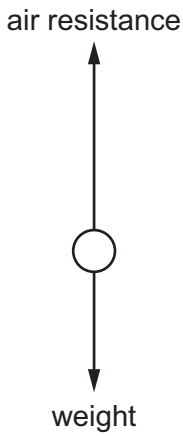
B



C



D



9 In which situation is total linear momentum **always** conserved?

A in all collisions between two objects

B in collisions between two objects moving at equal and opposite velocity

C in collisions between two objects that form an isolated system

D in collisions between two objects with the same mass

10 An object of mass m moving with velocity $9.0\,m\,s^{-1}$ has a head-on elastic collision with a stationary object of mass $2m$.

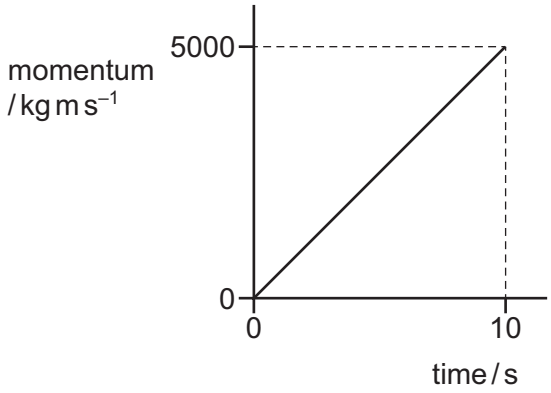
After the collision, both objects are moving. No external forces act on the system.

What is the velocity of the object of mass $2m$ after the collision?

A $-6.0\,m\,s^{-1}$ **B** $-3.0\,m\,s^{-1}$ **C** $4.5\,m\,s^{-1}$ **D** $6.0\,m\,s^{-1}$

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11 The graph shows how the momentum of a motorcycle changes with time.



What is the resultant force on the motorcycle?

A 500 N **B** 5000 N **C** 25 000 N **D** 50 000 N

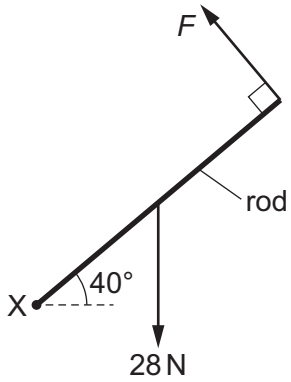
12 A rocket has a weight of W and an initial acceleration of a as the rocket leaves the ground vertically. The acceleration of free fall is g.

Which expression gives the initial upward force exerted on the rocket due to the engine?

A $\frac{Wa}{g} + W$ **B** $\frac{Wa}{g}$ **C** $\frac{Wa}{g} - W$ **D** $Wa + W$

13 A uniform rod of weight 28 N is supported at one end by force F.

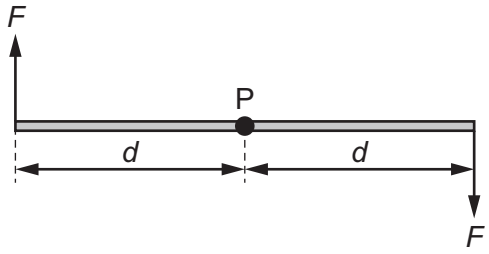
The rod is in equilibrium and attached to a frictionless hinge at end X.



What is the magnitude of F?

A 9.0 N **B** 11 N **C** 18 N **D** 21 N

14 Two forces, each of magnitude F, act in opposite directions on a rod.



Each force acts on the rod at a distance d from the pivot P.

What is the torque of this couple about P?

A 0 **B** $F \times d$ **C** $2F \times d$ **D** $2F \times 2d$

15 A man of weight 600 N stands with both feet flat on the ground.

What is a reasonable estimate of the pressure exerted on the ground by the weight of the man?

A 1×10^0 Pa **B** 1×10^2 Pa **C** 1×10^4 Pa **D** 1×10^6 Pa

16 The formula for the upthrust on a block of wood partially submerged in water is shown.

$$F = \rho g V$$

What do the symbols ρ and V represent?

	ρ	V
A	density of water	volume of whole block
B	density of water	volume of block below surface of water
C	density of wood	volume of whole block
D	density of wood	volume of block below surface of water

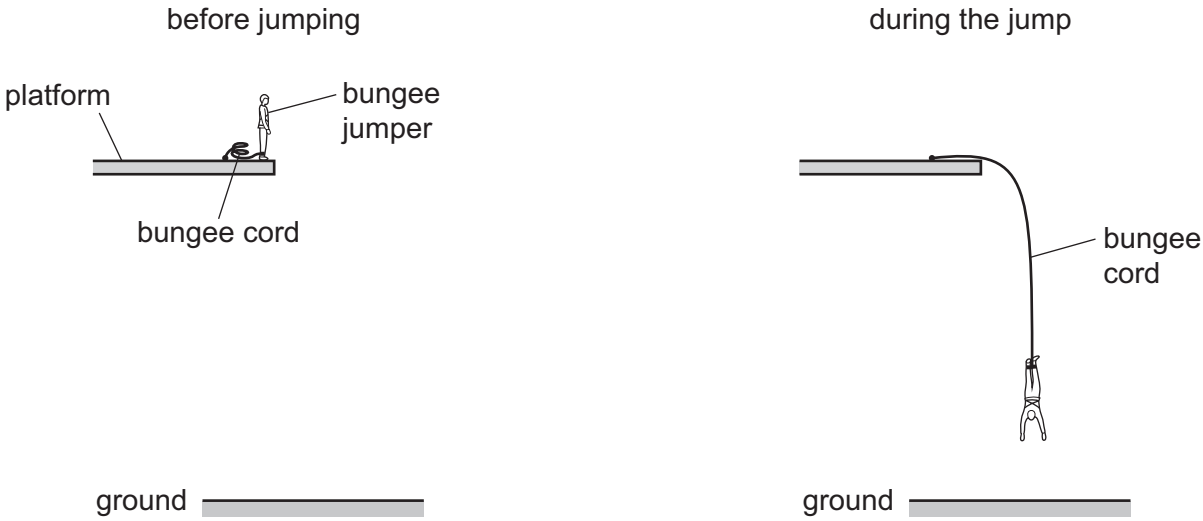
17 An object with a mass of 100 g falls a vertical distance of 10 m.

What is the change in the gravitational potential energy of the object?

A 1 J **B** 10 J **C** 100 J **D** 10 000 J

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18 A bungee jumper jumps from a platform and is decelerated by an elastic bungee cord, as shown.



NOT TO SCALE

When the jumper makes the jump, his initial gravitational potential energy relative to the ground is converted into his kinetic energy and into elastic potential energy in the cord.

At which part of the jump are all three types of energy non-zero?

A on the platform before the jump

B on the way down before the cord has started to extend

C on the way down as he decelerates

D at the bottom of the jump when he is stationary

19 A sailboat is pushed by the wind at a constant velocity *v* across a lake.

The force of the wind and the velocity of the sailboat are in the same direction.

Which additional information is required to determine the work done per unit time by the wind acting on the sailboat?

A the force exerted by the wind

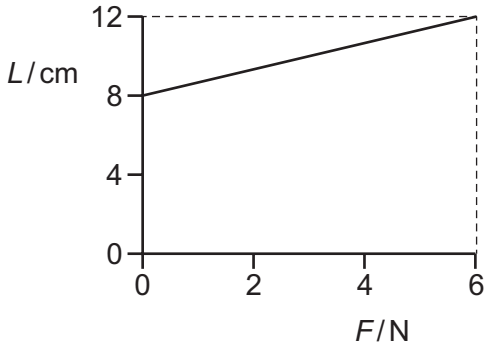
B the distance travelled by the sailboat per unit time

C the total energy input to the sailboat by the wind

D the weight of the sailboat

20 A spring is fixed at one end. The length L of the spring is increased by applying a tensile force F to the other end.

The graph shows the variation of L with F.



What is the elastic potential energy of the spring when F is 6.0 N?

A 0.12 J **B** 0.24 J **C** 0.36 J **D** 0.60 J

21 A wire has an unstretched length of 2.00 m.

A stress of 1.6×10^5 Pa is applied to the wire, and the new length of the wire is 2.10 m.

The wire obeys Hooke's law.

What is the Young modulus of the wire?

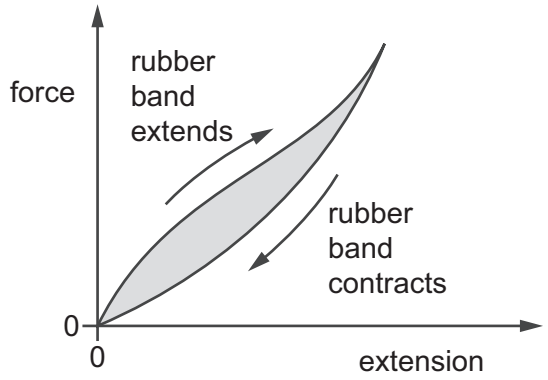
A 8.0×10^3 Pa **B** 7.8×10^4 Pa **C** 1.5×10^5 Pa **D** 3.2×10^6 Pa

22 What are the SI base units of stress?

A kg m s^{-2} **B** $\text{kg m}^{-1}\text{ s}^{-2}$ **C** $\text{kg m}^{-2}\text{ s}^{-2}$ **D** $\text{kg m}^{-3}\text{ s}^{-2}$

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23 The diagram shows a force–extension graph for a rubber band as the band is extended and then the stretching force is decreased to zero.



What can be deduced from the graph?

A The rubber band does **not** return to its original length when the force is decreased to zero.

B The rubber band obeys Hooke's law for the extensions shown.

C The rubber band remains elastic for the extensions shown.

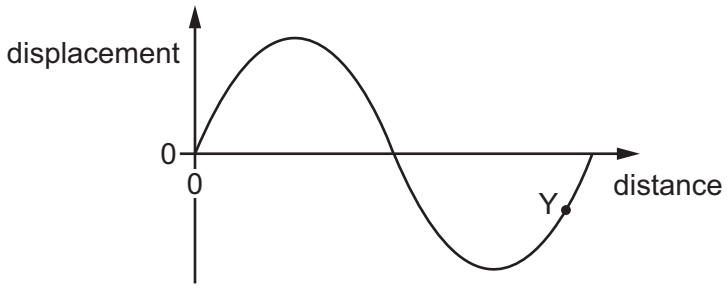
D The shaded area represents the work done in extending the rubber band.

24 Which row is correct for sound waves?

	can travel in a vacuum	can be polarised	can diffract through a gap
A	no	no	yes
B	yes	no	no
C	no	yes	no
D	no	yes	yes

25 A source emits a progressive sound wave in the horizontal direction. The wave travels away from the source in a direction towards the right.

The graph shows the variation of the displacement of the particles from their equilibrium positions with distance from the source at a particular instant in time. Displacements to the right of the equilibrium positions are shown as positive.



Position Y represents the displacement of a particle in the sound wave at a particular distance from the source at this instant.

Which statement about the motion of this particle is correct?

A The particle is moving away from the source.

B The particle is moving towards the source.

C The particle is moving upwards.

D The particle is moving downwards.

26 Two polarising filters are placed next to each other so that their planes are parallel.

The first polarising filter has its transmission axis at an angle of 50° to the vertical.

The second polarising filter has its transmission axis at an angle of 20° to the vertical. The angle between the transmission axes of the two polarising filters is 30°.

A beam of vertically polarised light of intensity $8.0\,W\,m^{-2}$ is incident normally on the first polarising filter.

What is the intensity of the light that is transmitted from the second polarising filter?

A zero **B** $2.5\,W\,m^{-2}$ **C** $2.9\,W\,m^{-2}$ **D** $6.0\,W\,m^{-2}$

27 A dolphin is swimming at a speed of $16.0\,m\,s^{-1}$ directly towards a stationary underwater microphone.

The dolphin emits a sound of frequency 122 kHz. The speed of sound in water is $1480\,m\,s^{-1}$.

What is the frequency of sound detected by the microphone?

A 1300 Hz **B** 1330 Hz **C** 121 kHz **D** 123 kHz

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28 A student carries out a double-slit experiment using a laser emitting red light of wavelength of 680 nm. The light is incident normally on a double slit.

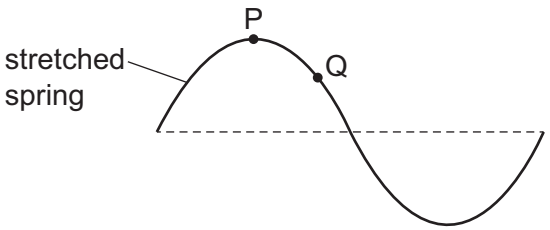
The diagram shows part of the pattern of bright fringes visible on a screen at a distance of 2.4 m from the slits. The distance across five bright fringes is measured as 34 mm.



What is the slit separation?

A 8.5×10^{-3} m **B** 2.4×10^{-4} m **C** 1.9×10^{-4} m **D** 4.8×10^{-5} m

29 The diagram shows a stationary wave on a stretched spring at an instant in time.



Two particles on the spring, P and Q, are shown.

Which statement about the vibrations of P and Q is correct?

A They have different frequencies.

B They have the same amplitudes.

C They have different periods.

D They are always in phase.

30 Green laser light passes through a diffraction grating and forms an interference pattern.

The diffraction grating contains 400 lines per mm.

The wavelength of the laser light is 550 nm.

What is the highest order diffraction maximum produced by the grating?

A 4 **B** 5 **C** 8 **D** 9

31 There is a potential difference V across a resistor of resistance R. The current in the resistor is I.

Which equation gives the power P dissipated by the resistor?

A $P = V^2 I$ **B** $P = \frac{V^2}{I}$ **C** $P = V^2 R$ **D** $P = \frac{V^2}{R}$

32 What is a possible charge on a particle?

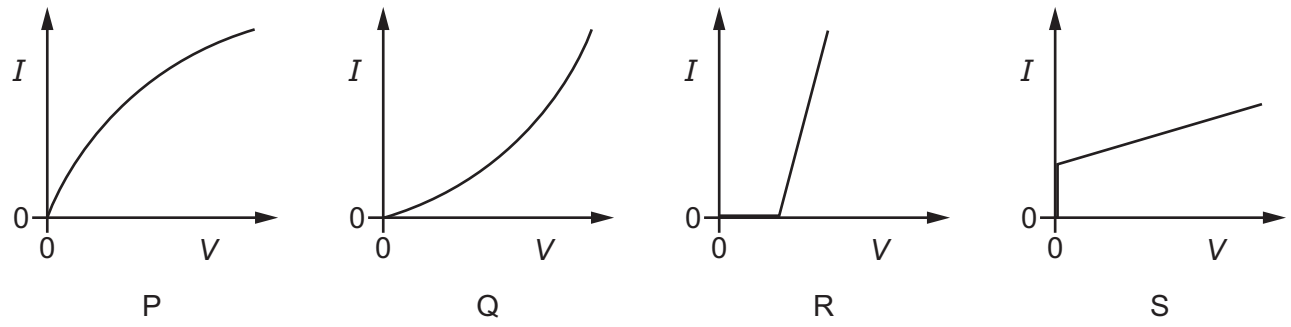
A 6.40×10^{-20} C

B 4.00×10^{-19} C

C 1.12×10^{-18} C

D 9.11×10^{-18} C

33 The graphs show possible current–voltage (I–V) characteristics for a filament lamp and for a semiconductor diode.



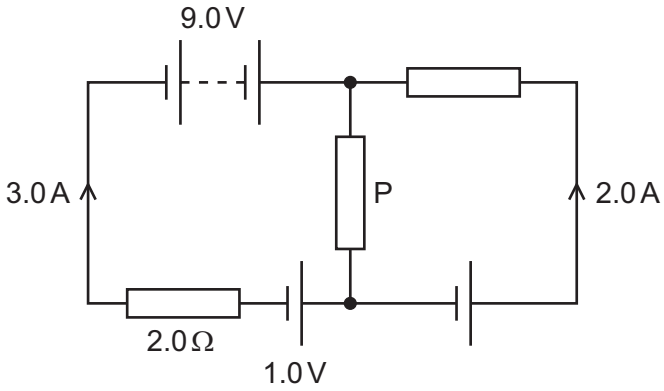
Which row identifies the I–V graphs for the lamp and for the diode?

	filament lamp	semiconductor diode
A	P	R
B	P	S
C	Q	R
D	Q	S

34 The diagram shows a circuit containing a battery and cells with negligible internal resistance.

Some values of current, electromotive force (e.m.f.) and resistance are shown.

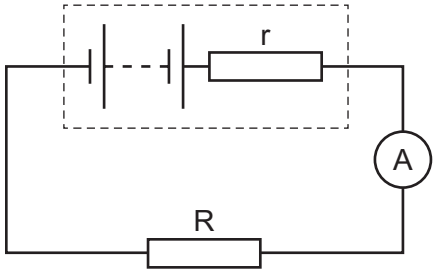
One resistor is labelled P.



What is the resistance of resistor P?

A 0.40 Ω **B** 0.80 Ω **C** 2.0 Ω **D** 2.8 Ω

35 The diagram shows a circuit with a battery connected to a resistor R. The battery has an internal resistance represented by resistor r.

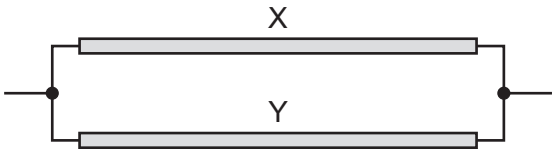


A second resistor, identical to R, is connected in parallel with R.

Which row describes the changes to the potential difference (p.d.) across r and the current shown on the ammeter when the second resistor is connected?

	p.d. across r	current shown on ammeter
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

36 Two identical wires X and Y, each of length L and radius r, are connected in parallel as shown.



The total resistance of this combination is R_1.

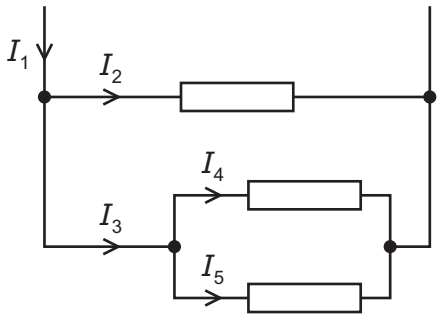
Wire X is replaced with a wire of the same material with length L and radius $2r$.

The total resistance of the new combination is R_2.

What is the ratio $\frac{R_1}{R_2}$?

A $\frac{2}{5}$ **B** $\frac{2}{3}$ **C** $\frac{3}{2}$ **D** $\frac{5}{2}$

37 The diagram shows the currents in part of an electric circuit.



The resistors are identical.

Which equation is **not** correct?

A $I_1 = I_2 + I_4 + I_5$

B $I_2 = I_1 - I_3$

C $I_2 = I_3 + I_4 + I_5$

D $I_4 = I_3 - I_5$

38 The diagram shows a sequence of radioactive decays involving three α-particles and a β^- particle.

$$^{237}_{93}\text{Np} \xrightarrow{\alpha} \text{Q} \xrightarrow{\beta^-} \text{R} \xrightarrow{\alpha} \text{S} \xrightarrow{\alpha} \text{T}$$

What is nuclide T?

A $^{225}_{88}\text{Ra}$ **B** $^{231}_{88}\text{Ra}$ **C** $^{225}_{90}\text{Th}$ **D** $^{229}_{90}\text{Th}$

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39 The number of electrons in a neutral atom of an isotope of plutonium, $^{239}_{94}Pu$, is changed to produce a charged atom (ion) W.

W has an overall charge of $+1.6 \times 10^{-19}$ C.

How many protons, neutrons and electrons are in W?

	protons	neutrons	electrons
A	94	145	93
B	94	145	95
C	238	94	239
D	239	94	238

40 Which particle is a lepton?

A meson

B positron

C proton

D quark

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PHYSICS **9702/11**

Paper 1 Multiple Choice **October/November 2025**

MARK SCHEME

Maximum Mark: 40

Published

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Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

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Question	Answer	Marks
1	D	**1**
2	D	**1**
3	B	**1**
4	C	**1**
5	D	**1**
6	D	**1**
7	B	**1**
8	C	**1**
9	C	**1**
10	D	**1**
11	A	**1**
12	A	**1**
13	B	**1**
14	C	**1**
15	C	**1**
16	B	**1**
17	B	**1**
18	C	**1**
19	A	**1**
20	A	**1**
21	D	**1**
22	B	**1**
23	C	**1**
24	A	**1**
25	B	**1**
26	B	**1**
27	D	**1**
28	C	**1**

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Question	Answer	Marks
29	D	**1**
30	A	**1**
31	D	**1**
32	C	**1**
33	A	**1**
34	A	**1**
35	D	**1**
36	D	**1**
37	C	**1**
38	A	**1**
39	A	**1**
40	B	**1**

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November 2025 · Paper 12

9702_w25_qp_12

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/12**

Paper 1 Multiple Choice **October/November 2025**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

7826120105

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What is a reasonable estimate of the weight of an adult human?

A 7×10^0 N **B** 7×10^2 N **C** 7×10^4 N **D** 7×10^6 N

2 What describes a set of data with a high precision?

A data measured using equipment with small scale divisions

B data that is close to the accepted value

C data with each value having a low uncertainty

D data with repeats that are close to each other

3 Which physical quantity could have units of $\text{N s}^2\,\text{m}^{-1}$?

A acceleration

B force

C mass

D momentum

4 A ball is released from rest. The distance the ball falls and the time the ball takes to fall that distance are both measured.

The percentage uncertainty in the measured distance is negligible. The percentage uncertainty in the measured time is 4%.

The distance and the time are then used to calculate the acceleration of free fall.

Air resistance is negligible.

What is the percentage uncertainty in the calculated value of the acceleration of free fall?

A 2% **B** 4% **C** 8% **D** 16%

5 A car has an initial velocity u. The car then moves with constant acceleration a in a straight line through a displacement d. The car reaches a final velocity v.

Which expression gives the initial velocity u of the car?

A $v - ad$ **B** $\sqrt{v^2 - 2ad}$ **C** $v + ad$ **D** $v - \frac{1}{2}ad^2$

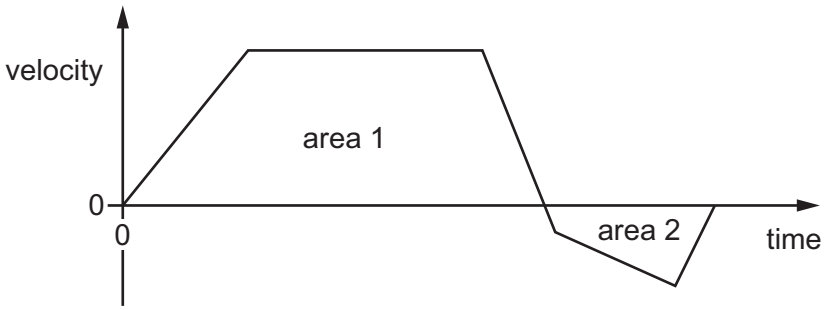
6 A bicycle brakes so that it undergoes uniform deceleration from a speed of $8\,\text{m s}^{-1}$ to $6\,\text{m s}^{-1}$ over a distance of 7 m.

The deceleration of the bicycle remains constant.

Which further distance will the bicycle travel before coming to rest?

A 7 m **B** 9 m **C** 16 m **D** 21 m

7 The velocity–time graph for an object is shown.



Which expression gives the total displacement of the object?

A area 1 – area 2

B $\frac{(\text{area 1} + \text{area 2})}{2}$

C area 1 + area 2

D area 2 – area 1

8 Which statement describes the weight of an object?

A It is equal to the mass of the object multiplied by its acceleration.

B It is equal to the resultant force when the object falls at terminal (constant) velocity.

C It is the force acting on the object due to a gravitational field.

D It is the property of the object that resists change in motion.

9 What is a statement of the principle of conservation of momentum?

A In an elastic collision, momentum is constant.

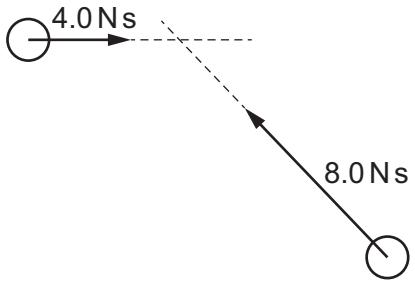
B Momentum is the product of mass and velocity.

C The force acting on a body is proportional to its rate of change of momentum.

D The momentum of an isolated system is constant.

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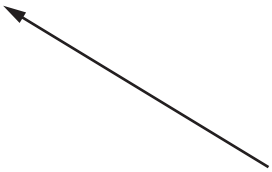
10 The diagram shows the view from above of two balls moving along a horizontal frictionless surface before they collide. The momentum of each ball is also shown.



The balls stick together during the collision.

Which vector diagram represents the combined momentum of the balls after the collision?

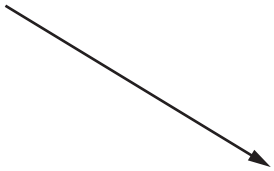
A



B



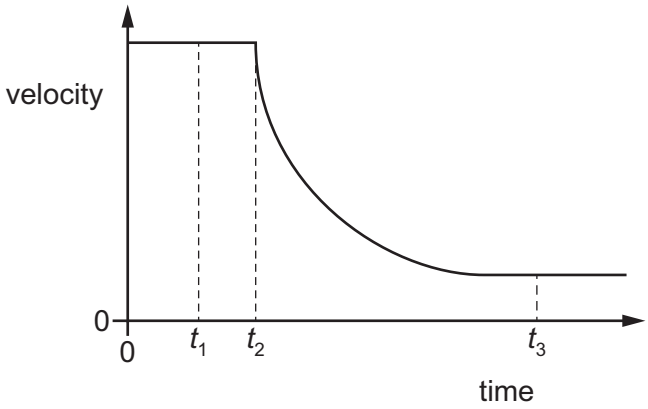
C



D



11 A skydiver is falling at constant velocity. She then opens her parachute. The graph shows the variation with time of her velocity.



Which statements about the motion of the skydiver are correct?

1 The magnitude of the acceleration is maximum at time t_2.

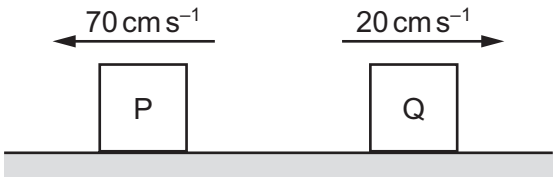
2 The magnitude of the drag force at time t_1 equals the magnitude of the drag force at time t_3.

3 The magnitude of the drag force is maximum at time t_1.

A 1 and 2 **B** 1 and 3 **C** 2 only **D** 3 only

12 Two different blocks, P and Q, slide towards each other on a horizontal frictionless surface. The blocks have an elastic collision.

The diagram shows the velocities of the two blocks immediately after the collision.

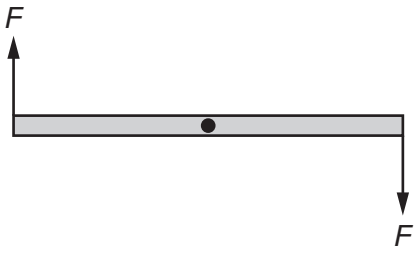


Which row gives possible velocities of the two blocks immediately before the collision?

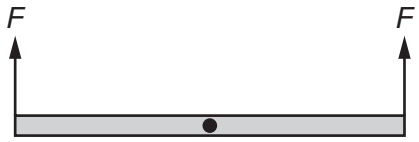
	velocity of P	velocity of Q
A	10 cm s^{-1} to the right	60 cm s^{-1} to the left
B	50 cm s^{-1} to the right	zero
C	20 cm s^{-1} to the left	70 cm s^{-1} to the right
D	60 cm s^{-1} to the right	30 cm s^{-1} to the left

13 Which diagram shows a couple?

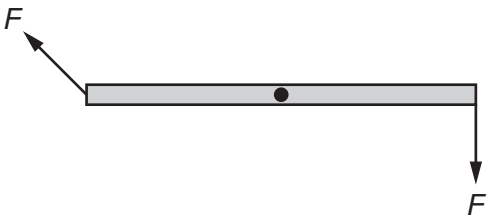
A



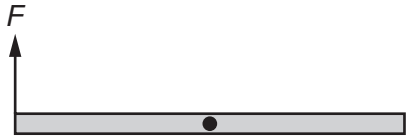
B



C

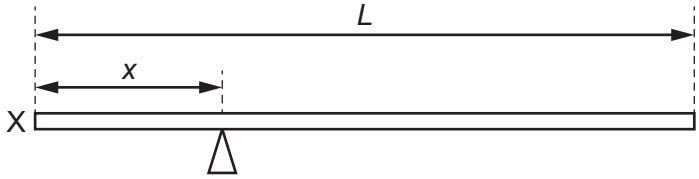


D



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14 The diagram shows a uniform bar of mass M and length L resting on a pivot at a distance x from end X.



An object of mass m is placed on the bar at distance y from the pivot so that the bar is in equilibrium.

What is an expression for y?

A $\frac{xM}{m}$ **B** $\frac{M}{m}(L-x)$ **C** $\frac{M}{2m}(L-2x)$ **D** $\frac{1}{m}(Lm-xM)$

15 The mass and volume of an object are varied.

Which two changes, when made together, **must** increase the density of the object?

	mass	volume
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

16 On Earth, a solid object that is fully submerged in a liquid experiences an upthrust U_E.

On Mars, the same object, fully submerged in the same liquid, experiences an upthrust U_M.

The acceleration of free fall on Mars is $3.7\,\text{m}\,\text{s}^{-2}$.

Assume that the liquid's density and the object's volume have the same values on Earth and Mars.

What is the ratio $\frac{U_M}{U_E}$?

A 0.38 **B** 1.0 **C** 2.7 **D** 3.6

17 A student attempts to derive the formula for kinetic energy E_K. She begins by considering an object of mass m that is initially at rest. A constant force F applied to the object causes it to accelerate to final velocity v in displacement s. The kinetic energy gained by the object is equal to the work done on the object by the force F.

Which equation does the student **not** need in order to derive the formula for E_K?

A $F = ma$ **B** $W = Fs$ **C** $E = \frac{1}{2}Fs$ **D** $v^2 = u^2 + 2as$

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18 A ball falls towards the ground from point X. Point X is 2.4 m above the ground.

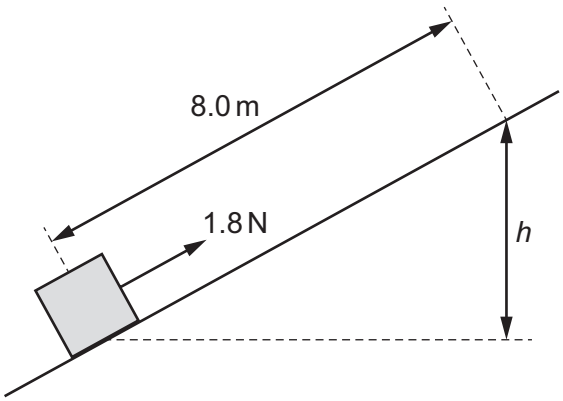
The ball rebounds from the ground and rises to point Y. Point Y is 1.8 m above the ground.

A single value of the change in height Δh is used to calculate the change in gravitational potential energy of the ball from X to Y.

What is the magnitude of Δh?

A 0.6 m **B** 1.8 m **C** 2.4 m **D** 4.2 m

19 The diagram shows a block on a slope.



A constant force of 1.8 N in a direction parallel to the slope is exerted on the block. This causes the block to move along the slope at a constant speed.

The block moves a distance of 8.0 m along the slope and gains height h. The work done against the frictional force acting on the block is 4.8 J.

The weight of the block is 4.0 N.

What is the value of h?

A 1.2 m **B** 2.4 m **C** 3.6 m **D** 4.8 m

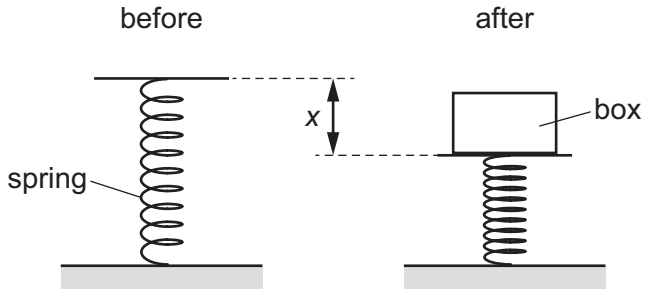
20 The motor of a crane lifts a load of mass 600 kg. The load rises vertically at a constant speed of 12 m per minute.

What is the useful power output of the motor?

A 0.12 kW **B** 1.2 kW **C** 7.2 kW **D** 71 kW

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21 A platform is supported by a spring. A box is placed onto the platform. The diagram shows the spring before and after the box is placed onto it.



What is the term for the quantity marked as *x*?

A compression

B load

C strain

D stress

22 A copper wire has length 1.7 m and uniform diameter 0.64 mm.

The Young modulus of copper is 1.2×10^{11} Pa.

What is the spring constant of the wire?

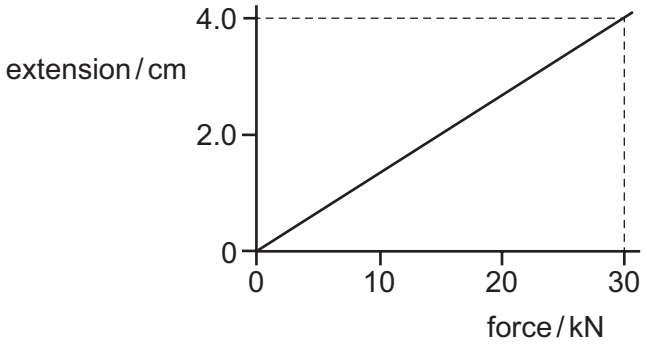
A $2.3 \times 10^{4}\,\text{N m}^{-1}$

B $9.1 \times 10^{4}\,\text{N m}^{-1}$

C $4.5 \times 10^{7}\,\text{N m}^{-1}$

D $7.1 \times 10^{7}\,\text{N m}^{-1}$

23 The graph shows how the extension of a spring varies with the force used to stretch it.



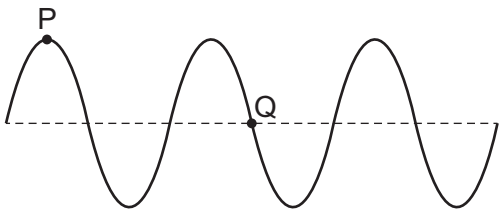
What is the elastic potential energy of the spring when the extension is 4.0 cm?

A 60 J **B** 120 J **C** 600 J **D** 1200 J

24 Which row describes a progressive longitudinal wave?

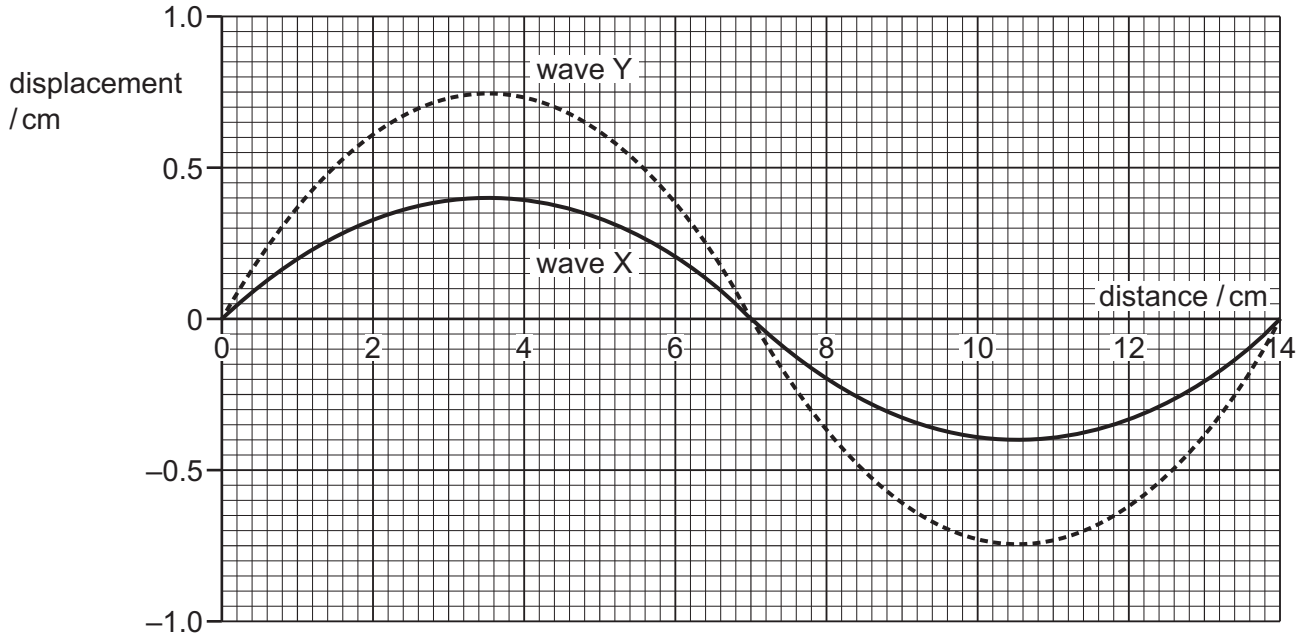
	transfers energy in a direction parallel to the oscillations	requires a medium to travel through	always travels at the same speed
A	false	false	true
B	false	true	true
C	true	false	false
D	true	true	false

25 What is the phase difference between points P and Q on the progressive wave shown in the diagram?



A 180° **B** 450° **C** 540° **D** 720°

26 The graph shows the variation of displacement with distance for two progressive waves X and Y of the same type in the same medium.



The intensity of wave X is I.

What is the intensity of wave Y?

A $0.28I$ **B** $0.53I$ **C** $1.9I$ **D** $3.5I$

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27 A moving source emits sound of frequency 1200 Hz.

A stationary observer hears sound of frequency 960 Hz. The speed of the sound is 340 $m\,s^{-1}$.

What could be the velocity of the source?

A 68 $m\,s^{-1}$ directly away from the observer

B 68 $m\,s^{-1}$ directly towards the observer

C 85 $m\,s^{-1}$ directly away from the observer

D 85 $m\,s^{-1}$ directly towards the observer

28 A light wave passes through a single slit and a diffraction pattern forms.

What happens to the frequency and the speed of the wave when it is diffracted?

	frequency	speed
A	decreases	increases
B	increases	decreases
C	no change	decreases
D	no change	no change

29 The equation $n\lambda = d\sin\theta$ can be used with a diffraction grating to find the wavelength of visible light.

Which quantity is **not** correct for use in this equation?

	symbol	quantity
A	d	distance from grating to screen
B	λ	wavelength of light
C	n	order of intensity maximum
D	θ	diffraction angle of intensity maximum

30 Two progressive sound waves move in opposite directions and superpose to form a stationary wave.

Which statement describes an antinode of this stationary wave?

A a position where the phase difference between the two waves is always 0°

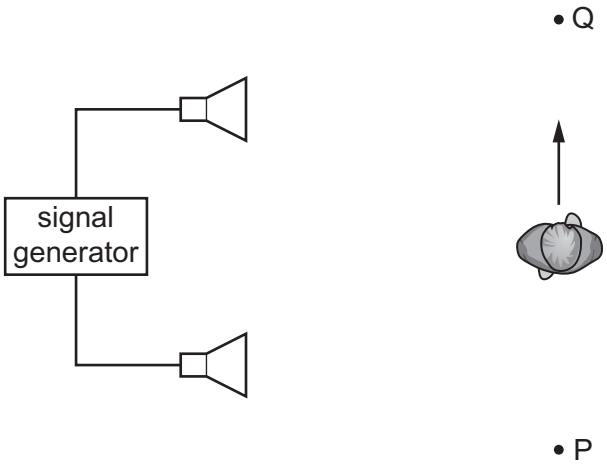
B a position where the phase difference between the two waves is always 180°

C a position where the stationary wave has maximum amplitude

D a position where the stationary wave has minimum amplitude

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31 A student connects two loudspeakers to a signal generator.



As the student walks from P to Q, he notices that the loudness of the sound rises and falls repeatedly.

What causes the loudness of the sound to vary?

A diffraction of the sound waves

B Doppler shift of the sound waves

C interference of the sound waves

D reflection of the sound waves

32 The current in a resistor of constant resistance is 8.0 mA. The power dissipated by the resistor is P.

The current in the resistor is increased to 12.0 mA.

What is the new power dissipated by the resistor?

A 0.44P **B** 0.67P **C** 1.5P **D** 2.3P

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33 A piece of wire X has resistivity ρ, length L and cross-sectional area A. Wire X has a resistance R.

A second piece of wire Y is made of a different metal. It has the same resistance as X but has twice the length of X.

Which row gives possible values for the resistivity and the cross-sectional area of Y?

	resistivity	cross-sectional area
A	$\frac{1}{2}\rho$	$\frac{1}{2}A$
B	$\frac{1}{2}\rho$	A
C	ρ	$\frac{1}{2}A$
D	2ρ	A

34 A cell of electromotive force (e.m.f.) E delivers a charge Q to an external circuit.

Which statement is correct?

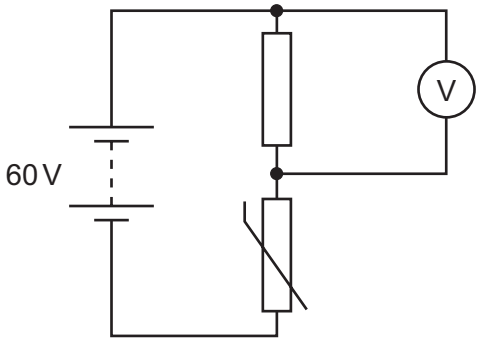
A The energy dissipated in the external circuit is EQ.

B The energy dissipated within the cell is EQ.

C The external resistance is EQ.

D The total energy dissipated in the cell and the external circuit is EQ.

35 The diagram shows a circuit with a battery of electromotive force (e.m.f.) 60 V and negligible internal resistance, a 20 kΩ resistor, a thermistor and a voltmeter. The resistance of the thermistor is 20 kΩ.



The temperature of the thermistor decreases, and this causes its resistance to change by 50%.

After this change, what is the reading on the voltmeter?

A 20 V **B** 24 V **C** 36 V **D** 40 V

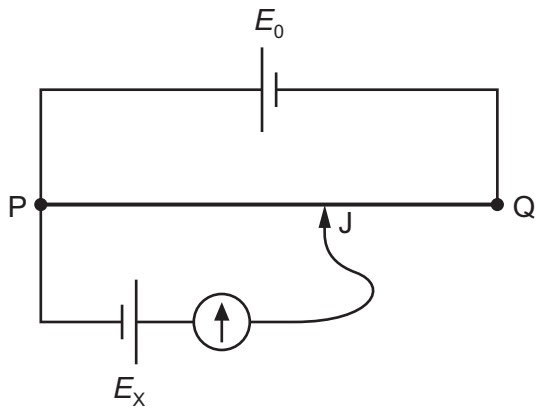
36 A potentiometer circuit is used to determine the electromotive force (e.m.f.) E_X of a cell. The circuit includes a second cell of known e.m.f. E_0 and negligible internal resistance, and a uniform resistance wire PQ of known length.

E_X is less than E_0.

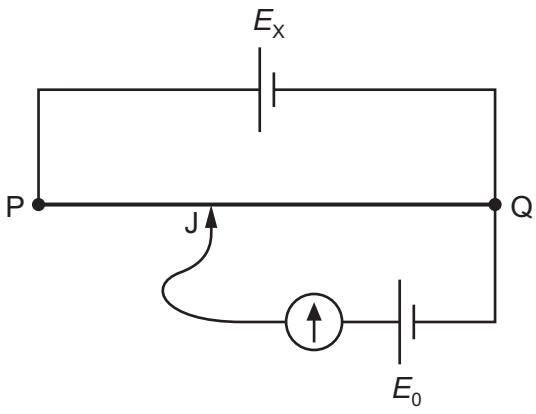
The movable connection J can be positioned anywhere along the length of the resistance wire.

Which circuit is suitable for determining E_X?

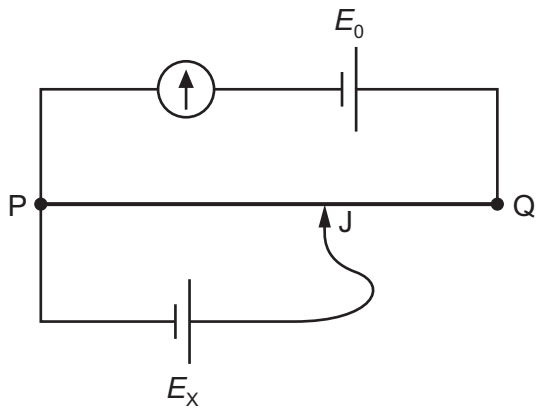
A



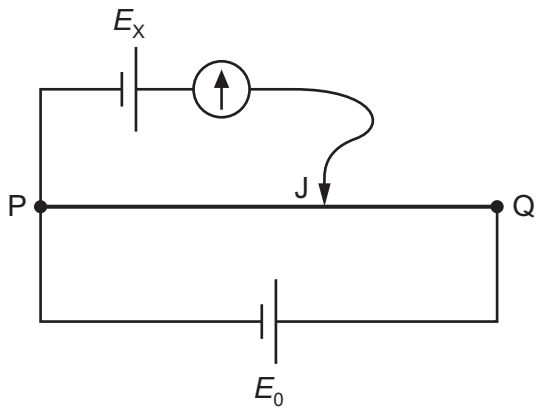
B



C



D



37 The principles of conservation of which two quantities are associated with Kirchhoff's first and second laws?

	first law	second law
A	charge	energy
B	charge	voltage
C	energy	charge
D	voltage	charge

38 An up quark in a hadron changes to a down quark.

The elementary charge is e.

What is the magnitude of the change in the charge of the hadron?

A $\frac{1}{3}e$ **B** $\frac{2}{3}e$ **C** e **D** zero

39 A nucleus of magnesium-23 decays to form a nucleus of sodium-23, emitting a β^+ particle and particle X.

What is particle X?

A an antineutrino

B an electron

C a neutron

D a neutrino

40 A nucleus of $^{238}_{92}\text{U}$ decays in stages by emitting α-particles and β^- particles, eventually forming a nucleus of $^{206}_{82}\text{Pb}$.

How many α-particles and how many β^- particles are emitted during the decay chain?

	α-particles	β^- particles
A	8	6
B	8	10
C	16	6
D	16	22

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PHYSICS **9702/12**

Paper 1 Multiple Choice **October/November 2025**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	B	**1**
2	D	**1**
3	C	**1**
4	C	**1**
5	B	**1**
6	B	**1**
7	A	**1**
8	C	**1**
9	D	**1**
10	B	**1**
11	A	**1**
12	D	**1**
13	A	**1**
14	C	**1**
15	C	**1**
16	A	**1**
17	C	**1**
18	A	**1**
19	B	**1**
20	B	**1**
21	A	**1**
22	A	**1**
23	C	**1**
24	D	**1**
25	B	**1**
26	D	**1**
27	C	**1**
28	D	**1**

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Question	Answer	Marks
29	A	**1**
30	C	**1**
31	C	**1**
32	D	**1**
33	B	**1**
34	D	**1**
35	B	**1**
36	D	**1**
37	A	**1**
38	C	**1**
39	D	**1**
40	A	**1**

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November 2025 · Paper 13

9702_w25_qp_13

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **October/November 2025**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

1700426015

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 The table shows some physical quantities.

Which row correctly identifies the quantities as scalars or vectors?

	acceleration	charge	kinetic energy	wavelength
A	scalar	vector	vector	scalar
B	vector	vector	scalar	scalar
C	scalar	scalar	scalar	vector
D	vector	scalar	scalar	scalar

2 Two quantities are measured.

$$L = 6.8 \pm 0.1\,\text{cm}$$

$$T = 2.42 \pm 0.08\,\text{s}$$

L and T are related to X by the equation shown.

$$X = \frac{4\pi^2 L}{T^2}$$

What is the calculated value and uncertainty of X?

A $45.8 \pm 0.2\,\text{cm s}^{-2}$

B $45.8 \pm 0.3\,\text{cm s}^{-2}$

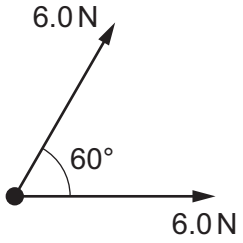
C $46 \pm 2\,\text{cm s}^{-2}$

D $46 \pm 4\,\text{cm s}^{-2}$

3 What are the SI base units of the watt?

A J s^{-1} **B** $\text{kg m}^2\text{ s}^{-3}$ **C** $\text{kg m}^2\text{ s}^{-1}$ **D** N m s^{-1}

4 The diagram shows two forces of 6.0 N acting on an object. The angle between the lines of action of the two forces is 60°.



What is the magnitude of the resultant force?

A 6.0 N **B** 7.9 N **C** 10 N **D** 12 N

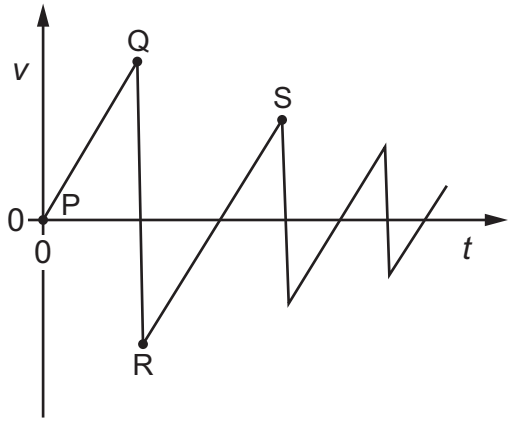
5 A goods train passes through a station at a constant speed of $10\,m\,s^{-1}$ at time $t = 0$. An express train is at rest at the station. The express train leaves the station with a uniform acceleration of $0.5\,m\,s^{-2}$ just as the goods train goes past. Both trains move in the same direction on straight, parallel tracks.

At which time t does the express train overtake the goods train?

A 6 s **B** 10 s **C** 20 s **D** 40 s

6 A ball is held above the ground and released. It falls to the ground and bounces several times.

The graph shows the variation with time t of the velocity v of the ball.



Four points on the graph are labelled P, Q, R and S.

Which statement is **not** correct?

A The area under line PQ represents the initial height of the ball.

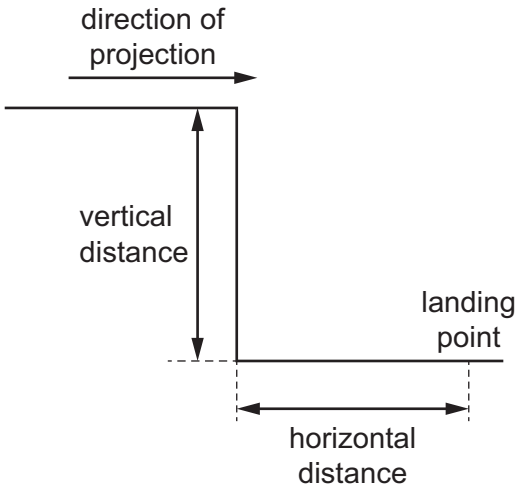
B The collisions of the ball with the ground are inelastic.

C The gradient of the line RS represents the acceleration due to free fall.

D The maximum upwards velocity of the ball is reached at point P.

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7 An object is projected horizontally. The object falls a vertical distance y and travels a horizontal distance x before landing on the ground.



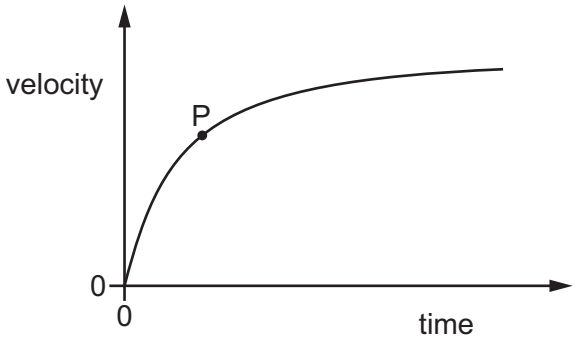
A second object is projected horizontally with the same initial velocity and falls a vertical distance $4y$ before landing on the ground.

Assume that air resistance is negligible.

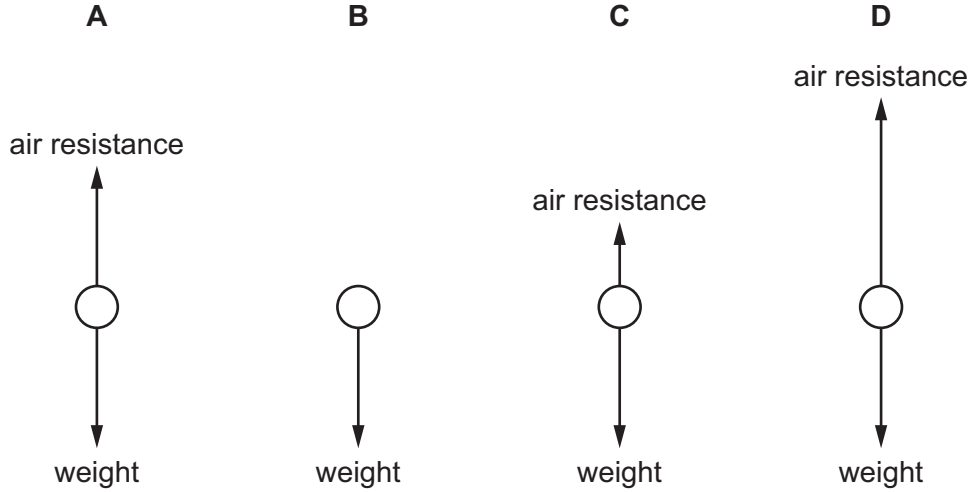
Which horizontal distance does the second object travel?

A x **B** $2x$ **C** $4x$ **D** $16x$

8 A sphere falls from rest through the air. The graph shows the variation with time of the sphere's velocity.



Which diagram shows the forces acting on the sphere when it is at the velocity corresponding to point P on the graph?



9 In which situation is total linear momentum **always** conserved?

A in all collisions between two objects

B in collisions between two objects moving at equal and opposite velocity

C in collisions between two objects that form an isolated system

D in collisions between two objects with the same mass

10 An object of mass m moving with velocity $9.0\,m\,s^{-1}$ has a head-on elastic collision with a stationary object of mass $2m$.

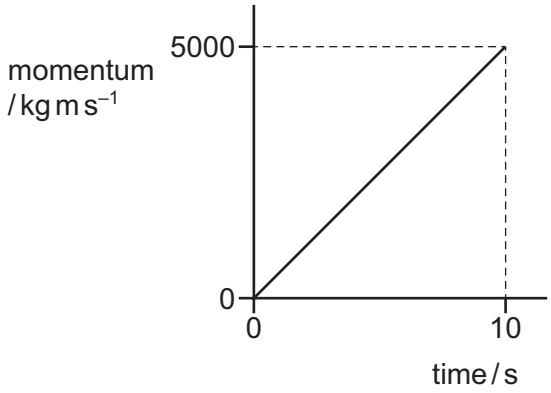
After the collision, both objects are moving. No external forces act on the system.

What is the velocity of the object of mass $2m$ after the collision?

A $-6.0\,m\,s^{-1}$ **B** $-3.0\,m\,s^{-1}$ **C** $4.5\,m\,s^{-1}$ **D** $6.0\,m\,s^{-1}$

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11 The graph shows how the momentum of a motorcycle changes with time.



What is the resultant force on the motorcycle?

A 500 N **B** 5000 N **C** 25 000 N **D** 50 000 N

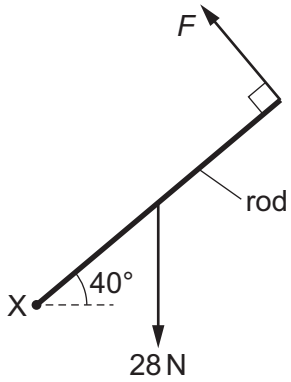
12 A rocket has a weight of W and an initial acceleration of a as the rocket leaves the ground vertically. The acceleration of free fall is g.

Which expression gives the initial upward force exerted on the rocket due to the engine?

A $\frac{Wa}{g} + W$ **B** $\frac{Wa}{g}$ **C** $\frac{Wa}{g} - W$ **D** $Wa + W$

13 A uniform rod of weight 28 N is supported at one end by force F.

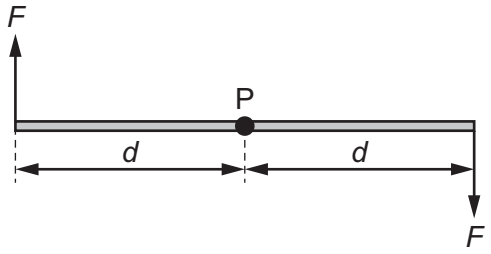
The rod is in equilibrium and attached to a frictionless hinge at end X.



What is the magnitude of F?

A 9.0 N **B** 11 N **C** 18 N **D** 21 N

14 Two forces, each of magnitude F, act in opposite directions on a rod.



Each force acts on the rod at a distance d from the pivot P.

What is the torque of this couple about P?

A 0 **B** $F \times d$ **C** $2F \times d$ **D** $2F \times 2d$

15 A man of weight 600 N stands with both feet flat on the ground.

What is a reasonable estimate of the pressure exerted on the ground by the weight of the man?

A 1×10^0 Pa **B** 1×10^2 Pa **C** 1×10^4 Pa **D** 1×10^6 Pa

16 The formula for the upthrust on a block of wood partially submerged in water is shown.

$$F = \rho g V$$

What do the symbols ρ and V represent?

	ρ	V
A	density of water	volume of whole block
B	density of water	volume of block below surface of water
C	density of wood	volume of whole block
D	density of wood	volume of block below surface of water

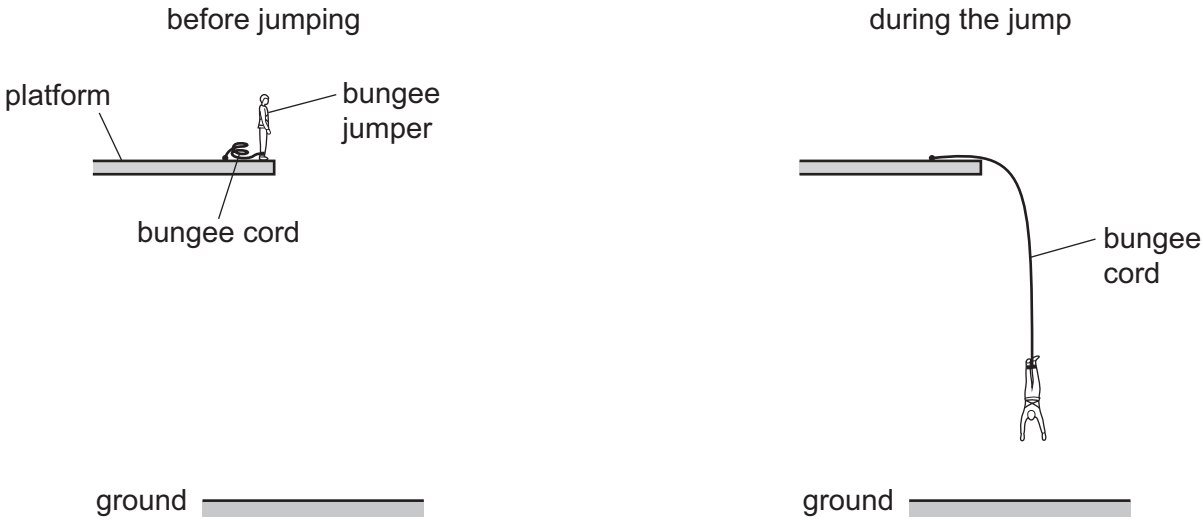
17 An object with a mass of 100 g falls a vertical distance of 10 m.

What is the change in the gravitational potential energy of the object?

A 1 J **B** 10 J **C** 100 J **D** 10 000 J

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18 A bungee jumper jumps from a platform and is decelerated by an elastic bungee cord, as shown.



When the jumper makes the jump, his initial gravitational potential energy relative to the ground is converted into his kinetic energy and into elastic potential energy in the cord.

At which part of the jump are all three types of energy non-zero?

A on the platform before the jump

B on the way down before the cord has started to extend

C on the way down as he decelerates

D at the bottom of the jump when he is stationary

19 A sailboat is pushed by the wind at a constant velocity *v* across a lake.

The force of the wind and the velocity of the sailboat are in the same direction.

Which additional information is required to determine the work done per unit time by the wind acting on the sailboat?

A the force exerted by the wind

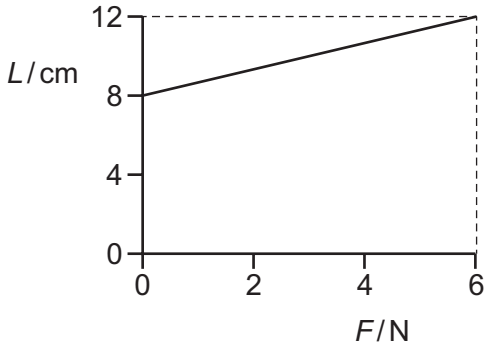
B the distance travelled by the sailboat per unit time

C the total energy input to the sailboat by the wind

D the weight of the sailboat

20 A spring is fixed at one end. The length L of the spring is increased by applying a tensile force F to the other end.

The graph shows the variation of L with F.



What is the elastic potential energy of the spring when F is 6.0 N?

A 0.12 J **B** 0.24 J **C** 0.36 J **D** 0.60 J

21 A wire has an unstretched length of 2.00 m.

A stress of 1.6×10^5 Pa is applied to the wire, and the new length of the wire is 2.10 m.

The wire obeys Hooke's law.

What is the Young modulus of the wire?

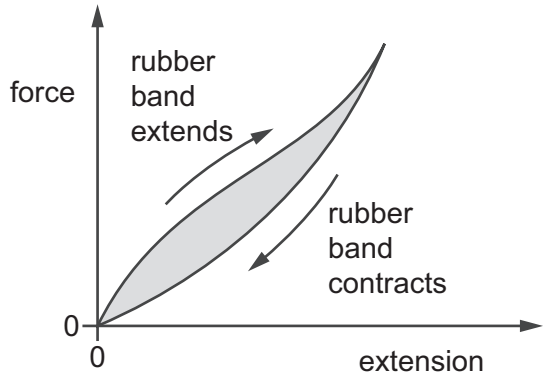
A 8.0×10^3 Pa **B** 7.8×10^4 Pa **C** 1.5×10^5 Pa **D** 3.2×10^6 Pa

22 What are the SI base units of stress?

A kg m s^{-2} **B** $\text{kg m}^{-1}\text{ s}^{-2}$ **C** $\text{kg m}^{-2}\text{ s}^{-2}$ **D** $\text{kg m}^{-3}\text{ s}^{-2}$

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23 The diagram shows a force–extension graph for a rubber band as the band is extended and then the stretching force is decreased to zero.



What can be deduced from the graph?

A The rubber band does **not** return to its original length when the force is decreased to zero.

B The rubber band obeys Hooke's law for the extensions shown.

C The rubber band remains elastic for the extensions shown.

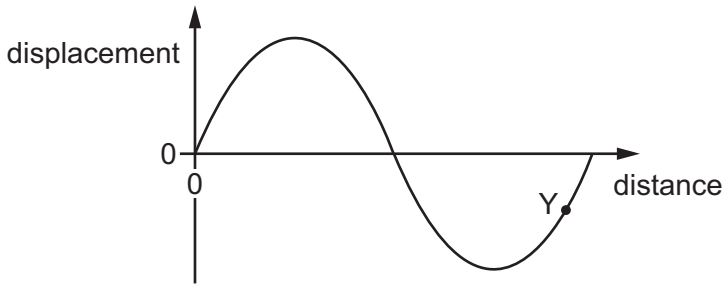
D The shaded area represents the work done in extending the rubber band.

24 Which row is correct for sound waves?

	can travel in a vacuum	can be polarised	can diffract through a gap
A	no	no	yes
B	yes	no	no
C	no	yes	no
D	no	yes	yes

25 A source emits a progressive sound wave in the horizontal direction. The wave travels away from the source in a direction towards the right.

The graph shows the variation of the displacement of the particles from their equilibrium positions with distance from the source at a particular instant in time. Displacements to the right of the equilibrium positions are shown as positive.



Position Y represents the displacement of a particle in the sound wave at a particular distance from the source at this instant.

Which statement about the motion of this particle is correct?

A The particle is moving away from the source.

B The particle is moving towards the source.

C The particle is moving upwards.

D The particle is moving downwards.

26 Two polarising filters are placed next to each other so that their planes are parallel.

The first polarising filter has its transmission axis at an angle of 50° to the vertical.

The second polarising filter has its transmission axis at an angle of 20° to the vertical. The angle between the transmission axes of the two polarising filters is 30°.

A beam of vertically polarised light of intensity $8.0\,W\,m^{-2}$ is incident normally on the first polarising filter.

What is the intensity of the light that is transmitted from the second polarising filter?

A zero **B** $2.5\,W\,m^{-2}$ **C** $2.9\,W\,m^{-2}$ **D** $6.0\,W\,m^{-2}$

27 A dolphin is swimming at a speed of $16.0\,m\,s^{-1}$ directly towards a stationary underwater microphone.

The dolphin emits a sound of frequency 122 kHz. The speed of sound in water is $1480\,m\,s^{-1}$.

What is the frequency of sound detected by the microphone?

A 1300 Hz **B** 1330 Hz **C** 121 kHz **D** 123 kHz

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28 A student carries out a double-slit experiment using a laser emitting red light of wavelength of 680 nm. The light is incident normally on a double slit.

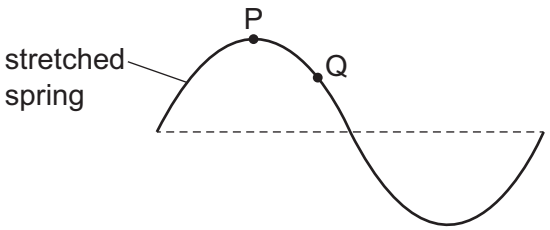
The diagram shows part of the pattern of bright fringes visible on a screen at a distance of 2.4 m from the slits. The distance across five bright fringes is measured as 34 mm.



What is the slit separation?

A 8.5×10^{-3} m **B** 2.4×10^{-4} m **C** 1.9×10^{-4} m **D** 4.8×10^{-5} m

29 The diagram shows a stationary wave on a stretched spring at an instant in time.



Two particles on the spring, P and Q, are shown.

Which statement about the vibrations of P and Q is correct?

A They have different frequencies.

B They have the same amplitudes.

C They have different periods.

D They are always in phase.

30 Green laser light passes through a diffraction grating and forms an interference pattern.

The diffraction grating contains 400 lines per mm.

The wavelength of the laser light is 550 nm.

What is the highest order diffraction maximum produced by the grating?

A 4 **B** 5 **C** 8 **D** 9

31 There is a potential difference V across a resistor of resistance R. The current in the resistor is I.

Which equation gives the power P dissipated by the resistor?

A $P = V^2 I$ **B** $P = \frac{V^2}{I}$ **C** $P = V^2 R$ **D** $P = \frac{V^2}{R}$

32 What is a possible charge on a particle?

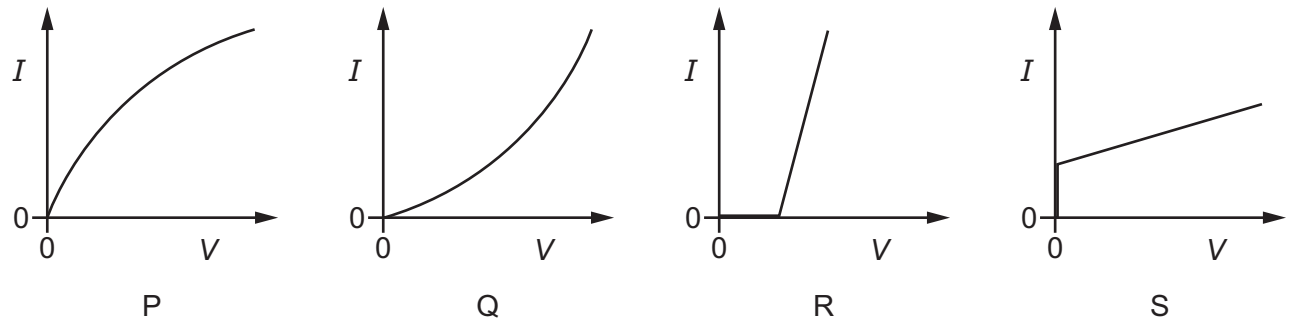
A 6.40×10^{-20} C

B 4.00×10^{-19} C

C 1.12×10^{-18} C

D 9.11×10^{-18} C

33 The graphs show possible current–voltage (I–V) characteristics for a filament lamp and for a semiconductor diode.



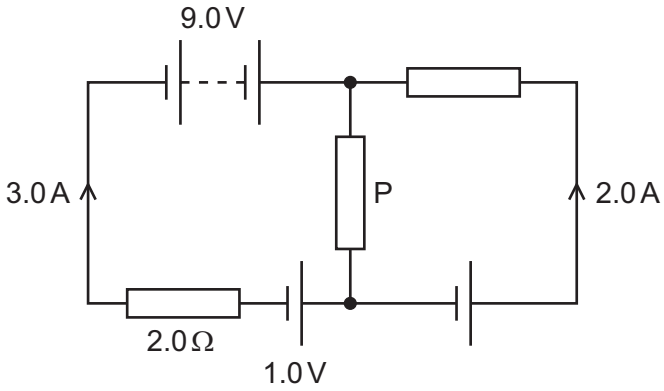
Which row identifies the I–V graphs for the lamp and for the diode?

	filament lamp	semiconductor diode
A	P	R
B	P	S
C	Q	R
D	Q	S

34 The diagram shows a circuit containing a battery and cells with negligible internal resistance.

Some values of current, electromotive force (e.m.f.) and resistance are shown.

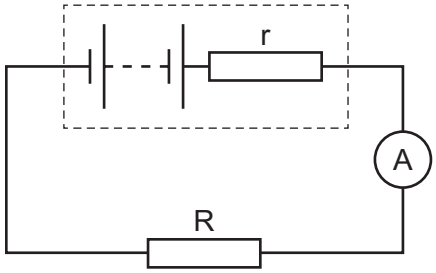
One resistor is labelled P.



What is the resistance of resistor P?

A 0.40 Ω **B** 0.80 Ω **C** 2.0 Ω **D** 2.8 Ω

35 The diagram shows a circuit with a battery connected to a resistor R. The battery has an internal resistance represented by resistor r.

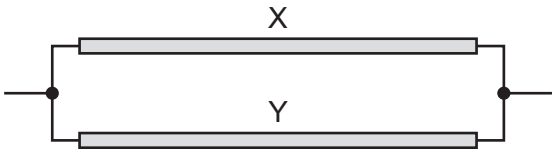


A second resistor, identical to R, is connected in parallel with R.

Which row describes the changes to the potential difference (p.d.) across r and the current shown on the ammeter when the second resistor is connected?

	p.d. across r	current shown on ammeter
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

36 Two identical wires X and Y, each of length L and radius r, are connected in parallel as shown.



The total resistance of this combination is R_1.

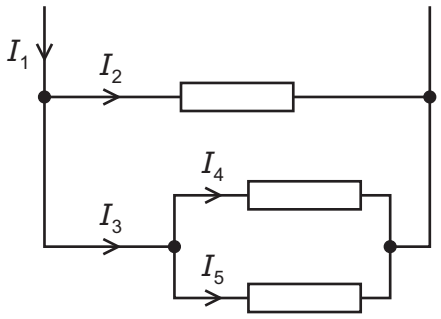
Wire X is replaced with a wire of the same material with length L and radius $2r$.

The total resistance of the new combination is R_2.

What is the ratio $\frac{R_1}{R_2}$?

A $\frac{2}{5}$ **B** $\frac{2}{3}$ **C** $\frac{3}{2}$ **D** $\frac{5}{2}$

37 The diagram shows the currents in part of an electric circuit.



The resistors are identical.

Which equation is **not** correct?

A $I_1 = I_2 + I_4 + I_5$

B $I_2 = I_1 - I_3$

C $I_2 = I_3 + I_4 + I_5$

D $I_4 = I_3 - I_5$

38 The diagram shows a sequence of radioactive decays involving three α-particles and a β^- particle.

$$^{237}_{93}\text{Np} \xrightarrow{\alpha} \text{Q} \xrightarrow{\beta^-} \text{R} \xrightarrow{\alpha} \text{S} \xrightarrow{\alpha} \text{T}$$

What is nuclide T?

A $^{225}_{88}\text{Ra}$ **B** $^{231}_{88}\text{Ra}$ **C** $^{225}_{90}\text{Th}$ **D** $^{229}_{90}\text{Th}$

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39 The number of electrons in a neutral atom of an isotope of plutonium, $^{239}_{94}Pu$, is changed to produce a charged atom (ion) W.

W has an overall charge of $+1.6 \times 10^{-19}$ C.

How many protons, neutrons and electrons are in W?

	protons	neutrons	electrons
A	94	145	93
B	94	145	95
C	238	94	239
D	239	94	238

40 Which particle is a lepton?

A meson

B positron

C proton

D quark

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Cambridge International AS & A Level

PHYSICS **9702/13**

Paper 1 Multiple Choice **October/November 2025**

MARK SCHEME

Maximum Mark: 40

Published

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Question	Answer	Marks
1	D	**1**
2	D	**1**
3	B	**1**
4	C	**1**
5	D	**1**
6	D	**1**
7	B	**1**
8	C	**1**
9	C	**1**
10	D	**1**
11	A	**1**
12	A	**1**
13	B	**1**
14	C	**1**
15	C	**1**
16	B	**1**
17	B	**1**
18	C	**1**
19	A	**1**
20	A	**1**
21	D	**1**
22	B	**1**
23	C	**1**
24	A	**1**
25	B	**1**
26	B	**1**
27	D	**1**
28	C	**1**

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Question	Answer	Marks
29	D	**1**
30	A	**1**
31	D	**1**
32	C	**1**
33	A	**1**
34	A	**1**
35	D	**1**
36	D	**1**
37	C	**1**
38	A	**1**
39	A	**1**
40	B	**1**

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November 2025 · Paper 14

9702_w25_qp_14

Cambridge Assessment International Education

Cambridge International AS & A Level

PHYSICS **9702/14**

Paper 1 Multiple Choice **October/November 2025**

1 hour 15 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.

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Data

acceleration of free fall	$g = 9.81\,\text{m s}^{-2}$
speed of light in free space	$c = 3.00 \times 10^{8}\,\text{m s}^{-1}$
elementary charge	$e = 1.60 \times 10^{-19}\,\text{C}$
unified atomic mass unit	$1\,\text{u} = 1.66 \times 10^{-27}\,\text{kg}$
rest mass of proton	$m_\text{p} = 1.67 \times 10^{-27}\,\text{kg}$
rest mass of electron	$m_\text{e} = 9.11 \times 10^{-31}\,\text{kg}$
Avogadro constant	$N_\text{A} = 6.02 \times 10^{23}\,\text{mol}^{-1}$
molar gas constant	$R = 8.31\,\text{J K}^{-1}\,\text{mol}^{-1}$
Boltzmann constant	$k = 1.38 \times 10^{-23}\,\text{J K}^{-1}$
gravitational constant	$G = 6.67 \times 10^{-11}\,\text{N m}^2\,\text{kg}^{-2}$
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\,\text{F m}^{-1}$ $\left(\frac{1}{4\pi\varepsilon_0} = 8.99 \times 10^{9}\,\text{m F}^{-1}\right)$
Planck constant	$h = 6.63 \times 10^{-34}\,\text{J s}$
Stefan–Boltzmann constant	$\sigma = 5.67 \times 10^{-8}\,\text{W m}^{-2}\,\text{K}^{-4}$

Formulae

uniformly accelerated motion	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
hydrostatic pressure	$\Delta p = \rho g \Delta h$
upthrust	$F = \rho g V$
Doppler effect for sound waves	$f_\text{o} = \frac{f_\text{s} v}{v \pm v_\text{s}}$
electric current	$I = Anvq$
resistors in series	$R = R_1 + R_2 + \ldots$
resistors in parallel	$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$

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1 What is essential to accurately represent all physical quantities?

A a base unit and a number

B a unit and a number expressed in standard form (scientific notation)

C a unit and a numerical magnitude

D an SI unit and a numerical magnitude

2 A steel rule can be read to the nearest millimetre. It is used to measure the length of a bar whose true length is 895 mm. Repeated measurements give the following readings.

length / mm 892, 891, 892, 891, 891, 892

Are the readings accurate and precise to within 1 mm?

	results are accurate to within 1 mm	results are precise to within 1 mm
A	no	no
B	no	yes
C	yes	no
D	yes	yes

3 A stone is released from rest and falls vertically to the ground.

The time taken to fall to the ground and the distance travelled are measured. The measurements are used to determine the acceleration of free fall.

The percentage uncertainty in the measured time is 0.05%. The percentage uncertainty in the measured distance fallen is 0.6%.

What is the percentage uncertainty in the calculated value of the acceleration of free fall?

A 0.5% **B** 0.7% **C** 1.1% **D** 1.3%

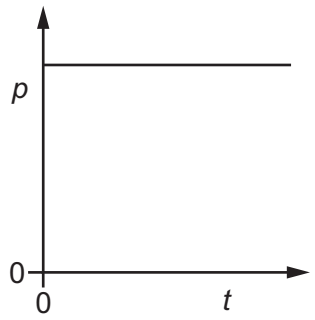
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4 An object falls from rest towards the ground.

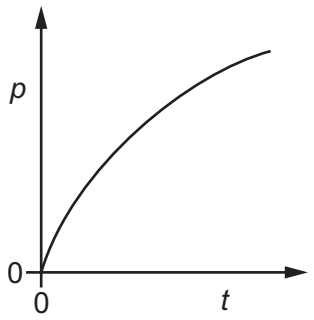
Air resistance is negligible.

Which graph shows the variation of the momentum p of the object with time t until it hits the ground?

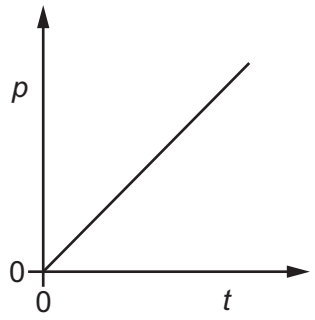
A



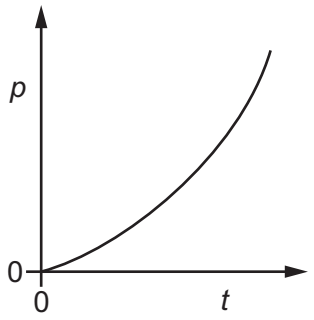
B



C



D



5 A projectile is fired at an angle of 45° upwards from horizontal ground. Air resistance is negligible.

Which row describes the horizontal motion and the vertical motion of the projectile after it is fired until immediately before it reaches the ground again?

	horizontal motion	vertical motion
A	constant velocity	constant acceleration
B	constant velocity	varying acceleration
C	varying velocity	constant acceleration
D	varying velocity	varying acceleration

6 An aircraft on a runway accelerates uniformly from rest to its take-off speed of $58\,m\,s^{-1}$.

The acceleration of the aircraft is $4.2\,m\,s^{-2}$, and the aircraft uses 74% of the length of the runway to reach its take-off speed.

What is the length of the runway?

A 300 m **B** 540 m **C** 590 m **D** 800 m

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7 How can the acceleration of an object be determined?

A from the area under a displacement–time graph

B from the area under a velocity–time graph

C from the gradient of a displacement–time graph

D from the gradient of a velocity–time graph

8 The acceleration of free fall on the Earth is different to the acceleration of free fall on the Moon.

How do the mass and weight of an object on the Earth compare to its mass and weight on the Moon?

	mass	weight
A	different	different
B	different	same
C	same	different
D	same	same

9 A sphere moves vertically downwards at terminal (constant) velocity in a liquid.

Which statement about the magnitude of the upthrust acting on the sphere is correct?

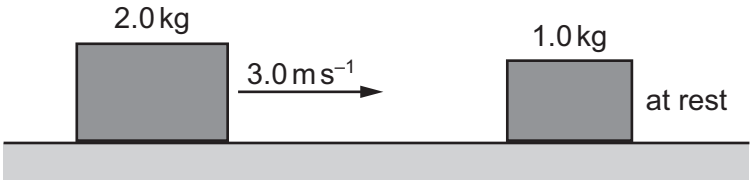
A It is proportional to the acceleration of free fall.

B It is equal to the weight of the sphere.

C It is proportional to the density of the sphere.

D It is proportional to the square of the radius of the sphere.

10 An object of mass 2.0 kg is travelling at a speed of 3.0 $m\,s^{-1}$ on a horizontal frictionless surface. This object collides head-on with a stationary object of mass 1.0 kg. The two objects stick together on impact.



How much kinetic energy is lost on impact?

A zero **B** 2.0 J **C** 2.4 J **D** 3.0 J

11 A moving object X collides with a stationary object Y.

The objects separate after the collision.

The collision is perfectly elastic and there are no external forces acting.

Which word equation is **not** correct?

A during the collision, (force acting on object X) + (force acting on object Y) = zero

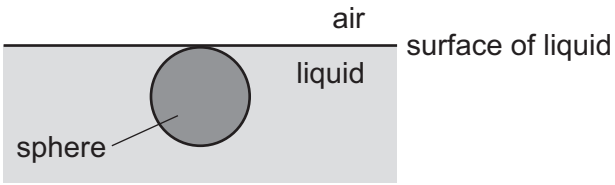
B (relative speed of approach of X and Y) + (relative speed of separation of X and Y) = zero

C (total kinetic energy before collision) = (total kinetic energy after collision)

D (total momentum before collision) = (total momentum after collision)

12 The diagram shows a stationary sphere that is just fully submerged in a liquid. The radius of the sphere is r and the density of the liquid is ρ. The acceleration of free fall is g.

The air exerts pressure P_A on the surface of the liquid.



What is the pressure at the lowest point of the sphere?

A $P_A + \frac{4}{3}\pi r^3 \rho g$

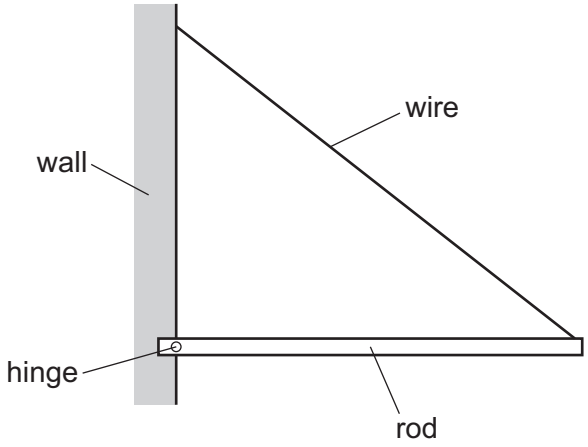
B $P_A - \frac{4}{3}\pi r^3 \rho g$

C $2r\rho g + P_A$

D $2r\rho g - P_A$

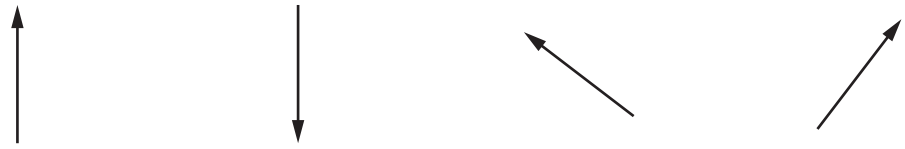
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13 A uniform rod is attached by a hinge at one end to a wall. The other end of the rod is supported by a wire so that the rod is horizontal and in equilibrium.

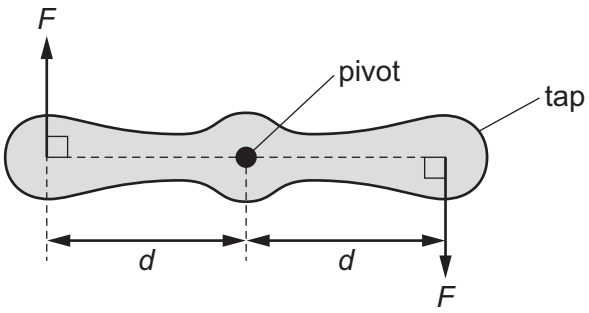


Which arrow shows the direction of the force on the rod from the hinge?

A **B** **C** **D**



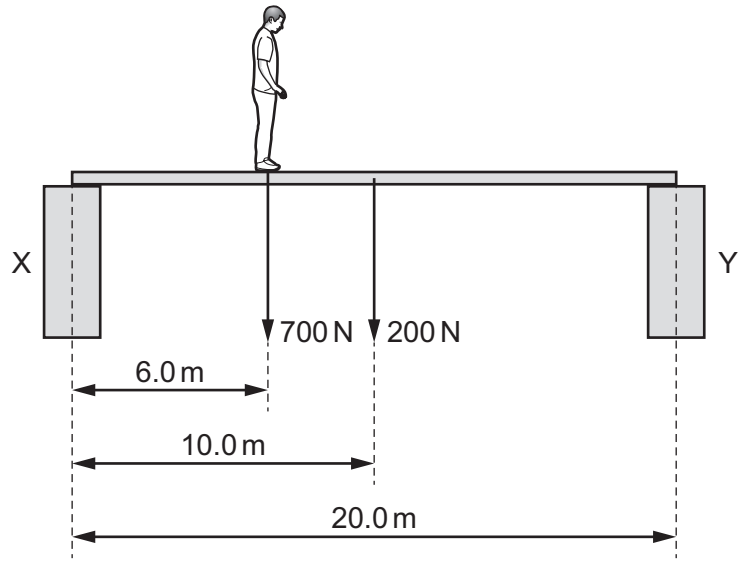
14 A couple is applied to a tap, as shown.



What is the torque of the couple?

A $\frac{Fd}{2}$ **B** Fd **C** $2Fd$ **D** $4Fd$

15 A uniform beam rests on two supports, X and Y, as shown.



not to scale

The beam has length 20.0 m and weight 200 N.

A man of weight 700 N stands on the beam at a distance of 6.0 m from support X.

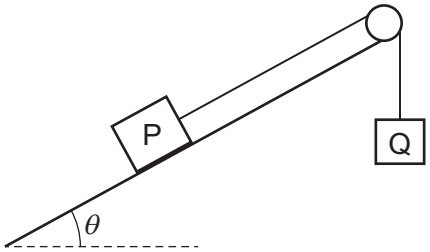
The beam is in equilibrium.

What is the contact force of support X on the beam?

A 310 N **B** 450 N **C** 590 N **D** 900 N

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16 The diagram shows a block P of mass M connected by a string over a frictionless pulley to a block Q of mass m.



Block P moves up the slope with a constant velocity v. The slope is at angle θ to the horizontal.

The acceleration of free fall is g.

The resistive forces on the blocks are negligible.

Which expressions give the energy transferred per unit time to block P?

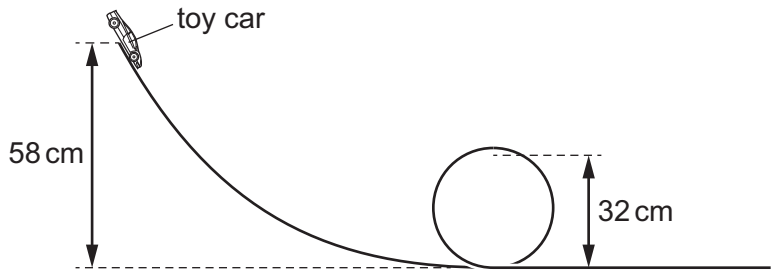
1 mgv

2 $Mgv\sin\theta$

3 $(M + m)gv$

A 1 and 2 **B** 1 only **C** 2 and 3 **D** 2 only

17 A toy car travels around a vertical loop track.



The toy car is released from rest at a height of 58 cm above the bottom of the vertical loop.

The car is at a height of 32 cm when it is at the top of the vertical loop.

Assume that no resistive forces act on the car.

What is the speed of the car at the top of the vertical loop?

A $0.87\,m\,s^{-1}$ **B** $2.3\,m\,s^{-1}$ **C** $2.5\,m\,s^{-1}$ **D** $3.4\,m\,s^{-1}$

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18 The total energy supplied to an electric motor is E. Energy Q is wasted and the remaining energy does useful work.

What is the efficiency of the motor?

A $\frac{Q}{E}$ **B** $\left(\frac{Q}{E}\right) - 1$ **C** $1 - \left(\frac{Q}{E}\right)$ **D** $\frac{(1-Q)}{E}$

19 An object travels between two points. The change in gravitational potential energy ΔE_p of the object is given by

$$\Delta E_p = mg\Delta h$$

where m is the mass of the object, Δh is its change in height and g is the acceleration of free fall.

What is a necessary condition in order for the above equation to be valid?

A The object must have a constant acceleration of $9.81\,m\,s^{-2}$.

B The object must be travelling in a uniform gravitational field.

C The object must be travelling only in a vertical direction.

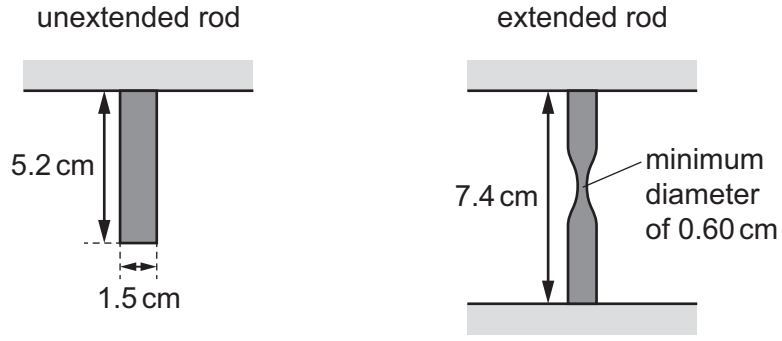
D The resultant force on the object must be equal to its weight.

20 A cylindrical steel rod with negligible weight has a diameter of 1.5 cm and a length of 5.2 cm.

The rod is firmly attached at the top end.

The rod is firmly attached to a machine at the other end that applies a constant force of 363 N to the rod.

This causes the rod to extend to a length of 7.4 cm and to have a minimum diameter of 0.60 cm in the position shown.

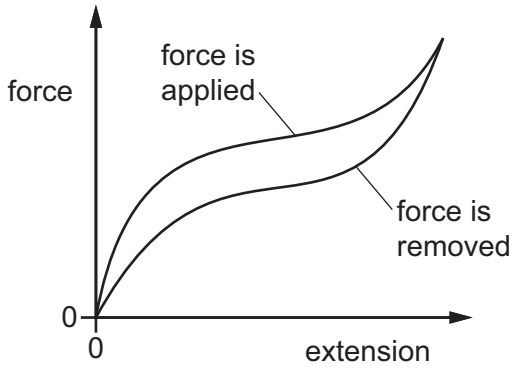


What is the maximum stress acting on the steel rod when the length is 7.4 cm?

A $2.1 \times 10^6\,Pa$ **B** $3.2 \times 10^6\,Pa$ **C** $5.7 \times 10^6\,Pa$ **D** $1.3 \times 10^7\,Pa$

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21 The graph shows the variation in extension with force for a sample of rubber.



The top line shows the variation in extension as a force is applied.

The bottom line shows the variation in extension as the force is removed.

What is represented by the area between the two lines?

A the work done as the force is applied

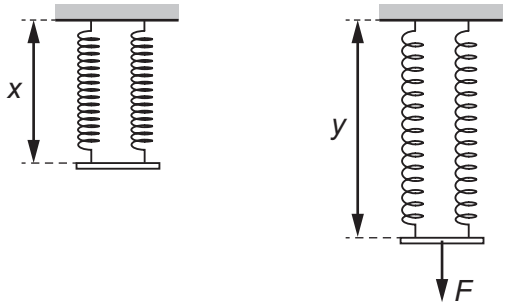
B the work done as the force is applied minus the work done as the force is removed

C the work done as the force is removed

D the work done as the force is removed plus the work done as the force is applied

22 Two identical springs have the same spring constant k. The springs are connected in parallel. The length of the unstretched springs is x.

A force F is applied to the spring combination. The length of the springs is now y.



Both springs are deformed within their limits of proportionality.

Which expression gives the elastic potential energy stored in **one** of the springs?

A $\frac{1}{4}F(y-x)$ **B** $\frac{1}{2}F(y-x)$ **C** $\frac{1}{4}k(y-x)^2$ **D** $k(y-x)^2$

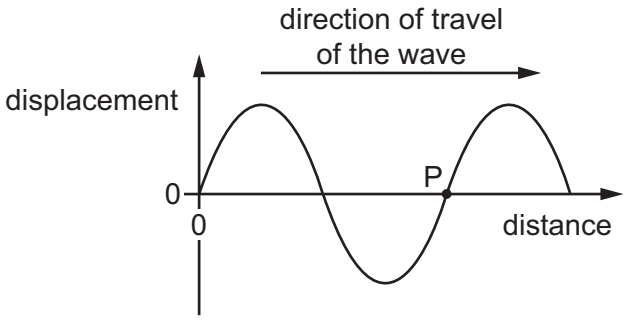
23 Which expression gives the formula for the spring constant?

A Fx **B** $2Fx^2$ **C** $\frac{F}{x}$ **D** $\frac{F}{x^2}$

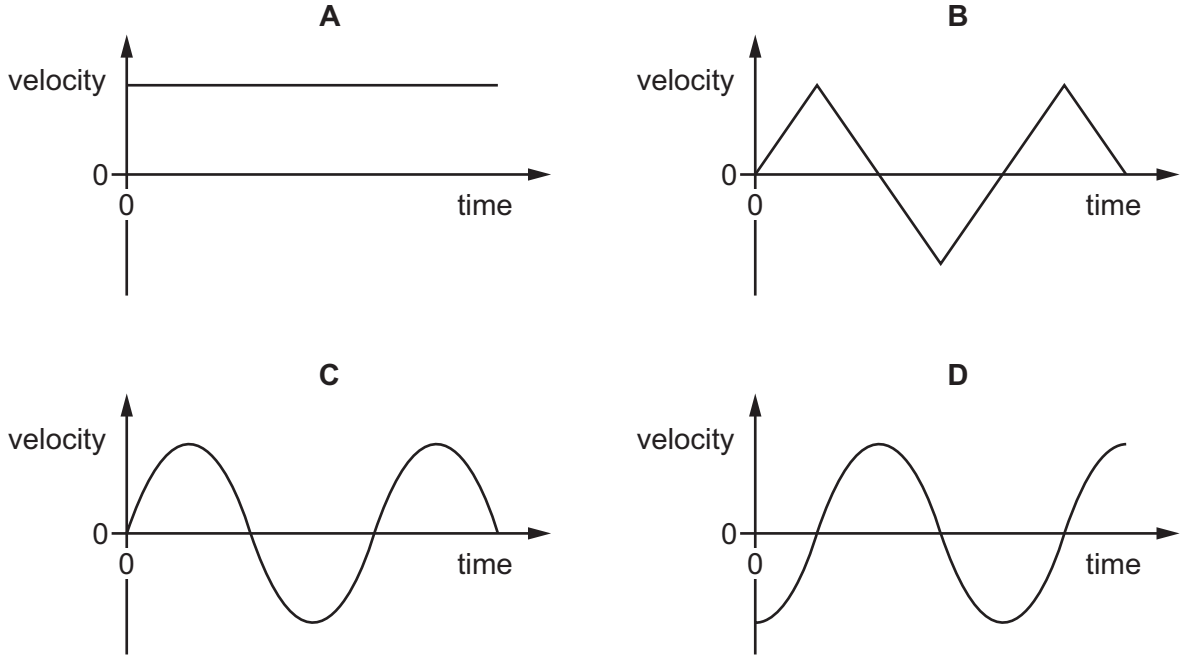
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24 A transverse progressive wave travels along a string.

The graph shows the variation with distance of the displacement of the string at time $t = 0$.



Which graph represents the variation with time of the velocity of point P on the string?



25 Which statement about electromagnetic waves is correct?

A A wave of wavelength 5.0×10^{-6} m is invisible to the human eye.

B They can all travel at different speeds in free space.

C They **cannot** be polarised.

D They consist of vibrating atoms.

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26 For a progressive transverse wave, what describes the term diffraction?

A the change in observed frequency of a wave due to a movement of the wave source

B the spreading of a wave as it passes through a gap or around an obstacle

C when oscillations of a wave are confined to one plane

D the resultant displacement of two waves is equal to the sum of the displacements of the individual waves when the waves meet

27 An aircraft flies at a velocity v_s directly away from a stationary observer.

The aircraft emits a sound of constant frequency.

The speed of sound in air is v.

The frequency of the sound heard by the observer on the ground is 500 Hz.

The speed of the aircraft is increased so that it flies away from the observer at a greater velocity.

The observer now hears a sound of frequency 250 Hz.

Which expression gives the new velocity of the aircraft?

A $2(v + v_s)$ **B** $\frac{v + v_s}{2}$ **C** $2v_s + v$ **D** $2v_s - v$

28 What may be observed with light waves but **not** with sound waves?

A diffraction

B interference

C polarisation

D reflection

29 A wire is fixed at both ends and is vibrated by a source of frequency f. A stationary wave is formed on the wire with a total of two antinodes.

The frequency of the source is increased to $3f$ and a new stationary wave is formed.

What is the total number of antinodes on the new wave?

A 4 **B** 5 **C** 6 **D** 9

30 A parallel beam of red light of wavelength 700 nm is incident normally on a diffraction grating that has 400 lines per millimetre.

What is the total number of intensity maxima from the grating?

A 6 **B** 7 **C** 8 **D** 9

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31 Two light sources are used to produce an interference pattern.

Interference fringes appear when the two sources emit waves that are coherent.

What is meant by coherent waves?

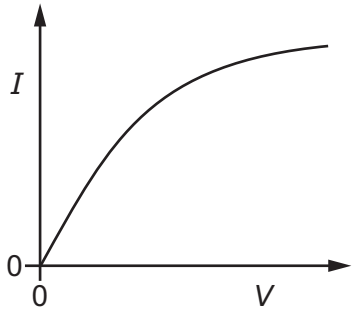
A The two waves are emitted with the same frequency.

B The two waves are emitted with constant phase difference.

C The two waves are emitted with the same intensity.

D The two waves are emitted with zero phase difference.

32 Which component has the I–V graph shown?



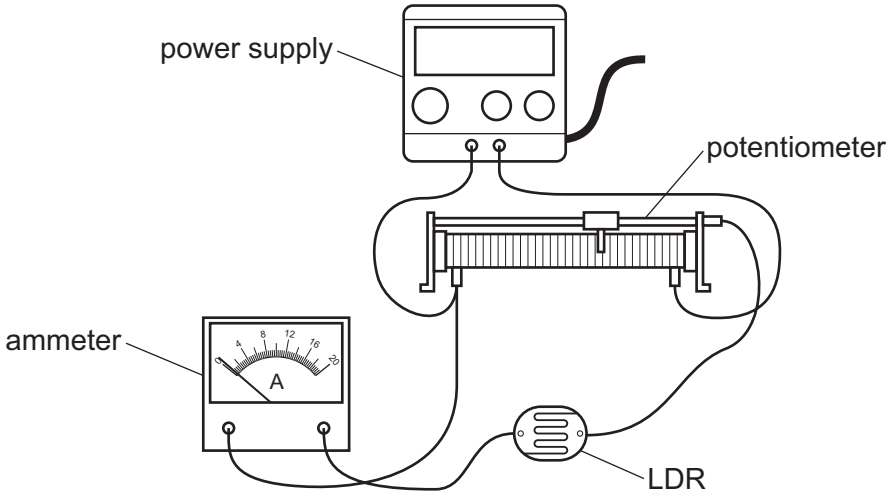
A filament lamp

B metallic conductor at constant temperature

C resistor of fixed resistance

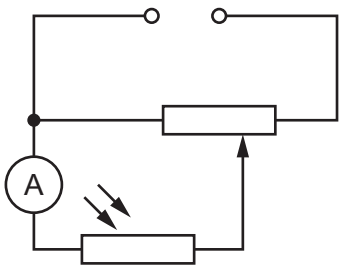
D semiconductor diode

33 The diagram shows an electrical circuit containing a light-dependent resistor (LDR).

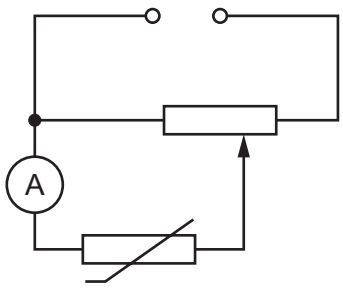


What is the circuit diagram for this circuit?

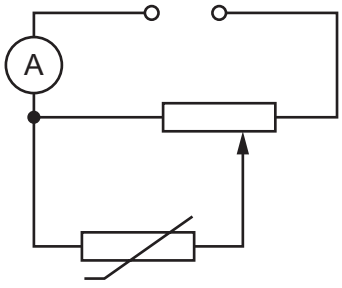
A



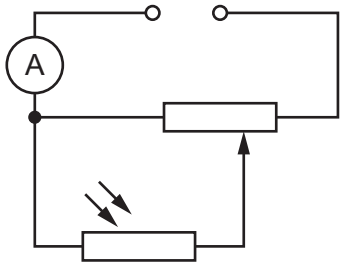
B



C



D



34 A copper wire of length 2 m has a circular cross-section. There is a constant current in the wire when it is connected to a mobile phone in order to charge the battery.

The wire has a fault. A 1 m length of the wire has half the diameter of the other 1 m length.

The power dissipated in the thicker length of wire is P.

What is the power dissipated in the thinner length of wire?

A $\frac{1}{4}P$ **B** $\frac{1}{2}P$ **C** $2P$ **D** $4P$

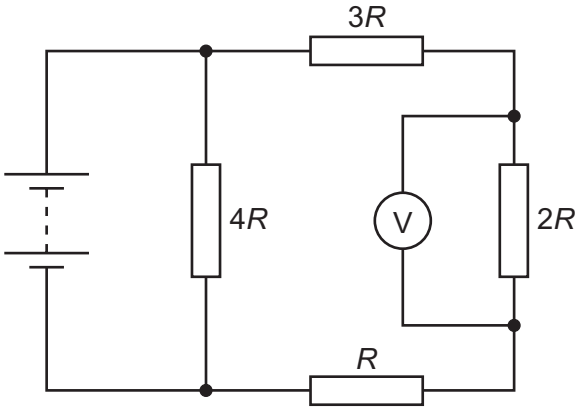
35 A wire carries a current of 5.6 A.

What is the number of conduction electrons that pass a point on the wire in a time of 20 s?

A 1.8×10^{18} **B** 2.2×10^{19} **C** 3.5×10^{19} **D** 7.0×10^{20}

36 Four resistors of resistance R, $2R$, $3R$ and $4R$ are connected to form a network.

A battery of negligible internal resistance and a voltmeter are connected to the resistor network as shown.

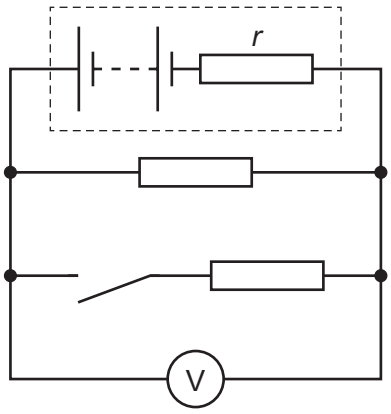


The voltmeter reading is 2 V.

What is the electromotive force (e.m.f.) of the battery?

A 2 V **B** 4 V **C** 6 V **D** 10 V

37 The circuit diagram shows a battery with internal resistance *r*, two resistors, a switch and a voltmeter. The switch is open.



The switch is now closed.

What happens to the current in the battery, and what happens to the reading on the voltmeter, when the switch is closed?

	current in battery	reading on voltmeter
A	increases	decreases
B	increases	remains the same
C	remains the same	decreases
D	remains the same	remains the same

38 What is the correct equation for β^+ decay?

A neutron $\rightarrow$ proton + electron + electron antineutrino

B neutron $\rightarrow$ proton + electron + electron neutrino

C proton $\rightarrow$ neutron + positron + electron antineutrino

D proton $\rightarrow$ neutron + positron + electron neutrino

39 A nucleus of polonium, $^{214}_{84}$Po, decays by emitting an α-particle to become a nucleus of lead.

The nucleus of lead is also unstable and decays by emitting a β^- particle to form a nucleus of bismuth which then decays by emitting a β^- particle to produce a nucleus X.

What is the number of neutrons in nucleus X?

A 126 **B** 128 **C** 130 **D** 210

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40 Which combination of three quarks has no overall charge?

A down, strange, strange

B up, charm, strange

C up, charm, top

D up, down, strange

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Paper 1 Multiple Choice **October/November 2025**

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Question	Answer	Marks
1	C	**1**
2	B	**1**
3	B	**1**
4	C	**1**
5	A	**1**
6	B	**1**
7	D	**1**
8	C	**1**
9	A	**1**
10	D	**1**
11	B	**1**
12	C	**1**
13	D	**1**
14	C	**1**
15	C	**1**
16	A	**1**
17	B	**1**
18	C	**1**
19	B	**1**
20	D	**1**
21	B	**1**
22	A	**1**
23	C	**1**
24	D	**1**
25	A	**1**
26	B	**1**
27	C	**1**
28	C	**1**

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Question	Answer	Marks
29	C	**1**
30	B	**1**
31	B	**1**
32	A	**1**
33	A	**1**
34	D	**1**
35	D	**1**
36	C	**1**
37	A	**1**
38	D	**1**
39	A	**1**
40	D	**1**

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